Advanced Combustion via Microgravity Experiments (ACME)

Software Interface Design Description

September, 2017





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SOFTWARE INTERFACE DESIGN DESCRIPTION

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REVISIONS
ACME
SOFTWARE INTERFACE DESIGN DESCRIPTION

Revision Letter	Publication Date	Change Description	Approval
–	11/2013	Baseline Release	
A	02/2016	Major expansion to include detailed information about commands, telmetry, files, and interfaces.	
B	05/2016	Filled in extinction detection commands. Changes to telemetry packet. Changes to registry values. ACME-ECO-611	EM 5/16/2016 BB 5/16/2016
C	08/2016	Added system script commands. Added 3rd order to EField control voltage equation. Changes to telemetry packet. Changes to registry values.	EM 8/30/2016 BB 8/30/2016
D	09/2017	Updates to commands and telemetry to reflect state of software during initial operations.	EM 9/25/2017 BB 9/25/2017

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1.0 SCOPE

1.1 Identification

The purpose of this document is to identify the design of the communication interfaces used by the Advanced Combustion via Microgravity Experiments (ACME).

1.2 System Overview

The Advanced Combustion via Microgravity Experiments (ACME) is mini-facility designed to support Principal Investigators (PI) interested in studying various aspects of advanced combustion in a microgravity environment. ACME will be integrated into the Fluids and Combustion Facility's (FCF) Combustion Integrated Rack (CIR) onboard the International Space Station in the US Lab Module.

ACME will perform microgravity experiments to study fire suppression/flame extinguishment via 5 experiments: s-flame, Flame design, CLD Flame, E-Field Flames, and Burning Rate Emulator Flames as described in the ACME Baseline Description Document, ACME-DOC-004. ACME will allow investigation using pure as well as mixtures of gaseous fuels at various chamber atmosphere pressures, compositions, and oxygen concentrations. The microgravity environment allows for observation and measurement of flame formation and extinguishment processes that cannot be made in sustained gravity. The modular design of ACME allows for flexibility in configuration for specific experiments and easy maintainability of the hardware.

The core elements of gaseous combustion are included in the ACME design as a basis for current and future gaseous combustion studies. These elements are establishment of gas flow through various exchangeable mass flow controllers, ignition of gas at the burner and the use of appropriate diagnostic measurement devices. The five identified experiments, and any future experiments, use these core elements to build a unique set of requirements for the study of various phenomena distinctive to gaseous combustion. The goal of ACME is to improve the fundamental understanding of combustion processes in order to enable increased efficiency and reduce pollutant emission in practical combustion.

1.3 Document Overview

This document is divided into interfaces used to communicate with the ISS (i.e., external communication interfaces) and interfaces used to communicate with devices within ACME that require a protocol (i.e., internal communication interfaces).

2.0 REFERENCED DOCUMENTS

This section shall list the number and title of all documents referenced in this description

Table 1 Referenced Documents

Document Number	Title
NPR 7150.2A	NASA Software Engineering Requirements
GRC-SW-TPLT-IDD	Interface Design Description Template
ACME-REQ-001	Engineering Requirements Document
FCF-ICD-0076A	FCF Software Interface Control Document
FCF-ICD-CIR-ACME	FCF CIR/ACME Interface Control Document

3.0 EXTERNAL COMMUNICATION INTERFACES

ACME commands originate from the ACME Ground Software (AGS) CSCI on a ground workstation at the Telescience Center (TSC) at Glenn Research Center (GRC). These commands are then forwarded through a variety of systems until they eventually end up at the CIR on the ISS.

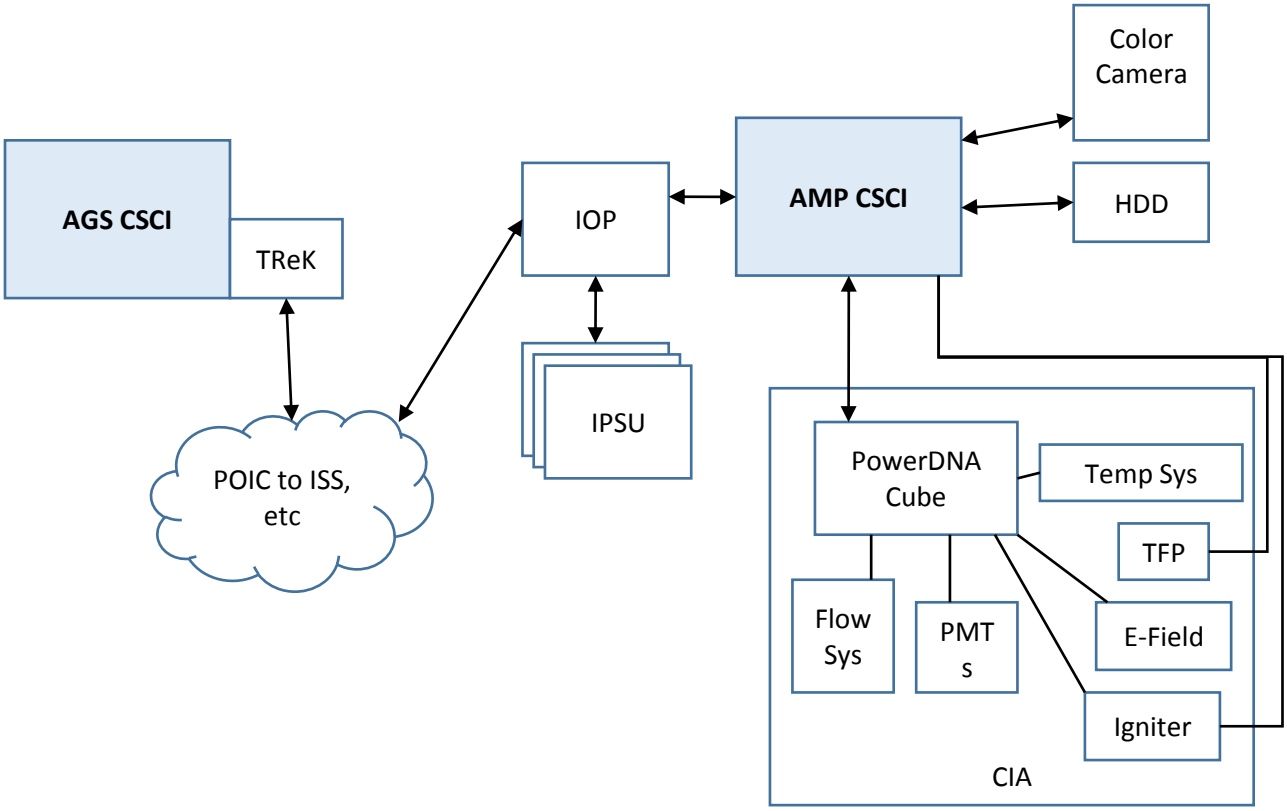


Figure 1 ACME System Diagram

3.1 CIR Input/Output Processor (IOP) Interface

Commands sent through the CIR IOP (Input Output Processor) are in the format required by FCF-ICD-0076A. See section A.1, FCF Command Protocol for a listing of the required command formats, including all required headers. All commands from the ground go to the IOP with the FCF Command Header and User Data filled in. Commands from ACME to the IPSUs and IPSU-Gs go through the FCFCommunications application (Base Classes) which fills in the Command Sequence Count and Time Fields in the FCF Command Header before submitting the packet to the IOP for routing.

3.1.1 ACME Commands

Because it is a CIR PI package, ACME Package Command IDs in the FCF Command Header are assigned the range 0xC000-0xCFFF. The table below lists the commands, their IDs, and mnemonics.

Detailed descriptions and parameter lists for each command follow the table. In the parameter list for each command the parameter name is the variable name in the code, which is useful to know when Qt scripts are being written. The name of the command, for example 'NullCommand', is also the name in code, in Qt scripts, and in log messages announcing the reception and completion of commands.

Regarding parameters of type String, the entire available length can be used for characters without a terminating zero character. If a string is less than the available length then a terminating zero character needs to be used. The ground command software will make this transparent to the normal user.

The Common Group is so-named because it is implemented in code shared between payloads.

Table 2 ACME Command ID Assignments

Cmd ID	Name	Description	Mnemonic
Common Group			
0xC001	NullCommand	Do nothing	ACME_COMMON_NULL
0xC002	ExternalCommand	Pass through to external module	ACME_COMMON_EXTERNAL_PASS_THRU
0xC003	GoToIdle	Set the system state to Idle	ACME_COMMON_STATE_IDLE
0xC004	GoToExp	Set the system state to Experiment	ACME_COMMON_STATE_EXPERIMENT
0xC005	GoToMaint	Set the system state to Maintenance	ACME_COMMON_STATE_MAINTENANCE
0xC006	GoToUpDownLink	Set the system state to Uplink/downlink	ACME_COMMON_STATE_UPLINK_DOWNLINK
0xC007	GoToSafed	Set the system state to Safed.	ACME_COMMON_STATE_SAFED
0xC008	SetRegistryItem	Select registry item to change	ACME_COMMON_REGISTRY_SET_ITEM
0xC009	SetRegistryValue	Change registry item value	ACME_COMMON_REGISTRY_SET_VALUE
0xC00A	SaveRegistry	Save registry changes	ACME_COMMON_REGISTRY_SAVE
0xC00B	Reboot	Reboot operating system	ACME_COMMON_SYSTEM_REBOOT

Cmd ID	Name	Description	Mnemonic
0xC00C	Shutdown	Shutdown processor	ACME_COMMON_SYSTEM_SHUTDOWN
0xC00D	RunCli	Run a command line interface command	ACME_COMMON_CLI_RUN
0xC00E	AbortCli	Abort a command line interface command	ACME_COMMON_CLI_ABORT
0xC00F	FtpLocalFile	Set FTP local file	ACME_COMMON_FTP_LOCAL_FILE
0xC010	FtpLocalDirectory	Set FTP local directory	ACME_COMMON_FTP_LOCAL_DIR
0xC011	FtpRemoteFile	Set FTP remote file	ACME_COMMON_FTP_REMOTE_FILE
0xC012	FtpRemoteDirectory	Set FTP remote directory	ACME_COMMON_FTP_REMOTE_DIR
0xC013	FtpLoginParameters	Set FTP login parameters	ACME_COMMON_FTP_LOGIN_PARAMETERS
0xC014	FtpAbort	Abort current FTP command	ACME_COMMON_FTP_ABORT
0xC015	FtpSendFile	Send file via FTP	ACME_COMMON_FTP_SEND_FILE
0xC016	FtpSendDirectory	Send directory via FTP	ACME_COMMON_FTP_SEND_DIRECTORY
0xC017	FtpGetFile	Get file via FTP	ACME_COMMON_FTP_GET_FILE
0xC018	FtpGetDirectory	Get directory via FTP	ACME_COMMON_FTP_GET_DIRECTORY
0xC019	ChangeDirectory	Linux ‘cd’ command	ACME_COMMON_FS_CHANGE_DIR
0xC01A	MakeDirectory	Linux ‘mkdir -P’ command	ACME_COMMON_FS_MAKE_DIR
0xC01B	RemoveSingleEmptyDirectory	Linux ‘rmdir command	ACME_COMMON_FS_REMOVE_SINGLE_EMPTY_DIR
0xC01C	RemoveMultipleEmptyDirectories	Linux ‘rmdir –p’ command	ACME_COMMON_FS_REMOVE_MULTIPLE_EMPTY_DIRS
0xC01D	MoveFiles	Linux ‘mv’ command	ACME_COMMON_MOVE_FILES
0xC01E	RenameFile	Rename file	ACME_COMMON_FS_RENAME_FILE
0xC01F	RenameDirectory	Rename directory	ACME_COMMON_FS_RENAME_DIR
0xC020	RemoveFiles	Linux ‘rm’ command	ACME_COMMON_FS_REMOVE_FILES
0xC021	RecursiveRemoveFiles	Linux ‘rm -r’ command	ACME_COMMON_FS_REMOVE_FILES_RECURSIVE
0xC022	CopyFiles	Linux ‘cp’ command	ACME_COMMON_FS_COPY_FILES
0xC023	RecursiveCopyFiles	Linux ‘cp -r command	ACME_COMMON_FS_COPY_FILES_RECURSIVE
0xC024	ListFileInfo	Linux ‘ls command	ACME_COMMON_FS_LIST_FILES
0xC025	SetSource	Set source for filesystem commands	ACME_COMMON_FS_SET_SOURCE
0xC026	SetDestination	Set destination for filesystem commands	ACME_COMMON_FS_SET_DESTINATION
0xC027	CompareCrcTwoFiles	Perform comparative CRC check	ACME_COMMON_FS_COMPARE_CRC_TWO_FILES

Cmd ID	Name	Description	Mnemonic
0xC028	AbortFilesystemCmd	Abort current filesystem command	ACME_COMMON_FS_ABORT_FS_COMMAND
0xC029	RunBashScript	Run a shell (BASH) script	ACME_COMMON_BASH_SCRIPT_RUN
0xC02A	SetBashScriptDirectory	Set BASH script directory	ACME_COMMON_BASH_SCRIPT_DIRECTORY
0xC02B	AbortBashScript	Abort current BASH script	ACME_COMMON_BASH_SCRIPT_ABORT
0xC02C	SetScriptDirectory	Set Qt script directory	ACME_COMMON_SCRIPT_ DIRECTORY
0xC02D	SetScriptParameterFileDirectory	Set directory for Qt script parameter files	ACME_COMMON_SCRIPT_PARAMETER_FILE_DIRECTOR Y
0xC02E	LoadScript	Load a Qt Script	ACME_COMMON_SCRIPT _LOAD
0xC02F	RunScript	Run a Qt Script	ACME_COMMON_SCRIPT _RUN
0xC030	AbortScript	Abort a Qt Script	ACME_COMMON_SCRIPT _ABORT
0xC031	PauseScript	Pause a Qt Script	ACME_COMMON_SCRIPT _PAUSE
0xC032	ResumeScript	Resume a Qt Script	ACME_COMMON_SCRIPT _RESUME
0xC033	ChangeScriptParameter	Change a Qt Script parameter	ACME_COMMON_SCRIPT _PARAMETER_CHANGE
0xC034	SetFileUplinkDirectory	Set file uplink directory	ACME_COMMON_FILE _UPLINK_DIRECTORY
0xC035	FileUplinkStart	Start file uplink	ACME_COMMON_FILE _UPLINK_ START
0xC036	FileUplinkBlock	Uplink a file block	ACME_COMMON_FILE _UPLINK_ BLOCK
0xC037	FileUplinkDone	Signal file uplink is done	ACME_COMMON_FILE _UPLINK_ DONE
0xC038	FileUplinkAbort	Abort a file uplink	ACME_COMMON_FILE _UPLINK_ ABORT
0xC039	MountDevice	Mount a remote disk locally	ACME_COMMON_MOUNT_DEVICE
0xC03A	UnmountDevice	Unmount remote disk	ACME_COMMON_UNMOUNT_DEVICE
0xC03B	DstatControl	Turn processor stats on/off	ACME _COMMON_DSTAT
0xC03C	SetAppLogging	Turn logging categories on/off	ACME _COMMON_SET_APP_LOGGING
0xC03D	SetFtpLogging	Set FTP logging verbosity	ACME _COMMON_SET_FTP_LOGGING
System Group			
0xC101	SystemDefineSetup	Define ACME hardware setup	ACME_ SYSTEM_DEFINE_SETUP
0xC102	SystemOpenTestPoint	Open a test point	ACME_ SYSTEM_TEST_POINT_OPEN
0xC103	SystemCloseTestPoint	Close current test point	ACME_ SYSTEM_TEST_POINT_CLOSE
0xC104	SystemResetMotionController	Reset motion controller for igniter arm and TFP	ACME_SYSTEM_RESET_MOTION_CONTROLLER
0xC105 – 0xC106		*Not used*	

Cmd ID	Name	Description	Mnemonic
0xC107	SystemCirFlowState	Report of CIR flow state	ACME_ SYSTEM_CIR_FLOW_STATE
0xC108	SystemLoadScript	Load a Qt Script	ACME_SYSTEM_SCRIPT_LOAD
0xC109	SystemRunScript	Run a Qt Script	ACME_SYSTEM_SCRIPT_RUN
0xC10A	SystemAbortScript	Aboart a Qt Script	ACME_SYSTEM_SCRIPT_ABORT
0xC10B	SystemPauseScript	Pause a Qt Script	ACME_SYSTEM_SCRIPT_PAUSE
0xC10C	SystemResumeScript	Resume a Qt Script	ACME_SYSTEM_SCRIPT_RESUME
Imaging Group			
0xC201	DataCameraSetBinning	Set data camera binning mode	ACME_DATA_CAMERA_SET_BINNING
0xC202	DataCameraSetPixelFormat	Set data camera pixel format	ACME_DATA_CAMERA_SET_PIXEL_FORMAT
0xC203	DataCameraSetRegionOfInterest	Set data camera region of interest	ACME_DATA_CAMERA_SET_ROI
0xC204	DataCameraSetAutomatic Exposure	Set data camera auto exposure mode	ACME_DATA_CAMERA_SET_AUTO_EXPOSURE
0xC205	DataCameraSetManualExposure	Set data camera manual exposure	ACME_DATA_CAMERA_SET_EXPOSURE
0xC206	DataCameraSetFrameRate	Set data camera frame rate	ACME_DATA_CAMERA_SET_FRAME_RATE
0xC207	DataCameraSetAutoGain	Set data camera auto gain	ACME_DATA_CAMERA_SET_AUTO_GAIN
0xC208	DataCameraSetManualGain	Set data camera manual gain	ACME_DATA_CAMERA_SET_MANUAL_GAIN
0xC209	DataCameraSetAutoWhiteBalance	Set data camera auto white balance	ACME_DATA_CAMERA_SET_AUTO_WHITE_BAL
0xC20A	DataCameraSetManualWhite Balance	Set data camera manual white balance	ACME_DATA_CAMERA_SET_MAN_WHITE_BAL
0xC20B	DataCameraLoadConfig	Load data camera configuration	ACME_DATA_CAMERA_LOAD_CONFIG
0xC20C	DataCameraSaveConfig	Save data camera configuration	ACME_DATA_CAMERA_SAVE_CONFIG
0xC20D	DataCameraSetPowerUpConfig	Set data camera power up configuration	ACME_DATA_CAMERA_SET_PWR_UP_CONFIG
0xC20E	DataCameraStartCapture	Data camera start capture	ACME_DATA_CAMERA_START_CAPTURE
0xC20F	DataCameraStopCapture	Data camera stop capture	ACME_DATA_CAMERA_STOP_CAPTURE
0xC210	DataCameraStartArchive	Data camera start archive	ACME_DATA_CAMERA_START_ARCHIVE
0xC211	DataCameraStopArchive	Data camera stop archive	ACME_DATA_CAMERA_STOP_ARCHIVE
0xC212	DataCameraGrabOneImage	Data camera grab one image	ACME_DATA_CAMERA_GRAB_ONE_IMAGE
0xC213	DataCameraMoveZoomAxis	Data camera move zoom axis	ACME_DATA_CAMERA_MOVE_ZOOM_AXIS
0xC214	DataCameraMoveIrisAxis	Data camera move iris axis	ACME_DATA_CAMERA_MOVE_IRIS_AXIS
0xC215	DataCameraMoveFocusAxis	Data camera move focus axis	ACME_DATA_CAMERA_MOVE_FOCUS_AXIS
0xC216	DataCameraMoveFilterAxis	Data camera move filter axis	ACME_DATA_CAMERA_MOVE_FILTER_AXIS

Cmd ID	Name	Description	Mnemonic
0xC217	DataCameraMoveWithParameters	Data camera move axis with parameters	ACME_DATA_CAMERA_MOVE_WITH_PARAMS
0xC218	DataCameraSetupMoveWith Parameters	Data camera setup move axis with parameters	ACME_DATA_CAMERA_SETUP_MOVE_WITH_PARAMS
0xC219	DataCameraExecuteSetupMove	Data camera execute setup move	ACME_DATA_CAMERA_EXECUTE_SETUP_MOVE
0xC21A	DataCameraSelectFilter	Move to selected filter	ACME_DATA_CAMERA_SELECT_FILTER
0xC21B	AnalogCameraSetAgc	Analog camera set AGC gain or disable	ACME_ANALOG_CAMERA_SET_AGC
0xC21C	AnalogCameraSetShutter	Analog camera set shutter	ACME_ANALOG_CAMERA_SET_SHUTTER
0xC21D	IlluminationConfigure	Turn lights on or off	ACME_ILLUMINATION_CONFIG
0xC21E	IlluminationCycle	Cycle lights on or off	ACME_ILLUMINATION_CYCLE
0xC21F	StartIpsus	Start imaging on IPSUs and IPSU-Gs	ACME_IPSUS_START
0xC220	StopIpsus	Stop imaging on IPSUs and IPSU-Gs	ACME_IPSUS_STOP
0xC221	DataCameraGetReport	Only when ACME, not IPSU-G is controlling camera	ACME_DATA_CAMERA_GET_REPORT
0xC222	IntensifiedCameraGate	Set gate parameters	ACME_INTENSIFIED_CAMERA_GATE
0xC223	IntensifiedCameraGainAmplitude	Set gain	ACME_INTENSIFIED_CAMERA_GAIN_AMPLITUDE
0xC224	DataCameraExposureBracketing	Specify bracketed exposure	ACME_DATA_CAMERA_EXPOSURE_BRACKETING
0xC225	IntensifiedCameraMoveFilter	Change the filter	ACME_INTENSIFIED_CAMERA_MOVE_FILTER
0xC226	IntensifiedCameraSafe	Put in a safe configuration	ACME_INTENSIFIED_CAMERA_SAFE
Igniter Group			
0xC301	IgniterDisable	Disable the igniter	ACME_IGNITER_DISABLE
0xC302	IgniterPulse	Pulse the igniter power	ACME_IGNITER_PULSE
0xC303	IgniterMove	Move the igniter	ACME_IGNITER_MOVE
0xC304	IgniterSetupMove	Prepare a move of the igniter	ACME_IGNITER_SETUP_MOVE
0xC305	IgniterSetupMoveWithParameters	Prepare a move of the igniter with parameters	ACME_IGNITER_SETUP_MOVE_WITH_PARAMS
0xC306	IgniterExecuteSetupMove	Execute a prepared igniter move	ACME_IGNITER_EXECUTE_SETUP_MOVE
0xC307	IgniterMotorSetState	Enable or disable igniter motor	ACME_IGNITER_MOTOR_SET_STATE
Flow Group			
0xC401	FlowEnableMfcs	Enable mass flow controllers	ACME_FLOW_ENABLE_MFCS

Cmd ID	Name	Description	Mnemonic
0xC402	FlowSetMfcRate	Set mass flow controller rate	ACME_FLOW_SET_MFC_RATE
0xC403	FlowSetAllMfcRates	Set all mass flow controller rates	ACME_FLOW_SET_ALL_MFC_RATES
0xC404	FlowZeroMfcs	Zero mass flow controllers	ACME_FLOW_ZERO_MFCS
0xC405	FlowMfcSerialCommand	Perform a mass flow controller raw serial command	ACME_FLOW_MFC_SERIAL_COMMAND
0xC406	FlowSetSolenoidValveState	Set solenoid valve state	ACME_FLOW_SET_SOLENOID_VALVE_STATE
0xC407	FlowSetAllSolenoidValveStates	Set all solenoid valve states	ACME_FLOW_SET_ALL_SOLENOID_VALVE_STATES
0xC408	FlowOpenAll	Open all ACME valves	ACME_FLOW_OPEN_ALL
0xC409	FlowCloseAll	Close all ACME valves	ACME_FLOW_CLOSE_ALL
0xC40A	FlowConfigForFill	Configure ACME valves for chamber fill	ACME_FLOW_CONFIG_FOR_FILL
0xC40B	FlowStepped	Run stepped flow rates	ACME_FLOW_STEPPED
0xC40C	FlowRamped	Run ramped flows rates	ACME_FLOW_RAMPED
0xC40D	FlowRectangularWave	Run rectangular wave flow rates	ACME_FLOW_RECTANGULAR_WAVE
0xC40E	FlowRampedWave	Run ramped wave flow rates	ACME_FLOW_RAMPED_WAVE
0xC40F	FlowStopWaveform	Stop flow waveform	ACME_FLOW_STOP_WAVEFORM
Diagnostic Group			
0xC501	DataRecordingStart	Start data recording	ACME_DATA_RECORDING_START
0xC502	DataRecordingStop	Stop data recording	ACME_DATA_RECORDING_STOP
0xC503	FiberMove	Move FIBER	ACME_FIBER_MOVE
0xC504	FiberSetupMove	Prepare a FIBER move	ACME_FIBER_SETUP_MOVE
0xC505	FiberExecuteSetupMove	Execute prepared FIBER move	ACME_FIBER_EXECUTE_SETUP_MOVE
0xC506	FiberRampedMove	Move FIBER with ramp	ACME_FIBER_RAMPED_MOVE
0xC507	FiberRampedWaveMove	Move FIBER with ramped wave	ACME_FIBER_RAMPED_WAVE_MOVE
0xC508	FiberHaltMove	Stop FIBER movement	ACME_FIBER_HALT_MOVE
0xC509	FiberMotorSetState	Enable or disable FIBER motor	ACME_FIBER_MOTOR_SET_STATE
0xC50A	PmtPower	Turn PMTs on or off	ACME_PMT_POWER
0xC50B	PmtSetGain	Set a PMT gain	ACME_PMT_SET_GAIN
0xC50C	PmtSetAllGains	Set all PMT gains	ACME_PMT_SET_ALL_GAINS
0xC50D	ExtinctionDetectionConfigure	Configure extinction detection	ACME_EXTINCTION_DETECTION_CONFIGURE
0xC50E	ExtinctionDetectionSetState	Set extinction state	ACME_EXTINCTION_DETECTION _SET_STATE
0xC50F	ExtinctionDetectionSetActionZero	Set extinction action to zero	ACME_EXTINCTION_DETECTION _SET_ACTION_ZERO

Cmd ID	Name	Description	Mnemonic
0xC510	ExtinctionDetectionSetActionSingleStep	Set extinction action to a single step	ACME_EXTINCTION_DETECTION_SET_ACTION_SINGLE_STEP
0xC511	ExtinctionDetectionSetActionSingleRamp	Set extinction action to a single ramp	ACME_EXTINCTION_DETECTION_SET_ACTION_SINGLE_RAMP
0xC512	ExtinctionDetectionSetActionStepping	Set extinction action to stepping	ACME_EXTINCTION_DETECTION_SET_ACTION_STEPPING
0xC513	ExtinctionDetectionSetActionRamping	Set extinction action to ramping	ACME_EXTINCTION_DETECTION_SET_ACTION_RAMPING
0xC514	DataRecordingSetSampleRate	Change the period between data samples	ACME_DATA_RECORDING_SAMPLE_RATE
E-Field Group			
0xC601	EFieldEnable	Enable or disable E-Field	ACME_EFIELD_ENABLE
0xC602	EFieldSetVoltage	Set E-Field voltage	ACME_EFIELD_SET_VOLTAGE
0xC603	EFieldSteppedSignal	E-Field stepped signal	ACME_EFIELD_STEPPED_SIGNAL
0xC604	EFieldRampedSignal	E-Field ramped signal	ACME_EFIELD_RAMPED_SIGNAL
0xC605	EFieldRectangularWave	E-Field rectangular wave	ACME_EFIELD_RECTANGULAR_WAVE
0xC606	EFieldRampedWave	E-Field ramped wave	ACME_EFIELD_RAMPED_WAVE

3.1.2 ACME Common Group Command Details

These commands are called ‘common’ because they are implemented using code shared between several payloads. The detailed descriptions below are derived from the original documentation for these commands in Appendix K of FCF-ICD-0076, Rev C or later. As stated in that appendix the high order bytes of the command ID are replaced by the command group number assigned to ACME common commands, which is 0xC0. In that appendix the command IDs range from 0x0001 to 0x0038. For ACME they range from 0xC001 to 0xC038.

3.1.2.1 NullCommand

The *NullCommand* does nothing. It can be used to test communication paths.

Command ID: 0xC001
Mnemonic: ACME_COMMON_NULL

3.1.2.2 ExternalCommand

The *ExternalCommand* command allows an avionics Main Processor (MP) CSCI to have no knowledge of this command and make use of the External Module feature. The command is passed to an External Module without intervention. This command enforces the requirement that an External Module requires a “module identifier” as the first parameter. The command group portion of the command ID, XX, depends on the module.

Command ID: 0xXX02
Mnemonic: ACME_COMMON_EXTERNAL_PASS_THRU

Parameter	# Bytes Type	Description
moduleIdentifier	2 Uint16	Identifier (e.g. camera number 1,2)
Command parameters	80	Area for module command-dependent parameters.

3.1.2.3 GoToIdle

The *GoToIdle* command requests an avionics MP CSCI to be set to the IDLE state.

Command ID: 0xC003
Mnemonic: ACME_COMMON_STATE_IDLE

3.1.2.4 GoToExp

The *GoToExp* command requests an avionics MP CSCI to be set to the EXPERIMENT state.

Command ID: 0xC004
Mnemonic: ACME_COMMON_STATE_EXPERIMENT

3.1.2.5 GoToMaint

The *GoToMaint* command requests an avionics MP CSCI to be set to the MAINTENANCE state.

Command ID: 0xC005
Mnemonic: ACME_COMMON_STATE_MAINTENANCE

3.1.2.6 GoToUpDownLink

The *GoToUpDownLink* command requests an avionics MP CSCI to be set to the UPLINK/DOWNLINK state.

Command ID: 0xC006
Mnemonic: ACME_COMMON_STATE_UPLINK_DOWNLINK

3.1.2.7 GoToSafed

The *GoToSafed* command requests an avionics MP CSCI to be set to the SAFED state.

Command ID: 0xC007
Mnemonic: ACME_COMMON_STATE_SAFED

3.1.2.8 SetRegistryItem

The *SetRegistryItem* command selects an item in the registry file, /payload/app/config/AcmeConfig.ini, for modification by the *SetRegistryValue* command. The full path including group is required. In **Table *32*, *AcmeConfig.ini Contents***, the full path is /Group/Key.

Command ID: 0xC008
Mnemonic: ACME_COMMON _REGISTRY_SET _ITEM

Parameter	# Bytes Type	Description
registryName	82 String	/Group/Key in AcmeConfig.ini (e.g. /EField/OutputFactorA)

3.1.2.9 SetRegistryValue

The *SetRegistryValue* command selects the key set via the *SetRegistryItem* command and modifies it with the supplied value. All software sub-systems are notified of the change and apply it used in the sub-system. This is just a temporary change. It must be saved with the *SaveRegistry* command to become permanent in the AcmeConfig.ini file. See **Table *32***.

Command ID: 0xC009
Mnemonic: ACME_COMMON_REGISTRY_SET_VALUE

Parameter	# Bytes Type	Description
registryValue	82 String	String value (e.g. 172.16.200.52)

3.1.2.10 SaveRegistry

The *SaveRegistry* command dumps the "in memory" registry to /payload/app/config/AcmeConfig.ini.

Command ID: 0xC00A
Mnemonic: ACME_COMMON_REGISTRY_SAVE

3.1.2.11 Reboot

The *Reboot* command initiates an avionics MP CSCI reboot. The system must placed in the Safed state using the *GoToSafed* command prior to issuing this command.

Command ID: 0xC00B
Mnemonic: ACME_COMMON_SYSTEM_REBOOT

3.1.2.12 Shutdown

The *Shutdown* command initiates an avionics MP CSCI shutdown. The system must placed in the Safed state using the *GoToSafed* command prior to issuing this command.

Command ID: 0xC00C
Mnemonic: ACME_COMMON_SYSTEM_SHUTDOWN

3.1.2.13 RunCli

The *RunCli* command executes the given linux command in the command shell. CLI stands for command line interface.

Command ID: 0xC00D
Mnemonic: ACME_COMMON_CLI_RUN

Parameter	# Bytes Type	Description
commandLine	82 String	Linux shell command (e.g. ls /payload/app/bin)

3.1.2.14 AbortCli

The *AbortCli* command terminates the command running in the shell started by the *RunCli* command. It terminates any child processes the command may have started and the shell itself. It restarts the shell (SH) with the /payload working directory as the current directory.

Command ID: 0xC00E
Mnemonic: ACME_COMMON_CLI_ABORT

3.1.2.15 FtpLocalFile

The *FtpLocalFile* command stores the path for later use with the *FtpSendFile* command. It may be an absolute file path or a relative file path. If a relative path is given it is combined with the absolute directory set via the *FtpLocalDirectory* command to form the complete path.

Command ID: 0xC00F
Mnemonic: ACME_COMMON_FTP_LOCAL_FILE

Parameter	# Bytes Type	Description
file	82 String	Local path (e.g. /payload/app/config/myFile *or* app/config/myFile *or* myFile)

3.1.2.16 FtpLocalDirectory

The *FtpLocalDirectory* command stores the path for later use with the *FtpSendFile*, *FtpGetFile*, *FtpSendDirectory*, and *FtpGetDirectory* commands. It must be an absolute directory path. If a relative path is provided in those commands the relative path is combined with this directory to form the complete path.

Command ID: 0xC010
Mnemonic: ACME_COMMON_FTP_LOCAL_DIR

Parameter	# Bytes Type	Description
directory	82 String	Local directory (e.g. /payload/app/config)

3.1.2.17 FtpRemoteFile

The *FtpRemoteFile* command stores the path for later use with the *FtpGetFile* command. It may be an absolute file path or a relative file path. If a relative path is given it is combined with the absolute directory set via the *FtpRemoteDirectory* command to form the complete path.

Command ID: 0xC011
Mnemonic: ACME_COMMON_FTP_REMOTE_FILE

Parameter	# Bytes Type	Description
file	82 String	Remote path (e.g. /payload/app/config/myFile *or* app/config/myFile *or* myFile)

3.1.2.18 FtpRemoteDirectory

The *FtpRemoteDirectory* command stores the path for later use with the *FtpSendFile*, *FtpGetFile*, *FtpSendDirectory*, and *FtpGetDirectory* commands. It must be an absolute directory path. If a relative path is given in those commands the relative path is combined with this directory to form the complete path.

Command ID: 0xC012
Mnemonic: ACME_COMMON_FTP_REMOTE_DIR

Parameter	# Bytes Type	Description
directory	82 String	Remote directory (e.g. /payload/app/config)

3.1.2.19 FtpLoginParameters

The *FtpLoginParameters* command stores the information required to FTP files to/from a remote machine for later use with the *FtpSendFile*, *FtpGetFile*, *FtpSendDirectory*, and *FtpGetDirectory* commands.

Command ID: 0xC013
Mnemonic: ACME_COMMON_FTP_LOGIN_PARAMETERS

Parameter	# Bytes Type	Description
hostname	25 String	Remote machine IP or hostname
user	25 String	Username
password	25 String	Password

3.1.2.20 FtpAbort

The *FtpAbort* command aborts the current FTP transfer.

Command ID: 0xC014
Mnemonic: ACME_COMMON_FTP_ABORT

3.1.2.21 FtpSendFile

The *FtpSendFile* command uses the information set via the *FtpLocalFile*, *FtpLocalDirectory*, and *FtpRemoteDirectory* commands. If an absolute path is provided in the *FtpLocalFile* command the result of the *FtpLocalDirectory* command is ignored, otherwise the two are combined to form an absolute file path for the source. The destination is provided as an absolute path from the *FtpRemoteDirectory* command. The file is sent from the local machine to the remote machine.

Command ID: 0xC015
Mnemonic: ACME_COMMON_FTP_SEND_FILE

3.1.2.22 FtpSendDirectory

The *FtpSendDirectory* command uses the absolute paths set via the *FtpLocalDirectory* command and the *FtpRemoteDirectory* command to copy a directory recursively from the local machine to the remote machine.

Command ID: 0xC016
Mnemonic: ACME_COMMON_FTP_SEND_DIRECTORY

3.1.2.23 FtpGetFile

The *FtpGetFile* command uses the information set via the *FtpRemoteFile* and *FtpRemoteDirectory* commands and the *FtpLocalDirectory* command. If an absolute path is provided in the *FtpRemoteFile* command the result of the *FtpRemoteDirectory* command is ignored, otherwise the two are combined to form an absolute file path for the source. The destination is provided as an absolute path from the *FtpLocalDirectory* command. The file is sent from the remote machine to the local machine.

Command ID: 0xC017
Mnemonic: ACME_COMMON_FTP_GET_FILE

3.1.2.24 FtpGetDirectory

The *FtpGetDirectory* command uses the absolute paths set via the *FtpRemoteDirectory* command and *FtpLocalDirectory* commands to copy a directory recursively from the remote machine to the local machine.

Command ID: 0xC018
Mnemonic: ACME_COMMON_FTP_GET_DIRECTORY

3.1.2.25 ChangeDirectory

The *ChangeDirectory* (linux ‘cd’) command moves the notion of the current directory for the common filesystem commands to the given path. The path may be an absolute path or a relative

path. Relative path *ChangeDirectory* commands can be chained to traverse a deep directory structure. The path must exist for the command to successfully complete.

Command ID: 0xC019
Mnemonic: ACME_COMMON_FS_CHANGE_DIR

Parameter	# Bytes Type	Description
path	82 String	Directory path

3.1.2.26 MakeDirectory

The *MakeDirectory* (linux ‘mkdir -P’) command creates a new directory(s). The path may be an absolute path or a relative path. All parent directories necessary to create the full path are made. If a relative path is given it is combined with the current working directory from a single or series of *ChangeDirectory* commands to form the complete path.

Command ID: 0xC01A
Mnemonic: ACME_COMMON_FS_MAKE_DIR

Parameter	# Bytes Type	Description
path	82 String	Directory path (e.g. /tmp/a/b/c *or* a/b/c)

3.1.2.27 RemoveSingleEmptyDirectory

The *RemoveSingleEmptyDirectory* (linux ‘rmdir’) command removes a directory. The given directory path must be empty for the command to succeed. The path may be an absolute path or a relative path. If a relative path is given it is combined with the current working directory from a single or series of *ChangeDirectory* commands to form the complete path.

Command ID: 0xC01B
Mnemonic: ACME_COMMON_FS_REMOVE_SINGLE_EMPTY_DIR

Parameter	# Bytes Type	Description
path	82 String	Directory path

3.1.2.28 RemoveMultipleEmptyDirectories

The *RemoveMultipleEmptyDirectories* (linux ‘rmdir -p’) command removes all parent directories in the path, provided they are empty. The path may be an absolute path or a relative path. If a relative path is given it is combined with the current working directory from a single or series of *ChangeDirectory* commands to form a complete path. This command is the opposite of the

MakeDirectory command. This command performs actions to the best of its ability and will stop if/when it finds a non-empty directory. Actions completed to that point are not rolled back.

Command ID: 0xC01C
Mnemonic: ACME_COMMON_FS_REMOVE_MULTIPLE_EMPTY_DIRS

Parameter	# Bytes Type	Description
path	82 String	Directory path

3.1.2.29 MoveFiles

The *MoveFiles* (linux 'mv SRC DIR') command moves SRC(s) to DIR. Source directory is set via the SetSource command and the destination directory is set via the SetDestination command. Wildcards * (any number of unknown characters) and ? (one unknown character) may be used in the filter. A blank filter or a filter consisting of a single * is interpreted as all files. If a single file move is desired, give filenames in the source and destination. A better solution is to use the *RenameFile* command. If moving an entire directory tree is desired, use the *RenameDirectory* command.

Command ID: 0xC01D
Mnemonic: ACME_COMMON_FS_MOVE_FILES

Parameter	# Bytes Type	Description
filter	82 String	Name filter (e.g. *.jpeg *or* image?003.jpeg)

3.1.2.30 RenameFile

The *RenameFile* command renames a source file set via the *SetSource* command to a destination file set via the *SetDestination* command.

Command ID: 0xC01E
Mnemonic: ACME_COMMON_FS_RENAME_FILE

3.1.2.31 RenameDirectory

The *RenameDirectory* command renames a source directory set via the *SetSource* command to a destination directory set via the *SetDestination* command.

Command ID: 0xC01F
Mnemonic: ACME_COMMON_FS_RENAME_DIR

3.1.2.32 RemoveFiles

The *RemoveFiles* (linux ‘rm’) command removes files from a directory. Source directory is set via the *SetSource* command. Wildcards * (any number of unknown characters) and ? (one unknown character) may be used in the filter. A blank filter or a filter consisting of a single * is interpreted as all files. If a single file remove is desired, give a filename in the source.

Command ID: 0xC020
Mnemonic: ACME_COMMON_FS_REMOVE_FILES

Parameter	# Bytes Type	Description
filter	82 String	Name filter (e.g. *.jpeg *or* image?003.jpeg)

3.1.2.33 RecursiveRemoveFiles

The *RecursiveRemoveFiles* (linux ‘rm –r’) command removes directories and their contents recursively. The source directory is set via the *SetSource* command.

Command ID: 0xC021
Mnemonic: ACME_COMMON_FS _REMOVE_FILES_RECURSIVE

3.1.2.34 CopyFiles

The *CopyFiles* (linux ‘cp SRC DIR’) command copies the SRC(s) to DIR. Source is set via the *SetSource* command and the destination is set via the *SetDestination* command. Wildcards * (any number of unknown characters) and ? (one unknown character) may be used in the filter. A blank filter or a filter consisting of a single * is interpreted as all files. If a single file copy is desired, give filenames in the source and destination. File timestamps may be preserved by setting the preserve flag.

Command ID: 0xC022
Mnemonic: ACME_COMMON_FS_COPY_FILES

Parameter	# Bytes Type	Values	Description
preserve	1 Uint8	0 = No 1 = Yes	Preserve original time stamp
filter	81 String		Name filter (e.g. *.jpeg *or* image?003.jpeg)

3.1.2.35 RecursiveCopyFiles

The *RecursiveCopyFiles* (linux ‘cp –r SRC DIR’) command copies the directory set via the SetSource command to the destination directory set via the *SetDestination* command. The destination directory, which is created, will contain the target directory. The destination directory

must not exist for this command to succeed. File timestamps may be preserved by setting the preserve flag.

Command ID: 0xC023
Mnemonic: ACME_COMMON_FS _COPY_FILES_RECURSIVE

Parameter	# Bytes Type	Values	Description
preserve	1 Uint8	0 = No 1 = Yes	Preserve original time stamp

3.1.2.36 ListFileInfo

The *ListFileInfo* (linux 'ls –a –l –r –t –R') command lists file information for the source directory set via the SetSource command. Wildcards * (any number of unknown characters) and ? (one unknown character) may be used in the filter. A blank filter or a filter consisting of a single * is interpreted as all files.

Command ID: 0xC024
Mnemonic: ACME_COMMON_FS_LIST_FILES

Parameter	# Bytes Type	Values	Description
all	1 Uint8	0 = No 1 = Yes	Include hidden files. (Those beginning with .) (ls –a option)
crc	1 Uint8		Display SHA256 of file.
longformat	1 Uint8	0 = No 1 = Yes	Long format (which includes file permissions) (ls –l option)
reverse	1 Uint8	0 = No 1 = Yes	Sort by reverse time. (ls –r option)
sortByTime	1 Uint8	0 = No 1 = Yes	Sort by time. (ls –t option)
Recursive	1 Uint8	0 = No 1 = Yes	Recursive. (ls –R option)
afterDate	4 time_t	0=don't use	List only file after this date, inclusive.
beforeDate	4 time_t	0=don't use	List only file before this date, inclusive.
filter	68 String		Name filter (e.g. *.jpeg *or* image?003.jpeg)

3.1.2.37 SetSource

The *SetSource* command provides the source for filesystem commands. The path may be an absolute path or a relative path. If a relative path is given it is combined with the current working directory from a single or series of *ChangeDirectory* commands. In some cases (COPY/MOVE/REMOVE) for a single file action, a filename may be included.

Command ID: 0xC025
Mnemonic: ACME_COMMON_FS_SET_SOURCE

Parameter	# Bytes Type	Description
path	82 String	Source path (e.g. /payload/app/config/IpsugConfig.ini *or* app/config *or* /payload/app/config)

3.1.2.38 SetDestination

The *SetDestination* command provides the destination for filesystem commands. The path may be an absolute path or a relative path. If a relative path is given it is combined with the current working directory from a single or series of *ChangeDirectory* commands to form a complete path. In some cases (COPY/MOVE/REMOVE) for a single file action, a filename may be included.

Command ID: 0xC026
Mnemonic: ACME_COMMON_FS_SET_DESTINATION

Parameter	# Bytes Type	Description
path	82 String	Destination path (e.g. /payload/app/config/IpsugConfig.ini *or* app/config *or* /payload/app/config)

3.1.2.39 CompareCrcTwoFiles

The *CompareCrcTwoFiles* command performs a SHA256 (Secure Hash Algorithm) on two files and compares the values. Source directory is set via the *SetSource* command and the destination directory is set via the *SetDestination* command. If a relative path is given for either source or destination it is combined with the current working directory from a single or series of *ChangeDirectory* commands to form the complete path. The file1 specified in the command is combined with the calculated source path and file2 is combined with the calculated destination path.

Command ID: 0xC027
Mnemonic: ACME_COMMON_FS_COMPARE_CRC_TWO_FILES

Parameter	# Bytes Type	Description
file1	41 String	File 1 filename
file2	41 String	File 2 filename

3.1.2.40 AbortFilesystemCmd

The *AbortFilesystemCmd* Command command aborts the currently running filesystem command.

Command ID: 0xC028
Mnemonic: ACME_COMMON_FS_ABORT_FS_COMMAND

3.1.2.41 RunBashScript

The *RunBashScript* command initiates execution of a BASH script. The path may be an absolute path or a relative path. If a relative path is given it is combined with the absolute directory set via the *SetBashScriptDirectory* command to form the complete path All bash scripts reside in /payloald/app/scripts/shell.

Command ID: 0xC029
Mnemonic: ACME_COMMON_BASH_SCRIPT_RUN

Parameter	# Bytes Type	Description
script	82 String	Bash Script path (e.g. /payload/app/scripts/shell/myScript.sh *or* myScript.sh)

3.1.2.42 SetBashScriptDirectory

The *SetBashScriptDirectory* command specifies a source directory of a BASH script. An absolute path must be provided. This path is used to form the complete path if a relative path is provided in the *RunBashScript* command. All bash scripts reside in /payload/app/scripts/shell.

Command ID: 0xC02A
Mnemonic: ACME_COMMON_BASH_SCRIPT_DIRECTORY

Parameter	# Bytes Type	Description
directory	82 String	Bash Script directory path

3.1.2.43 AbortBashScript

The *AbortBashScript* command terminates the command running in the SH shell started by the *RunBashScript* command. It terminates any child processes the command may have started and the shell itself.

Command ID: 0xC02B
Mnemonic: ACME_COMMON_BASH_SCRIPT_ABORT

3.1.2.44 SetScriptDirectory

The *SetScriptDirectory* command specifies a source directory for a Qt script. An absolute path must be provided. This path is used to form the complete path if a relative path is provided in the *LoadScript* command. All Qt Script scripts reside in /payload/app/scripts.

Command ID: 0xC02C
Mnemonic: ACME_COMMON_SCRIPT_DIRECTORY

Parameter	# Bytes Type	Description
directory	82 String	Script file directory

3.1.2.45 SetScriptParameterFileDirectory

The *SetScriptParameterFileDirectory* command specifies a source directory for a Qt script parameter file. The script parameter file feature is still in work and this command has no effect.

Command ID: 0xC02D
Mnemonic: ACME_COMMON_SCRIPT_PARAMETER_FILE_DIRECTORY

Parameter	# Bytes Type	Description
directory	82 String	Script file directory

3.1.2.46 LoadScript

Use the *SystemLoadScript* command instead for the reasons listed under that command description. The *LoadScript* command loads a script file into the Qt scripting engine. The path may be an absolute path or a relative path. If a relative path is given it is combined with the absolute path set via the *SetScriptDirectory* command to form the complete path.

Command ID: 0xC02E
Mnemonic: ACME_COMMON_SCRIPT_LOAD

Parameter	# Bytes Type	Description
script	82 String	Script source path (e.g. /payload/app/script/myScript.js *or* myScript.js)

3.1.2.47 RunScript

Use the *SystemRunScript* command instead. The *RunScript* command initiates execution of a script previously loaded into the Qt scripting engine via the *LoadScript* command.

Command ID: 0xC02F
Mnemonic: ACME_COMMON_SCRIPT_RUN

3.1.2.48 AbortScript

Use the *SystemAbortScript* command instead. The *AbortScript* command aborts the execution of the currently running Qt script.

Command ID: 0xC030
Mnemonic: ACME_COMMON_SCRIPT_ABORT

3.1.2.49 PauseScript

Use the *SystemPauseScript* command instead. The *PauseScript* command pauses the execution of the currently running Qt script.

Command ID: 0xC031
Mnemonic: ACME_COMMON_SCRIPT_PAUSE

3.1.2.50 ResumeScript

Use the *SystemResumeScript* command instead. The *ResumeScript* command resumes the execution of the currently paused Qt script.

Command ID: 0xC032
Mnemonic: ACME_COMMON_SCRIPT_RESUME

3.1.2.51 ChangeScriptParameter

The *ChangeScriptParameter* command updates a single parameter value for a loaded script.

Command ID: 0xC033
Mnemonic: ACME_COMMON_SCRIPT_PARAMETER_CHANGE

Parameter	# Bytes Type	Description
parameterInfo	82 String	Parameter to update (like JSON e.g. “parameterName”:value

3.1.2.52 SetFileUplinkDirectory

The *SetFileUplinkDirectory* command specifies a source directory of a file to uplink. An absolute path must be provided. This path is used to form the complete path if a relative path is provided in the *FileUplinkStart* command.

Command ID: 0xC034
Mnemonic: ACME_COMMON_FILE_UPLINK_DIRECTORY

Parameter	# Bytes Type	Description
directory	82 String	File uplink directory path

3.1.2.53 FileUplinkStart

The *FileUplinkStart* command initiates a file upload via the commanding stream. The path may be an absolute path or a relative path. If a relative path is given it is combined with the path set via the *SetFileUplinkDirectory* command to form the complete path. The path is created if it does not exist.

Command ID: 0xC035
Mnemonic: ACME_COMMON_FILE_UPLINK_START

Parameter	# Bytes Type	Description
fileSizeInBytes	4 Uint32	File size in bytes
path	78 String	File uplink destination path

3.1.2.54 FileUplinkBlock

The *FileUplinkBlock* command transfers a block of file data identified with a sequential id. All block data sent will be aggregated upon receiving the *FileUplinkDone* command.

Command ID: 0xC036
Mnemonic: ACME_COMMON_FILE_UPLINK_BLOCK

Parameter	# Bytes Type	Description
blockIndex	4 Uint32	Index of Block
size	2 Uint16	Size in Bytes
data	76 Byte array	File block data

3.1.2.55 FileUplinkDone

The *FileUplinkDone* command verifies the block data received by comparing the SHA of the received block data with the SHA provided in the command. If the given SHA matches the block data SHA the file is written in entirety to the destination set via the *FileUplinkStart* command.

Command ID: 0xC037
Mnemonic: ACME_COMMON_FILE_UPLINK_DONE

Parameter	# Bytes Type	Description
SHA256	32 Byte array	SHA256 for compare

3.1.2.56 FileUplinkAbort

The *FileUplinkAbort* command aborts the current file uplink transfer.

Command ID: 0xC038
Mnemonic: ACME_COMMON_FILE_UPLINK_ABORT

3.1.2.57 MountDevice

The *MountDevice* command mounts a remote file system to the local file system. When done unmount with the *UnmountDevice* command.

Command ID: 0xC039
Mnemonic: ACME_COMMON_MOUNT_DEVICE

Parameter	# Bytes Type	Description
ipAddr	15 String	IP address
localPath	15 String	Local path
remotePath	52 String	Remote path

3.1.2.58 UnmountDevice

The *UnmountDevice* command unmounts a device mounted with the *MountDevice* command.

Command ID: 0xC03A
Mnemonic: ACME_COMMON_UNMOUNT_DEVICE

Parameter	# Bytes Type	Description
localPath	82 String	Local path

3.1.2.59 DstatControl

The *DstatControl* command stops or starts processor usage statistics from being placed in the telemetry slush pile. The linux command that is run is: ‘dstat –cdnm –noheader –nocolor’. The ‘zt.dstat’ logging category must be enabled using the *SetAppLogging* command. Ground software can direct output to a separate window by looking for this category.

Command ID: 0xC03B
Mnemonic: ACME _COMMON_DSTAT

Parameter	# Bytes Type	Description
Onoff	1 Uint8	0=off/stop, 1=on/start

3.1.2.60 SetAppLogging

The *SetAppLogging* command turns on or off the logging of messages based on their category. See the comments in /payload/app/config/AcmeLogging.ini for available categories and how to use filters.

Command ID: 0xC03C
Mnemonic: ACME _COMMON_SET_APP_LOGGING

Parameter	# Bytes Type	Description
type	1 Uint8	0=Debug, 1=Warning, 2=Critical/System, 3=Fatal
enable	1 Uint8	0=Disable, 1=Enable
category	80 String	Category name or filter

3.1.2.61 SetFtpLogging

The *SetFtpLogging* command sets the verbosity level for FTP operations.

Command ID: 0xC03D
Mnemonic: ACME _COMMON_SET_FTP_LOGGING

Parameter	# Bytes Type	Description
ftploglevel	1 Uint8	0=Not Verbose, 1=Verbose, 2=Very Verbose

3.1.3 ACME System Group Command Details

3.1.3.1 SystemDefineSetup

The *SystemDefineSetup* command specifies the experimental configuration. All information communicated by this command is recorded in the header of each recorded data file. The burner type is used to determine igniter limits. See the IgniterMove command. The useIpsu parameters are used to determine which IPSUs to record images on. See the *StartIpsus* command. This

command must be issued before the software is commanded to the Experiment state. See the *GoToExp* command.

The parameters contained in this command are also made available to scripts. For example:

```
if ( SystemDefineSetup.radiomConfig == 1 )
{
  // Do some BRE thing
}
```

Command ID: 0xC101
Mnemonic: ACME_SYSTEM_DEFINE_SETUP

Parameter	# Bytes Type	Values	Description
experimentType	1 Uint8	0 = Flame Design 1 = Coflow 2 = Spherical 3 = E-Field 4 = BRE	
mfc1GasType	1 Uint8	0 Oxygen/Nitrogen (21/79) 1 Oxygen/Nitrogen (30/70) 2 Oxygen/Nitrogen (40/60) 3 Oxygen/Nitrogen (85/15) 4 Oxygen/Helium (30/70)	
mfc2GasType	1 Uint8	0 Nitrogen 1 Helium	
mfc3GasType	1 Uint8	0 Methane 1 Methane/Nitrogen (30/70) 2 Methane/Nitrogen (40/60) 3 Methane/Nitrogen (45/55) 4 Ethylene 5 Ethylene/Nitrogen (30/70) 6 Ethylene/Nitrogen (50/50) 7 Hydrogen 8 Hydrogen/Methane (44.4/55.6) 9 Hydrogen/Methane (55.6/44.4) 10 Hydrogen/Methane (66.7/33.3) 11 Hydrogen/Methane/Helium (20/25/55) 12 Hydrogen/Methane/Helium (25/20/55) 13 Hydrogen/Methane/Helium (30/15/55) 14 Hydrogen/Helium (45/55) 15 Methane/Helium (45/55) 16 Ethylene/Helium (20/80) 17 Ethylene/Helium (25/75) 18 Ethylene/Helium (30/70)	

Parameter	# Bytes Type	Values	Description
efieldConfig	1 Uint8	0 = Mesh Removed 1 = Mesh in 35 mm position 2 = Mesh in 40 mm position	Distance from the chamber window centerline.
efieldPolarity		0 = Supply Not Installed 1 = Negative 2 = Positive	
igniterArm		0 = 0 mm Position 1 = 10 mm Position 2 = 30 mm Position	
burnerType	1 Uint8	0 = Spherical (1/4) 1 = Spherical(3/8) 2 = Spherical(1/2) 3 = Coflow 4 = Gas Jet(1.3mm), 10 mm position 5 = Gas Jet(2.1mm), 10 mm position 6 = BRE (25mm) 7 = BRE (50mm)	
thermoConfig	1 Uint8	0 = Near 1 = Far 2 = Removed	
radiomConfig	1 Uint8	0 = Standard 1 = BRE	
useIpsu1	1 Uint8	0 = No 1 = Yes	
useIpsu2	1 Uint8	0 = No 1 = Yes	
useIpsu3	1 Uint8	0 = No 1 = Yes	
useIpsu4	1 Uint8	0 = No 1 = Yes	
useIpsug5	1 Uint8	0 Do not archive images 1 Archive images during test point (ACME camera and/or Intensified Camera)	
useIpsug6	1 Uint8	0 = No 1 = Yes	
fiberConfig	1 Uint8	0 Not installed 1 Installed	

3.1.3.2 SystemOpenTestPoint

The *SystemOpenTestPoint* command prepares the software for recording data. A directory is created at /payload/sci/testpointName to accept files, where testpointName is the supplied parameter. If that directory already exists then '_x' is appended to the name, where x is 1 or 2 or 3 or more until a non-existent directory is found. In this directory a file is created named testpointName_x_data.csv, where testpointName_x matches the directory name. Two lines of header are written to the file and it is left open ready to accept data once an *DataRecordingStart*

command is received. See the details under the *DataRecordingStart* command for more information about the headers.

Command ID: 0xC102
Mnemonic: ACME_SYSTEM_TEST_POINT_OPEN

Parameter	# Bytes Type	Values	Description
testpointName	30 String		Any whitespace characters are trimmed from the beginning and end of the string.

3.1.3.3 SystemCloseTestPoint

The *SystemCloseTestPoint* command marks the end of the test point started with the *SystemOpenTestPoint* command. Data recording is stopped, if it has not been stopped already using the DataRecordingStop command. The recorded data file, current command log file, and current message log file are compressed into the file /payload/sci/testpointName_x/testpointName_x.zip, where testpointName_x is as described under the *SystemOpenTestPoint* command.

Command ID: 0xC103
Mnemonic: ACME_SYSTEM_TEST_POINT_CLOSE

3.1.3.4 SystemResetMotionController

The *SystemResetMotionController* command resets the motion control board in the CIA that controls the igniter arm and TFP motors. Any current motion and waveforms are stopped, the board is reset, and the board version is reported in the log. This is a 'soft reset' as referenced in the Prodigy PC-104 Users Guide. Hard resets of the individual motor stepper drivers are achieved using the *IgniterMotorSetState* and *TfpMotorSetState* commands.

Command ID: 0xC104
Mnemonic: ACME_SYSTEM_RESET_MOTION_CONTROLLER

3.1.3.5 SystemCirFlowState

The *SystemCirFlowState* command is designed to be sent from the IOP to inform the ACME software about the state of the CIR manifolds. Scripts can access this information as shown:

```
If ( SystemCirFlowState.diluentManifoldState == 1)
{
        // Diluent manifold is opened
}
```

Command ID: 0xC107
Mnemonic: ACME_SYSTEM_CIR_FLOW_STATE

Parameter	# Bytes Type	Values	Description
fuelManifoldState	1 Uint8	0 = closed 1 = open	State of CIR fuel manifold.
diluentManifoldState	1 Uint8	0 = closed 1 = open	State of CIR diluent manifold.
oxygenManifoldState	1 Uint8	0 = closed 1 = open	State of CIR oxygen manifold.
nitrogenManifoldState	1 Uint8	0 = closed 1 = open 2 = bypass	State of CIR nitrogen manifold.

3.1.3.6 SystemLoadScript

The *SystemLoadScript* command loads a script file into the Qt scripting engine. The path may be an absolute path or a relative path. If a relative path is given it is combined with the absolute path set via the *SetScriptDirectory* command to form the complete path. Use this, and the other system script commands below, instead of the common *LoadScript* command in order to get accurate script state and line numbers in telemetry and better messaging during aborts.

Command ID: 0xC108
Mnemonic: ACME_SYSTEM_SCRIPT_LOAD

Parameter	# Bytes Type	Description
script	82 String	Script source path (e.g. /payload/app/script/myScript.js *or* myScript.js)

3.1.3.7 SystemRunScript

The *SystemRunScript* command initiates execution of a script previously loaded into the Qt scripting engine via the *LoadScript* command. Use instead of common *RunScript* command.

Command ID: 0xC109
Mnemonic: ACME_ SYSTEM _SCRIPT_RUN

3.1.3.8 SystemAbortScript

The *SystemAbortScript* command aborts the execution of the currently running Qt script. Use instead of common *AbortScript* command.

Command ID: 0xC10A
Mnemonic: ACME_ SYSTEM _SCRIPT_ABORT

3.1.3.9 SystemPauseScript

The *SystemPauseScript* command pauses the execution of the currently running Qt script. Use instead of common *PauseScript* command.

Command ID: 0xC10B
Mnemonic: ACME_ SYSTEM _SCRIPT_PAUSE

3.1.3.10 SystemResumeScript

The *SystemResumeScript* command resumes the execution of the currently paused Qt script. Use instead of common *ResumeScript* command.

Command ID: 0xC10C
Mnemonic: ACME_ SYSTEM _SCRIPT_RESUME

3.1.4 ACME Imaging Group Command Details

3.1.4.1 DataCameraSetBinning

The *DataCameraSetBinning* command sends a command to the IPSU-G to set the horizontal (X) and vertical (Y) binning. Binning is summing of charge of adjacent pixels.

Command ID: 0xC201
Mnemonic: ACME_DATA_CAMERA_SET_BINNING

Parameter	# Bytes Type	Values	Description
binningX	4 Uint32	1-8	
binningY	4 Uint32	1-14	

3.1.4.2 DataCameraSetPixelFormat

The *DataCameraSetPixelFormat* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC202
Mnemonic: ACME_DATA_CAMERA_SET_PIXEL_FORMAT

Parameter	# Bytes Type	Values	Description
pixelFormat	4 Uint32	Only settable when not capturing images 0 = Mono8 1 = BayerRG8 2 = BayerRG12 (desired format) 3 = BayerGR12Packed 4 = RGB8Packed 5 = BGR8Packed 6 = YUV411Packed 7 = YUV422Packed 8 = YUV444Packed	

3.1.4.3 DataCameraSetRegionOfInterest

The *DataCameraSetRegionOfInterest* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC203
Mnemonic: ACME_DATA_CAMERA_SET_ROI

Parameter	# Bytes Type	Values	Description
xCoordinate	4 Uint32	0-1360	OffsetX
yCoordinate	4 Uint32	0-1024	OffsetY
height	4 Uint32	1-1024	
width	4 Uint32	1-1360	

3.1.4.4 DataCameraSetAutomaticExposure

The *DataCameraSetAutomaticExposure* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC204
Mnemonic: ACME_DATA_CAMERA_SET_AUTO_EXPOSURE

Parameter	# Bytes Type	Values	Description
algorithm	4 Uint32	0x0 = Mean 0x1 = FitRange	
tolerance	4 Uint32	0-50 %	
maxTime	4 Uint32	Microseconds	

Parameter	# Bytes Type	Values	Description
minTime	4 Uint32	Microseconds	
outliers	4 Uint32	.01% units	
rate	4 Uint32	%	
target	4 Uint32	% (0 = black, 100 = white)	

3.1.4.5 DataCameraSetManualExposure

The *DataCameraSetManualExposure* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC205
Mnemonic: ACME_DATA_CAMERA_SET_EXPOSURE

Parameter	# Bytes Type	Values	Description
integrationTimeUs	4 Uint32	10 - 60,000,000	Microseconds

3.1.4.6 DataCameraSetFrameRate

The *DataCameraSetFrameRate* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC206
Mnemonic: ACME_DATA_CAMERA_SET_FRAME_RATE

Parameter	# Bytes Type	Values	Description
frameRate	4 Float	0.0127329 – 30.0111	

3.1.4.7 DataCameraSetAutoGain

The *DataCameraSetAutoGain* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC207
Mnemonic: ACME_DATA_CAMERA_SET_AUTO_GAIN

Parameter	# Bytes Type	Values	Description
algorithm	4 Uint32	0=Mean, 1=FitRange	

Parameter	# Bytes Type	Values	Description
tolerancePercent	4 Uint32		0-50 %
maxDb	4 Uint32		dB
minDb	4 Uint32		dB
outliers	4 Uint32		.01% units
ratePercent	4 Uint32		%
targetPercent	4 Uint32		%

3.1.4.8 DataCameraSetManualGain

The *DataCameraSetManualGain* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC208
Mnemonic: ACME_DATA_CAMERA_SET_MANUAL_GAIN

Parameter	# Bytes Type	Values	Description
gainValueDb	4 Uint32	0 - 33	

3.1.4.9 DataCameraSetAutoWhiteBalance

The *DataCameraSetAutoWhiteBalance* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC209
Mnemonic: ACME_DATA_CAMERA_SET_AUTO_WHITE_BAL

Parameter	# Bytes Type	Values	Description
tolerance	4 Uint32		
rate	4 Uint32		

3.1.4.10 DataCameraSetManualWhiteBalance

The *DataCameraSetManualWhiteBalance* command sends a command to the IPSU-G. Sets the gain of the red and blue channels. The green channel gain is always 1.00.

Command ID: 0xC20A
Mnemonic: ACME_DATA_CAMERA_SET_MAN_WHITE_BAL

Parameter	# Bytes Type	Values	Description
redPercent	4 Uint32	0.8 – 3.0	Red gain as % of camera default
bluePercent	4 Uint32	0.8 – 3.0	Blue gain as % of camera default

3.1.4.11 DataCameraLoadConfig

The *DataCameraLoadConfig* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC20B
Mnemonic: ACME_DATA_CAMERA_LOAD_CONFIG

Parameter	# Bytes Type	Values	Description
index	4 Uint32	0 - 5	

3.1.4.12 DataCameraSaveConfig

The *DataCameraSaveConfig* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC20C
Mnemonic: ACME_DATA_CAMERA_SAVE_CONFIG

Parameter	# Bytes Type	Values	Description
index	4 Uint32	0 - 5	

3.1.4.13 DataCameraSetPowerUpConfig

The *DataCameraSetPowerUpConfig* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC20D
Mnemonic: ACME_DATA_CAMERA_SET_PWR_UP_CONFIG

Parameter	# Bytes Type	Values	Description
index	4 Uint32	0 - 5	

3.1.4.14 DataCameraStartCapture

The *DataCameraStartCapture* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC20E
Mnemonic: ACME_DATA_CAMERA_START_CAPTURE

3.1.4.15 DataCameraStopCapture

The *DataCameraStopCapture* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC20F
Mnemonic: ACME_DATA_CAMERA_STOP_CAPTURE

3.1.4.16 DataCameraStartArchive

The *DataCameraStartArchive* command sends a command to the specified IPSU-G to start recording images using the specified cameras. More details TBD.

Command ID: 0xC210
Mnemonic: ACME_DATA_CAMERA_START_ARCHIVE

Parameter	# Bytes Type	Values	Description
fileName	81 bytes String		
IPSU-G Number	1 Uint8	5 or 6	
Camera 1	1 Uint8	1-4	
Camera 2	1 Uint8	1-4, or 0 for no camera 2-4	
Camera 3	1 Uint8	1-4, or 0 for no camera 3-4	
Camera 4	1 Uint8	1-4, or 0 for no camera 4	

3.1.4.17 DataCameraStopArchive

The *DataCameraStopArchive* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC211
Mnemonic: ACME_DATA_CAMERA_STOP_ARCHIVE

3.1.4.18 DataCameraGrabOneImage

The *DataCameraGrabOneImage* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC212
Mnemonic: ACME_DATA_CAMERA_GRAB_ONE_IMAGE

Parameter	# Bytes Type	Values	Description
fileName	81 bytes String		

3.1.4.19 DataCameraMoveZoomAxis

The *DataCameraMoveZoomAxis* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC213
Mnemonic: ACME_DATA_CAMERA_MOVE_ZOOM_AXIS

Parameter	# Bytes Type	Values	Description
position	4 Uint32		

3.1.4.20 DataCameraMoveIrisAxis

The *DataCameraMoveIrisAxis* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC214
Mnemonic: ACME_DATA_CAMERA_MOVE_IRIS_AXIS

Parameter	# Bytes Type	Values	Description
position	4 Uint32		

3.1.4.21 DataCameraMoveFocusAxis

The *DataCameraMoveFocusAxis* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC215
Mnemonic: ACME_DATA_CAMERA_MOVE_FOCUS_AXIS

Parameter	# Bytes Type	Values	Description
position	4 Uint32		

3.1.4.22 DataCameraMoveFilterAxis

The *DataCameraMoveFilterAxis* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC216
Mnemonic: ACME_DATA_CAMERA_MOVE_FILTER_AXIS

Parameter	# Bytes Type	Values	Description
position	4 Uint32		

3.1.4.23 DataCameraMoveWithParameters

The *DataCameraMoveWithParameters* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC217
Mnemonic: ACME_DATA_CAMERA_MOVE_WITH_PARAMS

Parameter	# Bytes Type	Values	Description
axisNumber	1 Uint8		
position	4 Uint32		
velocity	4 Uint32		
acceleration	4 Uint32		
deceleration	4 Uint32		
jerk	4 Uint32		

3.1.4.24 DataCameraSetupMoveWithParameters

The *DataCameraSetupMoveWithParameters* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC218
Mnemonic: ACME_DATA_CAMERA_SETUP_MOVE_WITH_PARAMS

Parameter	# Bytes Type	Values	Description
axisNumber	1 Uint8		
position	4 Uint32		
velocity	4 Uint32		
acceleration	4 Uint32		
deceleration	4 Uint32		
jerk	4 Uint32		

3.1.4.25 DataCameraExecuteSetupMove

The *DataCameraExecuteSetupMove* command sends a command to the IPSU-G. More details TBD.

Command ID: 0xC219
Mnemonic: ACME_DATA_CAMERA_EXECUTE_SETUP_MOVE

Parameter	# Bytes Type	Values	Description
axisNumber	1 Uint8	1-4	

3.1.4.26 DataCameraSelectFilter

The *DataCameraSelectFilter* command sends a command to the IPSU-G to move the filter barrel to the position of the specified filter. Positions are defined in AcmeConfig.ini using the keys AvtCameraMotion/FilterXPosition, where X is 1 through 4.

Command ID: 0xC21A
Mnemonic: ACME_DATA_CAMERA_SELECT_FILTER

Parameter	# Bytes Type	Values	Description
filter	1 Uint8	1-4	Filter barrel filter number.

3.1.4.27 AnalogCameraSetAgc

The *AnalogCameraSetAgc* command disables automatic gain control or enables it and sets the maximum gain on the operations camera located in the ACME CIA. For more details about the camera see the Hitachi KP-D20B web page: http://www.hitachikokusai.com/Products/IndustrialVisionSystems/InterlaceScanCameras/POC_007075.html

Command ID: 0xC21B
Mnemonic: ACME_ANALOG_CAMERA_SET_AGC

Parameter	# Bytes Type	Values	Description
gain	1 Uint8	0 = disabled 1 = 6 dB 2 = 12 dB 3 = 21 dB 4 = 31 dB	

3.1.4.28 AnalogCameraSetShutter

The *AnalogCameraSetShutter* command sets the shutter speed on the operations camera located in the ACME CIA. See the *AnalogCameraEnableAgc* command for more details about the camera.

Command ID: 0xC21C
Mnemonic: ACME_ANALOG_CAMERA_SET_SHUTTER

Parameter	# Bytes Type	Values	Description
shutter	1 Uint8	0 = 1/60 sec 1 = 1/100 sec 2 = 1/250 sec 3 = 1/500 sec 4 = 1/1000 sec 5 = 1/2000 sec 6 = 1/4000 sec 7 = 1/10000 sec 8 = 1/20000 sec 9 = 1/30000 sec	

3.1.4.29 IlluminationConfigure

The *IlluminationConfigure* command turns on or off the illumination LEDs in the CIA on the analog camera assembly.

Command ID: 0xC21D
Mnemonic: ACME_ILLUMINATION_CONFIG

Parameter	# Bytes Type	Values	Description
sideLEDBankState	1 Uint8	0 = Off 1 = On	This is the bank of 3 LEDs to the side of the camera axis and turning mirror, oriented in a row parallel to the camera axis.
endLEDBankState	1 Uint8	0 = Off 1 = On	This is the bank of 3 LEDs inline with the axis of the analog camera, just past the turning mirror, at the end of the analog camera assembly. It is oriented in a row perpendicular to the camera axis.

3.1.4.30 IlluminationCycle

The *IlluminationCycle* command cycles one or both banks of illumination LEDs in the CIA on and off for specified durations and number of cycles. The final state is always both banks off.

Hint: An ongoing cycle of illumination can be cancelled and the lights turned off by sending this command with bank states of 0, a cycle count of 1, and minimum on and off durations.

Command ID: 0xC21E
Mnemonic: ACME_ILLUMINATION_CYCLE

Parameter	# Bytes Type	Values	Description
sideLEDBankState	1 Uint8	0 = Off 1 = Cycle On/Off	Include the side bank in the on and off cycling or not.
endLEDBankState	1 Uint8	0 = Off 1 = Cycle On/Off	Include the end bank in the on and off cycling or not.
onDuration	4 Float	0.03 - 999.99	Duration in seconds that the LED(s) are on.
offDuration	4 Float	0.03 - 999.99	Duration in seconds that the LED(s) are off.
cycleCount	4 Uint32	1-9999	Number of times to cycle LED(s) on and off.

3.1.4.31 StartIpsus

The *StartIpsus* command initiates a sequence of commands to the CIR IPSUs and IPSU-Gs to start recording images under the specifed test point name and for the IPSU-Gs using the specified cameras. Command sequences are only sent to those IPSUs and IPSU-Gs that were specified as active in the *SystemDefineSetup* command. If sequence is running it must be stopped using the *StopIpsus* command before a new start sequence can be started.

The command sequence is a follows where each row in the table corresponds to a commanding phase.

IPSUs	IPSU-Gs

Test Point Open, ID 0x4020	Test Point Open(testpointName), ID 0x4001
Enable Archiving, ID 0x4025	(Waiting for IPSU response to any Enable Archiving commands)
Disable Live Video, ID 0x4036	(Waiting for IPSU response to any Disable Live Video commands)
Start Test Point(testpointName), ID 0x4022	Camera Start Test Point Archive, ID 0x420D

When all active IPSUs and IPSU-Gs respond to their phase command, then the next command phase begins. If a response is not received within a timeout period then a message is logged identifying the offending IPSU(s), the system is sent to the Idle state, and no further commands are sent to any IPSUs or IPSU-Gs. Any test points that were successfully started must be stopped by hand. The response timeout period is the registry value, Ipsu/IpsuCommandResponseTimoutMsec.

Command ID: 0xC21F
Mnemonic: ACME_IPSUS_START

Parameter	# Bytes Type	Values	Description
testpointName	9 String		
Ipsu 5 Number Cameras	1 Uint8	0-4	
Ipsu 5 Camera 1	1 Uint8	1-4	
Ipsu 5 Camera 2	1 Uint8	1-4	
Ipsu 5 Camera 3	1 Uint8	1-4	
Ipsu 5 Camera 4	1 Uint8	1-4	
Ipsu 6 Number Cameras	1 Uint8	0-4	
Ipsu 6 Camera 1	1 Uint8	1-4	
Ipsu 6 Camera 2	1 Uint8	1-4	
Ipsu 6 Camera 3	1 Uint8	1-4	
Ipsu 6 Camera 4	1 Uint8	1-4	

3.1.4.32 StopIpsus

The *StopIpsus* command initiates a sequence of commands to the CIR IPSUs and IPSU-Gs to stop recording images and close a currently running test point. Command sequences are only

sent to those IPSUs and IPSU-Gs that were specified as active in the *SystemDefineSetup* command.

The command sequence is a follows where each row in the table corresponds to a commanding phase.

IPSUs	IPSU-Gs
Test Point Stop, ID 0x4023	Test Point Stop, ID 0x420E
Test Point Close, ID 0x4021	Test Point Close, ID 0x4002

When all active IPSUs and IPSU-Gs respond to their phase command, then the next command phase begins. If a response is not received within a timeout period then a message is logged identifying the offending IPSU(s), the system is sent to the Idle state, and no further commands are sent to any IPSUs or IPSU-Gs. The response timeout period is the registry value, Ipsu/IpsuCommandResponseTimoutMsec.

Command ID: 0xC220
Mnemonic: ACME_IPSUS_STOP

3.1.4.33 DataCameraGetReport

The *DataCameraGetReport* command, when the ACME avionics package is controlling the ACME camera instead of the IPSU-G, causes a report of camera parameters to be placed in the telemetry log.

Command ID: 0xC221
Mnemonic: ACME_DATA_CAMERA_GET_REPORT

3.1.4.34 IntensifiedCameraGate

The *IntensifiedCameraGate* command sends associated commands to the IPSUG to set the intensifier gate type and width. If the gateType parameter is continuous the pulsedGateWidthCounts parameter is not used. Raw gate width of 0-49999 corresponds to 20 to 1,000,000 nS.

Command ID: 0xC222
Mnemonic: ACME_INTENSIFIED_CAMERA_GATE

Parameter	# Bytes Type	Values	Description
gateType	1 Uint8	0 = Pulsed (safe) 1= Continuous	
pulsedGateWidthCounts	4 Uint32	0-49999	The raw gate width (not nanoseconds)

3.1.4.35 IntensifiedCameraGainAmplitude

The *IntensifiedCameraGainAmplitude* command sends an associated command to IPSUG to change the intensifier gain.

Command ID: 0xC223
Mnemonic: ACME_INTENSIFIED_CAMERA_GAIN_AMPLITUDE

Parameter	# Bytes Type	Values	Description
gainAmplitudeCounts	4 Uint32	2048 (safe gain) to 4095	Intensifier gain

3.1.4.36 DataCameraExposureBracketing

The *DataCameraExposureBracketing* command issues an associated command to the IPSUG to perform (WHEN IMPLEMETED on the IPSUG) exposure bracketing on the ACME camera. Several images are taken at an exposure before increasing the exposure and taking several more, repeating for a specified number of exposures, then going back to the initial exposure. For example, with imagesPerExposure=2, exposures=3, exposureStartUsec=50000 usec, and exposureIncrementUsec=5000 the following image frames and exposures will be acquired:

1 50000
2 50000
3 55000
4 55000
5 60000
6 60000
7 50000
8 50000
9 55000 …

Command ID: 0xC224
Mnemonic: ACME_DATA_CAMERA_EXPOSURE_BRACKETING

Parameter	# Bytes Type	Values	Description
enable	1 Uint8	0=disable, 1=enable	Turn bracketing on or off
imagesPerExposure	2 Uint16	1-65535	Number of images at an exposure at given frame rate before moving to next exposure
exposures	2 Uint16	2-65535	Number of different exposures
exposureStartUsec	4 Int32		First exposure
exposureIncrementUsec	4 Int32		Amount to increment exposure time each time.

3.1.4.37 IntensifiedCameraMoveFilter

The *IntensifiedCameraMoveFilter* command moves the filter on the intensified camera to the specified position. This is implemted by sending the associated command to IPSUG.

Command ID: 0xC225
Mnemonic: ACME_INTENSIFIED_CAMERA_MOVE_FILTER

Parameter	# Bytes Type	Values	Description
position	4 Uint32		Filter position

3.1.4.38 IntensifiedCameraSafe

The *IntensifiedCameraSafe* command places the intensified camera in a state that is safe for turning on lights without harming the camera by sending commands to the IPSUG. The filter is commanded to home position and a fixed delay occurs to allow this to finish (without checking that it is finished). Then the filter is sent to a safe position with another fixed delay. The gain is set to 2048 (reporting back 2500). Gate type is set to pulsed and gate width is set to zero (reporting back 20). The delays and safe parameters are defined in the Intensifier group in AcmeConfig.ini.

Command ID: 0xC226
Mnemonic: ACME_INTENSFIED_CAMERA_SAFE

3.1.5 ACME Igniter Group Command Details

3.1.5.1 IgniterDisable

The *IgniterDisable* command shuts down power to the igniter. Specifically, it pulls low the primary control (PC) pin on the DC-DC converter that supplies the igniter. For more information see the data sheet for the Vicor V24C12C100BL2 DC-DC converter.

Command ID: 0xC301
Mnemonic: ACME_IGNITER_DISABLE

3.1.5.2 IgniterPulse

The *IgniterPulse* command turns on the igniter for a specified duration and power and then shuts it off. Specifically, it sets the secondary control (SC) pin on the DC-DC converter that supplies the igniter to the specfied power voltage and then sets to primary control (PC) pin to 5V which enables output to the igniter. After the specified pulse duration the PC pin is set to 0V, disabling output to the igniter.

The power parameter is converted to a control voltage using the equation,

Vcontrol = B * power + A, where A and B are ConversionFactorA and ConversionFactorB in the Igniter group of AcmeConfig.ini. See **Table *32***.

The power parameter is limited by MinPowerVolts and MaxPowerVolts in the Igniter group in AcmeConfig.ini. These are control voltages not voltages at the igniter.

Command ID: 0xC302
Mnemonic: ACME_IGNITER_PULSE

Parameter	# Bytes Type	Values	Description
power	4 Float	0.46V to 12V	Igniter voltage corresponding to -0.6 to 0.8V control voltage.
durationMs	4 Uint32		Pulse duration in milliseconds.

3.1.5.3 IgniterMove

The *IgniterMove* command moves the igniter arm to the specified absolute postion. The velocity, acceleration, and deceleration values used for the move are specified in AcmeConfig.ini. See ***Table 32***, Group: Igniter, Keys: VelocityCountsPerSec, AccelerationCountsPerSecSquared, and DecelerationCountsPerSecSquared. To go to the home (retracted) position specify zero as the position. In this case the keys used are HomeVelocityCountsPerSec, HomeAccelerationCountsPerSecSquared, and DecelerationCountsPerSecSquared.

The igniter arm is moved using a trapezoidal velocity versus time profile. If the acceleration parameter is greater than zero then movement begins with a ramp up until the specified velocity is achieved. If the deceleration parameter is greater than zero then there is a ramp down to zero velocity. The duration of the movement is calculated to reach the desired position.

This command takes a few seconds to execute. To shorten the duration between command and movement use the *IgniterSetupMove* or *IgniterSetupMoveWithParameters* command followed by the *IgniterExecuteSetupMove* command when ready to move.

The motor is a Haydon Kerk 25544-05-004ENG captive linear actuator. There is no encoder.

Command ID: 0xC303
Mnemonic: ACME_IGNITER_MOVE

Parameter	# Bytes Type	Values	Description
position	4 Uint32	0 = home (retracted) 1 to maximum position.	Move igniter arm motor to this absolute position in counts. Maximum position is specified in AcmeConfig.ini and depends on the installed burner. See ***Table 32***, Group: Igniter, Keys: BurnerLimits and MaxPosition.

3.1.5.4 IgniterSetupMove

An *IgniterSetupMove* command followed by an *IgniterSetupMove* command is the same as the *IgniterMove* command. By separating the setup, which takes a few seconds, and the execute, which occurs quickly, the beginning of movement can be commanded or scripted closer to an event or other command. See the *IgniterMove* command for information about what velocity, acceleration, and deceleration are used.

Command ID: 0xC304
Mnemonic: ACME_IGNITER_SETUP_MOVE

Parameter	# Bytes Type	Values	Description
position	4 Uint32	0 = home (retracted) 1 to maximum position.	Prepare to move igniter arm motor to this absolute position in counts. Maximum position is specified in AcmeConfig.ini and depends on the installed burner. See ***Table 32***, Group: Igniter, Keys: BurnerLimits and MaxPosition.

3.1.5.5 IgniterSetupMoveWithParameters

The *IgniterSetupMoveWithParameters* command is the same as the *IgniterSetupMove* command except that velocity, acceleration, and deceleration are specified in the command not in AcmeConfig.ini. Follow with an *IgniterExecuteSetupMove* command when ready to start motion.

Command ID: 0xC305
Mnemonic: ACME_IGNITER_SETUP_MOVE_WITH_PARAMS

Parameter	# Bytes Type	Values	Description
position	4 Uint32	0 = home (retracted) 1 to maximum position.	Prepare to move igniter arm motor to this absolute position in counts. Maximum position is specified in AcmeConfig.ini and depends on the installed burner. See ***Table 32***, Group: Igniter, Keys: BurnerLimits and MaxPosition.
velocity	4 Uint32	Software limit: 0 to 128,000,000 Hardware limit TBD	Prepare to move igniter arm using this velocity in counts/sec. The range of values at the PMDSetVelocity function is 0 to 2,147,483,647 following software limits.
acceleration	4 Uint32	Software limit: 0 to 4,294,967,295, Hardware limit TBD	Prepare to move igniter arm using this acceleration in counts/sec^2. The range of values at the PMDSetAcceleration function is 0 to 18,446,744 following software limits.
deceleration	4 Uint32	Software limit: 0 to 4,294,967,295, Hardware limit TBD	Prepare to move igniter arm using this deceleration in counts/sec^2. The range of values at the PMDSetDeceleration function is 0 to 18,446,744 following software limits.

3.1.5.6 IgniterExecuteSetupMove

The *IgniterExecuteSetupMove* command executes a move previously set up with the *IgniterSetupMove* or *IgniterSetupMoveWithParameters* commands.

Command ID: 0xC306
Mnemonic: ACME_IGNITER_EXECUTE_SETUP_MOVE

3.1.5.7 IgniterMotorSetState

The *IgniterMotorSetState* command disables or enables the igniter motor by setting the RST line on the stepper driver to 0 V or 5 V respectively. In normal operations this command should not be necessary because the motor is automatically enabled as soon as the cube is active (the signal originates in the cube), and is disabled when the software shuts down. For reference, the stepper driver module is an Intelligent Motion System (IMS) XIM481H installed on an IMS INT-481 board with MS0, MS2, and MS3 in the on position.

To reset the motion controller board instead use the *SystemResetMotionController* command.

Command ID: 0xC307

Mnemonic: ACME_IGNITER_MOTOR_SET_STATE

Parameter	# Bytes Type	Values	Description
state	1 Uint8	0 = Reset 1 = On	Sets IGN_MOTOR_RST (**Table *18***) to 0 V (Reset) or 5 V (On).

3.1.6 ACME Flow Group Command Details

For flow group commands refer to the latest revision of drawing S1101MFS004, "ACME/CIR Integrated Pneumatic Schematic" to see how flow devices are connected to each other. A screen shot of the ACME CIA portion of this drawing (Rev A) is shown below. Gas from the CIR comes in at the bottom of the diagram; fuel through QD11, nitrogen bypass through QD12, and static mixer through QD13.

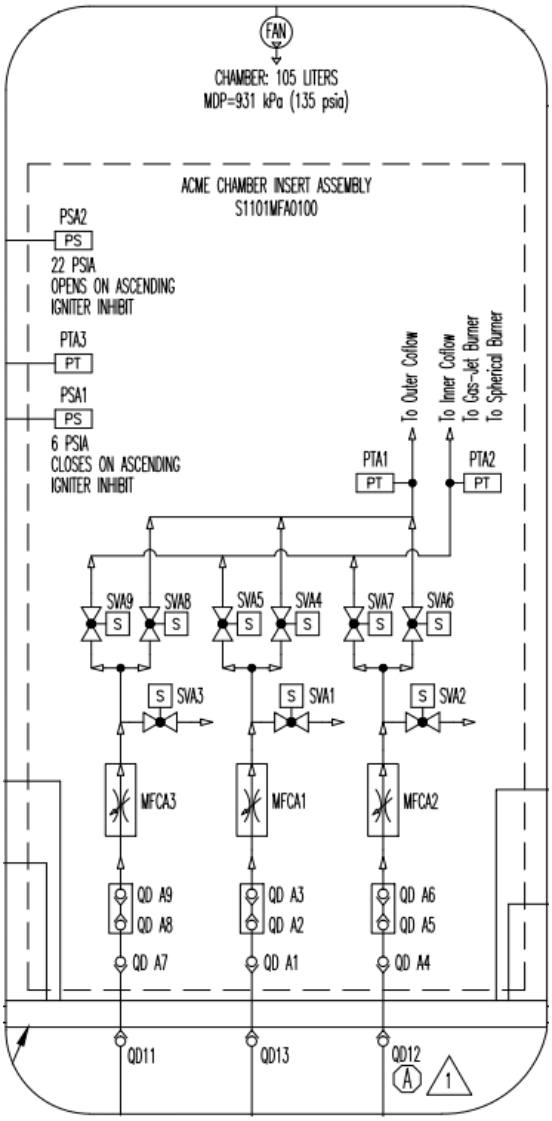


Figure 2 ACME CIA Flow Schematic

3.1.6.1 FlowEnableMfcs

The *FlowEnableMfcs* command turns the mass flow controllers on or off. Initially they are all turned on as soon as the cube boots. After this first power-on the MFCs are queried for their configuration, including full scale flow which appears in telemetry. Subsequent commanded power-ons do not cause a query of configuration.

The current state of a MFC is shown in telemetry by the Statix Mixer MFC State, Bypass MFC State, and Fuel MFC State items.

The mass flow controllers are Teledyne Hastings HFC-D-302 with different options for flow range depending on position, MFCA1, MFCA2, or MFCA3, and on experiment hardware configuration.

Command ID: 0xC401
Mnemonic: ACME_FLOW_ENABLE_MFCS

Parameter	# Bytes Type	Values	Description
MFCA1State	1 Uint8	0 = Off 1 = On	Sets MFCA1_PWR (**Table *18***) to 0 V (Off) or 5 V (On).
MFCA2State	1 Uint8	0 = Off 1 = On	Sets MFCA2_PWR (**Table *18***) to 0 V (Off) or 5 V (On).
MFCA3State	1 Uint8	0 = Off 1 = On	Sets MFCA3_PWR (**Table *18***) to 0 V (Off) or 5 V (On).

3.1.6.2 FlowSetMfcRate

The *FlowSetMfcRate* command sets the flow rate setpoint for the selected mass flow controller. The flow rate specified in the command is in standard liters per minute which is converted to 0 to 5V that is applied to the analog input control on the mass flow controller according to the equation:

Analog input control (V) = flowSlm / fullScaleValue * 5.0

,where fullScaleValue depends on the installed mass flow controller and is retrieved from the unit during the configuration query after power-on and is reported in telemetry for each MFC.

Command ID: 0xC402
Mnemonic: ACME_FLOW_SET_MFC_RATE

Parameter	# Bytes Type	Values	Description
massFlowController	1 Uint8	1 = MFCA1, Static Mixer 2 = MFCA2, Bypass 3 = MFCA3, Fuel	
flowSlm	4 Float	0 to maximum flow.	Desired flow setpoint in standard liters per minute. Maximum flow is the Full Scale value shown in telemetry for the selected MFC.

3.1.6.3 FlowSetAllMfcRates

The *FlowSetAllMfcRates* command sets the flow rate setpoint for all three mass flow controllers. See the *FlowSetMfcRate* command for more information about flow setpoints.

Command ID: 0xC403
Mnemonic: ACME_FLOW_SET_ALL_MFC_RATES

Parameter	# Bytes Type	Values	Description
MFCA1FlowSlm	4 Float	0 to maximum flow.	Desired flow setpoint in standard liters per minute. Maximum flow is the Full Scale value shown in telemetry for this MFC.
MFCA2FlowSlm	4 Float	0 to maximum flow.	Desired flow setpoint in standard liters per minute. Maximum flow is the Full Scale value shown in telemetry for this MFC.
MFCA3FlowSlm	4 Float	0 to maximum flow.	Desired flow setpoint in standard liters per minute. Maximum flow is the Full Scale value shown in telemetry for this MFC.

3.1.6.4 FlowZeroMfcs

The *FlowZeroMfcs* command issues the ZRO serial command to the selected mass flow controllers. From the Teledyne Hastings Digital 300 Software Guide the description of this command is as follows:

> This command adjusts the offset value of the flow meter such that the indicated flow reads zero flow at the present flow rate. Updates user offset value. This command should be executed periodically on flow meters when the flow rate is known to be zero. If this command is sent when the flow rate is non-zero this can cause significant flow reading errors. Flow controllers typically have the "autozero" function enabled and will not need to be re-zeroed.

ACME mass flow controllers have the autozero function TBD.

Command ID: 0xC404
Mnemonic: ACME_FLOW_ZERO_MFCS

Parameter	# Bytes Type	Values	Description
zeroMFCA1	1 Uint8	0 = Do not zero 1 = Zero this MFC	Whether or not to zero this MFC
zeroMFCA2	1 Uint8	0 = Do not zero 1 = Zero this MFC	Whether or not to zero this MFC
zeroMFCA3	1 Uint8	0 = Do not zero 1 = Zero this MFC	Whether or not to zero this MFC

3.1.6.5 FlowMfcSerialCommand

The *FlowMfcSerialCommand* command sends the specified raw serial command to the daisy-chained mass flow controllers. The start of the string is the address that determines the destination controller: '*00' = MFCA1 Static Mixer, '*01' = MFCA2 Bypass, '*02' = MFCA3 Fuel. The command follows the address, for example '*01ZRO' to zero MFCA2, '*02MS' to get the state of MFCA3, '*00sfl' to get a status of all configuration items (the command sent after power-on) from MFCA1. See the Teledyne Hastings Digital 300 Software Guide for available commands. Replies to commands are placed in the telemetry text area.

Serial commands are queued. A command will not be issued until a response is received from any prior command or until the prior command response times out. The response time out is specified by the registry value: Flow/MfcSerialCommandTimeoutMsec. See **Table *32***.

Command ID: 0xC405
Mnemonic: ACME_FLOW_MFC_SERIAL_COMMAND

Parameter	# Bytes Type	Values	Description
serialCommand	81 String		

3.1.6.6 FlowSetSolenoidValveState

The *FlowSetSolenoidValveState* command opens or closes the specified ACME solenoid valve. A valve is opened (allowing flow) by applying 5 V. A valve is closed (stopping flow) by applying 0 V. The valves are the normally-closed type, that is they do not allow flow when un-powered.

Command ID: 0xC406
Mnemonic: ACME_FLOW_SET_SOLENOID_VALVE_STATE

Parameter	# Bytes Type	Values	Description
solenoidValve	1 Uint8	1 to 9	Valve SVAx, where x=1 to 9.
state	1 Uint8	0 = closed (stop flow) 1 = open (allow flow)	Close or open the valve by applying 0V or 5V.

3.1.6.7 FlowSetAllSolenoidValveStates

The *FlowSetAllSolenoidValveStates* command sets the state, open or closed, of all nine ACME solenoid valves at once.

Command ID: 0xC407
Mnemonic: ACME_FLOW_SET_ALL_SOLENOID_VALVE_STATES

Parameter	# Bytes Type	Values	Description
SVA1State	1 Uint8	0 = closed (stop flow) 1 = open (allow flow)	
SVA2State	1 Uint8	0 = closed (stop flow) 1 = open (allow flow)	
SVA3State	1 Uint8	0 = closed (stop flow) 1 = open (allow flow)	
SVA4State	1 Uint8	0 = closed (stop flow) 1 = open (allow flow)	
SVA5State	1 Uint8	0 = closed (stop flow) 1 = open (allow flow)	
SVA6State	1 Uint8	0 = closed (stop flow) 1 = open (allow flow)	
SVA7State	1 Uint8	0 = closed (stop flow) 1 = open (allow flow)	
SVA8State	1 Uint8	0 = closed (stop flow) 1 = open (allow flow)	
SVA9State	1 Uint8	0 = closed (stop flow) 1 = open (allow flow)	

3.1.6.8 FlowOpenAll

The *FlowOpenAll* command sets MFCA1, MFCA2, and MFCA3 to full scale flow and opens solenoid valves SVA1 through SVA9.

Command ID: 0xC408
Mnemonic: ACME_FLOW_OPEN_ALL

3.1.6.9 FlowCloseAll

The *FlowCloseAll* command sets MFCA1, MFCA2, and MFCA3 to zero flow and closes solenoid valves SVA1 through SVA9. Any flow waveforms started with the *FlowStepped*, *FlowRamped*, *FlowRectangularWave*, or *FlowRampedWave* commands are stopped.

Command ID: 0xC409
Mnemonic: ACME_FLOW_CLOSE_ALL

3.1.6.10 FlowConfigForFill

The *FlowConfigForFill* command configures ACME mass flow controllers and solenoid valves for filling the chamber. MFCA1, MFCA2, and MFCA3 are set to full scale flow and SVA1, SVA2, and SVA3 solenoid valves are opened. SVA4 through SVA9 are closed.

Command ID: 0xC40A
Mnemonic: ACME_FLOW_CONFIG_FOR_FILL

3.1.6.11 FlowStepped

The *FlowStepped* command steps the flow of each mass flow controller up or down, or not at all, by a specified step size, waits a specified duration, and steps again, repeating until each specified maximum flow (or minimum flow if stepping down) is reached, and then leaving the mass flow controller at the last flow.

For positive step sizes the setpoints start at the minimum flow and step up until the maximum is reached, and does not go above. For negative step sizes the setpoints start at the maximum and step down until the minimum is reached, and does not go below.

Stepping can be stopped with the *FlowStopWaveform* command, which leaves the flow setpoints where they were last, or with the *FlowCloseAll* command, which sets all flows to zero and closes all valves, with the *FlowSetMfcRate* or *FlowSetAllMfcRates* commands which leave the flow at the setpoint specified by the command. Stepping is stopped for a MFC by any other flow waveform command if the parameters say to control that MFC, otherwise ongoing stepping continues until complete.

Command ID: 0xC40B
Mnemonic: ACME_FLOW_STEPPED

Parameter	# Bytes Type	Values	Description
stepDurationSecs	4 Float	0.1 to 999.9	Seconds to wait between steps
MFCA1StepSizeSlm	4 Float	-1*full scale to full scale 0 = do not step this MFC	Static mixer MFC step size. Full scale depends on the MFC and is in telemetry.
MFCA1MinimumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA1MaximumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA2StepSizeSlm	4 Float	-1*full scale to full scale 0 = do not step this MFC	Bypass MFC step size. Full scale depends on the MFC and is in telemetry.
MFCA2MinimumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA2MaximumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA3StepSizeSlm	4 Float	-1*full scale to full scale 0 = do not step this MFC	Fuel MFC step size. Full scale depends on the MFC and is in telemetry.
MFCA3MinimumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA3MaximumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.

3.1.6.12 FlowRamped

The *FlowRamped* command ramps the flow of each mass flow controller up or down, or not at all, at a specified ramp rate for a specified duration, waits a specified hold duration, and ramps again, repeating until each specified maximum flow (or minimum flow if ramping down) is reached, and then leaving the mass flow controller at the last flow.

For positive ramp rates the setpoints start at the minimum flow and ramp up until the maximum is reached, and does not go above. For negative ramp rates the setpoints start at the maximum and ramp down until the minimum is reached, and does not go below. During the ramp the control voltage of the mass flow controller is updated every millisecond.

Ramping can be stopped with the *FlowStopWaveform* command, which leaves the flow setpoints where they were last, with the *FlowCloseAll* command, which sets all flows to zero and closes all valves, or with the *FlowSetMfcRate* or *FlowSetAllMfcRates* commands which leave the flow at the setpoint specified by the command. Ramping is stopped for a MFC by any other flow waveform command if the parameters say to control that MFC, otherwise ongoing ramping continues until complete.

Command ID: 0xC40C
Mnemonic: ACME_FLOW_RAMPED

Parameter	# Bytes Type	Values	Description
rampDurationSec	4 Float	0.1 – 999.9	Seconds to ramp each MFC.
holdDurationSec	4 Float	0.1 – 999.9	Seconds to hold between ramps.
MFCA1RampRateSlmPerSec	4 Float	0 = do not ramp this MFC	Static mixer MFC ramp rate.
MFCA1MinimumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA1MaximumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA2RampRateSlmPerSec	4 Float	0 = do not ramp this MFC	Bypass MFC ramp rate.
MFCA2MinimumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA2MaximumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA3RampRateSlmPerSec	4 Float	0 = do not ramp this MFC	Fuel MFC ramp rate.
MFCA3MinimumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA3MaximumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.

3.1.6.13 FlowRectangularWave

The *FlowRectangularWave* command steps the flow of each mass flow controller a value, A, up, down, or not at all, waits a specified A duration, steps another value, B, up, down, or not at all, and waits a specified B duration. This sequence is repeated until the maximum or minimum flow is reached or until stopped by a command.

Rectangular waves can be stopped with the *FlowStopWaveform* command, which leaves the flow setpoints where they were last, with the *FlowCloseAll* command, which sets all flows to zero and closes all valves, or with the *FlowSetMfcRate* or *FlowSetAllMfcRates* commands which leave the flow at the setpoint specified by the command. Rectangular waves are stopped for a MFC by any other flow waveform command if the parameters say to control that MFC.

Command ID: 0xC40D
Mnemonic: ACME_FLOW_RECTANGULAR_WAVE

Parameter	# Bytes Type	Values	Description
stepADurationSec	4 Float	0.1 – 999.9	Seconds to wait after A steps.
stepBDurationSec	4 Float	0.1 – 999.9	Seconds to wait after B steps.
MFCA1StepASlm	4 Float	-1*full scale to full scale 0 = do not step A this MFC	Static mixer MFC step A size. Full scale depends on the MFC and is in telemetry.
MFCA1StepBSlm	4 Float	-1*full scale to full scale 0 = do not step B this MFC	Static mixer MFC step B size. Full scale depends on the MFC and is in telemetry.
MFCA1MinimumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA1MaximumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA2StepASlm	4 Float	-1*full scale to full scale 0 = do not step A this MFC	Bypass MFC step A size. Full scale depends on the MFC and is in telemetry.
MFCA2StepBSlm	4 Float	-1*full scale to full scale 0 = do not step B this MFC	Bypass MFC step B size. Full scale depends on the MFC and is in telemetry.
MFCA2MinimumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA2MaximumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA3StepASlm	4 Float	-1*full scale to full scale 0 = do not step A this MFC	Fuel MFC step A size. Full scale depends on the MFC and is in telemetry.
MFCA3StepBSlm	4 Float	-1*full scale to full scale 0 = do not step B this MFC	Fuel MFC step B size. Full scale depends on the MFC and is in telemetry.

Parameter	# Bytes Type	Values	Description
MFCA3MinimumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA3MaximumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.

3.1.6.14 FlowRampedWave

The *FlowRampedWave* command ramps the flow of each mass flow controller up or down, or not at all, at a specified ramp rate A for a specified duration A, waits a specified hold duration A, then ramps up or down or not at all at a specifried ramp rate B for a specified duration B, waits for a hold duration B, and repeats the sequence until specified maximum or minimum flow is reached or until stopped by command.

Ramped waves can be stopped with the *FlowStopWaveform* command, which leaves the flow setpoints where they were last, with the *FlowCloseAll* command, which sets all flows to zero and closes all valves, or with the *FlowSetMfcRate* or *FlowSetAllMfcRates* commands which leave the flow at the setpoint specified by the command. Ramped waves are stopped for a MFC by any other flow waveform command if the parameters say to control that MFC.

Command ID: 0xC40E
Mnemonic: ACME_FLOW_RAMPED_WAVE

Parameter	# Bytes Type	Values	Description
rampADurationSec	4 Float	0.1 – 999.9	
holdADurationSec	4 Float	0.1 – 999.9	
rampBDurationSec	4 Float	0.1 – 999.9	
holdBDurationSec	4 Float	0.1 – 999.9	
MFCA1RampARateSlmPerSec	4 Float	0 = do not ramp A this MFC	
MFCA1RampBRateSlmPerSec	4 Float	0 = do not ramp B this MFC	
MFCA1MinimumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA1MaximumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA2RampARateSlmPerSec	4 Float	0 = do not ramp A this MFC	
MFCA2RampBRateSlmPerSec	4 Float	0 = do not ramp B this MFC	
MFCA2MinimumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.

Parameter	# Bytes Type	Values	Description
MFCA2MaximumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA3RampARateSlmPerSec	4 Float	0 = do not ramp A this MFC	
MFCA3RampBRateSlmPerSec	4 Float	0 = do not ramp B this MFC	
MFCA3MinimumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.
MFCA3MaximumFlowSlm	4 Float	0 to full scale flow	Full scale depends on the MFC and is in telemetry.

3.1.6.15 FlowStopWaveform

The *FlowStopWaveform* command.

Command ID: 0xC40F
Mnemonic: ACME_FLOW_STOP_WAVEFORM

Parameter	# Bytes Type	Values	Description
stopMFCA1Waveform	1 Uint8	0 = Do not stop 1 = Stop waveform	Whether or not to stop the waveform for this MFC
stopMFCA2Waveform	1 Uint8	0 = Do not stop 1 = Stop waveform	Whether or not to stop the waveform for this MFC
stopMFCA2Waveform	1 Uint8	0 = Do not stop 1 = Stop waveform	Whether or not to stop the waveform for this MFC

3.1.7 ACME Diagnostic Group Command Details

3.1.7.1 DataRecordingStart

The *DataRecordingStart* command starts recording analog input and output data to a file at a rate defined by the Diagnostics/SampleTimeMsec registry entry, see **Table *32***, which defaults to 8 msec for 125 Hz sampling. Prior to this command the *SystemOpenTestPoint* command must be issued to define the file name, otherwise the command fails. The testpointName parameter in the *SystemOpenTestPoint* command results in the data being save to a path and file named /payload/sci/testpointName/testpointName_data.csv. See section 8.1 for the file contents.

Command ID: 0xC501
Mnemonic: ACME_DATA_RECORDING_START

3.1.7.2 DataRecordingStop

The *DataRecordingStop* command stops acquistion of data started with the *DataRecordingStart* command. The resultant data file is left open until the *SystemCloseTestPoint* command is received. Issuing another *DataRecordingStart* command instead will append data to the existing

file. However, there is no indication in the file between the two acquisition periods except for differences in the data, which, for example, could include a tell-tale signature in the E-Field control voltage.

Command ID: 0xC502
Mnemonic: ACME_DATA_RECORDING_STOP

3.1.7.3 FiberMove

The *FiberMove* command translates the thin filament pyrometry (Fiber) fiber array to the specified position at the specified velocity or translation rate. Movement acceleration and deceleration are registry values specified in AcmeConfig.ini, see **Table *32***, items Fiber/AccelerationCountsPerSecSquared and Fiber/DecelerationCountsPerSecSquared. The response time to start movement is on the order of seconds. To start movement closer to an event use the *FiberSetupMove* command followed by the *FiberExecuteSetupMove* command when ready to move.

If the specified position exceeds the limit the command is not performed. The position limit is the Fiber/MaxPosition registry value minus the Fiber/FiberToBurnerOffsets registry value (a comma-separated string of offsets in order of burner type number) for the burner type specified in the *SystemDefineSetup* command. See the descriptions of these registry values in **Table *32*** for more information.

Motion can be stopped prematurely using the *FiberHaltMove* command.

The motor is an Ultramotion HT17 stepper motor linear actuator with 6 inch stroke and encoder.

Command ID: 0xC503
Mnemonic: ACME_FIBER_MOVE

Parameter	# Bytes Type	Values	Description
position	4 Uint32	0 to maximum position.	Counts. Position 0 is retracted away from burner. See paragraphs above for max position calculation.
velocity	4 Uint32		Counts per second.

3.1.7.4 FiberSetupMove

The *FiberSetupMove* command sets up a move of the Fiber fiber array. Movement is not started until a *FiberExecuteSetupMove* command is received. This command pair can be used to place the start of movement closer to an event since setup takes on the order of seconds to complete.

The rest of the details of this command are the same as for the *FiberMove* command except that acceleration and deceleration are command parameters.

Command ID: 0xC504

Mnemonic: ACME_FIBER_SETUP_MOVE

Parameter	# Bytes Type	Values	Description
position	4 Uint32	0 to maximum position.	Counts. Position 0 is retracted away from burner. See paragraphs under *FiberMove* command for max position calculation.
velocity	4 Uint32		Counts per second.
acceleration	4 Uint32		Counts per second squared.
deceleration	4 Uint32		Counts per second squared.

3.1.7.5 FiberExecuteSetupMove

The *FiberExecuteSetupMove* command starts movement after a *FiberSetupMove* command.

Motion can be stopped prematurely using the *FiberHaltMove* command.

Command ID: 0xC505
Mnemonic: ACME_FIBER_EXECUTE_SETUP_MOVE

3.1.7.6 FiberRampedMove

The *FiberRampedMove* command moves the Fiber fiber array from its current position to a position the specified segment distance away, which can be positive or negative, waits the specified hold duration, and repeats this sequence the specified number of cycles.

The command fails and is not performed at all if the calculated end position after all cycles exceeds the limit. See the *FiberMove* command description for limit calculation. The starting position is retrieved from the encoder.

Motion can be stopped prematurely using the *FiberHaltMove* command.

Command ID: 0xC506
Mnemonic: ACME_FIBER_RAMPED_MOVE

Parameter	# Bytes Type	Values	Description
segmentDistance	4 Int32	-1 * max position to max position, except zero.	Counts. Negative moves away from burner. Positive moves toward burner.
velocity	4 Uint32		
acceleration	4 Uint32		

Parameter	# Bytes Type	Values	Description
deceleration	4 Uint32		
holdDuration	4 Float	0.00 – 999.99	Seconds.
cycles	4 Uint32	1 - 999	1 cycle is 1 translation and 1 hold.

3.1.7.7 FiberRampedWaveMove

The *FiberRampedWaveMove* command moves the Fiber fiber array from its current position to a position the specified segment distance A away, which can be positive or negative, waits the specified hold duration A, moves a relative segment distance B, waits the specified hold duration B, and repeats this sequence the specified number of cycles.

A triangle wave, essentially moving back and forth, is achieved by making the A and B segment distances equal but opposite in sign, and both hold durations zero.

The command fails and is not performed at all if the calculated end position after all cycles exceeds the limit. See the *FiberMove* command description for limit calculation. The starting position is retrieved from the encoder.

Motion can be stopped prematurely using the *FiberHaltMove* command.

Command ID: 0xC507
Mnemonic: ACME_FIBER_RAMPED_WAVE_MOVE

Parameter	# Bytes Type	Values	Description
segmentDistanceA	4 Uint32	-1 * max position to max position, except zero.	Counts. Negative moves away from burner. Positive moves toward burner.
velocityA	4 Uint32		
accelerationA	4 Uint32		
decelerationA	4 Uint32		
holdDurationA	4 Float	0.00 – 999.99	
segmentDistanceB	4 Uint32	-1 * max position to max position, except zero.	Counts. Negative moves away from burner. Positive moves toward burner.
velocityB	4 Uint32		

Parameter	# Bytes Type	Values	Description
accelerationB	4 Uint32		
decelerationB	4 Uint32		
holdDurationB	4 Float	0.00 – 999.99	
cycles	4 Uint32	1 - 999	

3.1.7.8 FiberHaltMove

The *FiberHaltMove* command stops any Fiber fiber array motion and aborts any ramps or ramped waves in progress.

Command ID: 0xC508
Mnemonic: ACME_FIBER_HALT_MOVE

3.1.7.9 FiberMotorSetState

The *FiberMotorSetState* command disables or enables the Fiber motor by setting the RST line on the stepper driver to 0 V or 5 V respectively. In normal operations this command should not be necessary because the motor is automatically enabled as soon as the cube is active (the signal originates in the cube), and is disabled when the software shuts down. For reference, the stepper driver module is an Intelligent Motion System (IMS) XIM481H installed on an IMS INT-481 board with MS0, MS2, and MS3 in the on position.

To reset the motion controller board instead use the *SystemResetMotionController* command.

Command ID: 0xC509
Mnemonic: ACME_FIBER_MOTOR_SET_STATE

Parameter	# Bytes Type	Values	Description
state	1 Uint8	0 = Reset 1 = On	Sets Fiber_MOTOR_RST (**Table *18***) to 0 V (Reset) or 5 V (On).

3.1.7.10 PmtPower

The *PmtPower* command turns all three photo multiplier tubes (PMTs) on or off.

The PMTs are Hamamatsu part number H10722-10-MOD2. PMT1 has a 310nm bandpass interference filter. PMT2 has a UV fused silica broadband filter. PMT3 has a 430nm bandpass interference filter.

Command ID: 0xC50A
Mnemonic: ACME_PMT_POWER

Parameter	# Bytes Type	Values	Description
state	1 Uint8	0 = Off 1 = On	

3.1.7.11 PmtSetGain

The *PmtSetGain* command sets the gain of the specified photo multipler tube.

The gain parameter in this command specifies the volts out of the cube DNx-AO-333 board channels AO-25, AO-26, and AO-27, see **Table *18***. These voltages are fed through a voltage divider of 10 Kohms and 1.21 Kohms before entering the control input on the PMT. According to the PMT data sheet the recommended control voltage is 0.5 V to 1.1 V, corresponding to 4.63 V to 10.19 V for the gain command parameter, which is limited to a maximum of 10 V.

The equation to convert PMT control input, x, to PMT gain is:

PMTGain = 2,000,000 * x ^ 8.643856

,which gives a range of PMTGain from 5000 to 3,500,000 for 0.5 V to 1.1V control input, or 4.63 V to 10.19 V command gain parameter.

The equation to convert command gain parameter, gain, to PMT gain is:

PMTGain = 2,000,000 * (gain/9.2645) ^ 8.643856

The equation to convert PMT gain to command gain parameter, gain, is:

gain = 10 ^ (0.115689 * log(PMTGain) + 0.23786213)

Command ID: 0xC50B
Mnemonic: ACME_PMT_SET_GAIN

Parameter	# Bytes Type	Values	Description
pmtNumber	1 Uint8	1 = PMT1 2 = PMT2 3 = PMT3	
gain	4 Float	0 - 10	Volts

3.1.7.12 PmtSetAllGains

The *PmtSetAllGains* command sets the gain of each photo multipler tube. See the *PmtSetGain* command for determination of gain parameters.

Command ID: 0xC50C
Mnemonic: ACME_PMT_SET_ALL_GAINS

Parameter	# Bytes Type	Values	Description
PMT1Gain	4 Float	0 - 10	Volts
PMT2Gain	4 Float	0 - 10	Volts
PMT3Gain	4 Float	0 - 10	Volts

3.1.7.13 ExtinctionDetectionConfigure

The *ExtinctionDetectionConfigure* command defines the limits, averaging duration, and method of extinction detection. Detection of extinction using this configuration does not begin until the *ExtinctionDetectionSetState* command enables detection. It ends when extinction is detected or when the *ExtinctionDetectionSetState* command is used to force detection or disable detection.

During detection the instruments that have limits in this command have their values continuously averaged over the specified duration. These averages are compared to the specified limts. The numberOfInstruments parameter is used to indicate how many instrument averages must go below their limits before extinction is detected. Instruments can be assigned no influence on the detection state by specifying negative one limit.

Command ID: 0xC50D
Mnemonic: ACME_EXTINCTION_CONFIGURE

Parameter	# Bytes Type	Values	Description
averagingDurationSecs	4 Float	0.0 – 99.99	Seconds
numberOfInstruments	1 Uint8	1-8	1 = any 8 = all
plus60RadiometerLimit	4 Float	-1 = Don’t use. 0 – 5V	
plus30RadiometerLimit	4 Float	-1 = Don’t use. 0 – 5V	
centerRadiometerLimit	4 Float	-1 = Don’t use. 0 – 5V	
minus30RadiometerLimit	4 Float	-1 = Don’t use. 0 – 5V	
minus60RadiometerLimit	4 Float	-1 = Don’t use. 0 – 5V	
PMT1Limit	4 Float	-1 = Don’t use. 0 – 4V	
PMT2Limit	4 Float	-1 = Don’t use. 0 – 4V	
PMT3Limit	4 Float	-1 = Don’t use. 0 – 4V	

3.1.7.14 ExtinctionDetectionSetState

The *ExtinctionDetectionSetState* command enables or disables extinction detection or forces the state to Detected. The Detected state can be used by the operator to inform the system when extinction has occurred or to test the resultant actions which are set up by the *ExtinctionDetectionSetAction…* commands.

Command ID: 0xC50E
Mnemonic: ACME_EXTINCTION_DETECTION_SET_STATE

Parameter	# Bytes Type	Values	Description
state	1 Uint8	0 = Disabled 1 = Enabled 2 = Detected	

3.1.7.15 ExtinctionDetectionSetActionZero

The *ExtinctionDetectionSetActionZero* command sets the action when extinction is detected to zero all mass flow controller flow setpoints after the specified wait duration.

Command ID: 0xC50F
Mnemonic: ACME_EXTINCTION_DETECTION_SET_ACTION_ZERO

Parameter	# Bytes Type	Values	Description
waitDuration	4 Float	0.0 – 99.99	Seconds

3.1.7.16 ExtinctionDetectionSetActionSingleStep

The *ExtinctionDetectionSetActionSingleStep* command sets the action when extinction is detected to step each mass flow controller flow setpoint up or down or not at all by the specified amount.

Command ID: 0xC510
Mnemonic: ACME_EXTINCTION_DETECTION_SET_ACTION_SINGLE_STEP

Parameter	# Bytes Type	Values	Description
MFCA1StepSlm	4 Float	-MFCA1FullScale to +MFCA1FullScale	SLPM
MFCA2StepSlm	4 Float	-MFCA2FullScale to +MFCA2FullScale	SLPM
MFCA3StepSlm	4 Float	-MFCA3FullScale to +MFCA3FullScale	SLPM

3.1.7.17 ExtinctionDetectionSetActionSingleRamp

The *ExtinctionDetectionSetActionSingleRamp* command sets the action when extinction is detected to ramp each mass flow controller flow setpoint up or down or not at all by the specified step size within the specified duration.

Command ID: 0xC511
Mnemonic: ACME_EXTINCTION_DETECTION_SET_ACTION_SINGLE_RAMP

Parameter	# Bytes Type	Values	Description
rampDuration	4 Float	0.0 – 99.99	Seconds
MFCA1RampRateSlmPerSec	4 Float		SLPM per second
MFCA2RampRateSlmPerSec	4 Float		SLPM per second
MFCA3RampRateSlmPerSec	4 Float		SLPM per second

3.1.7.18 ExtinctionDetectionSetActionStepping

The *ExtinctionDetectionSetActionStepping* command sets the action when extinction is detected to stepping each mass flow controller flow setpoint up or down or not all by the specified amount, holding those setpoints for the specified duration, then setting all setpoints to zero.

Command ID: 0xC512
Mnemonic: ACME_EXTINCTION_DETECTION_SET_ACTION_STEPPING

Parameter	# Bytes Type	Values	Description
stepDuration	4 Float	0.0 – 99.99	Seconds
MFCA1StepSlm	4 Float		SLPM
MFCA1MaxSlm	4 Float		SLPM
MFCA2StepSlm	4 Float		SLPM
MFCA2MaxSlm	4 Float		SLPM
MFCA3StepSlm	4 Float		SLPM
MFCA3MaxSlm	4 Float		SLPM
zeroWhenDone	1 Quint8	0 = Don’t zero when done 1 = Zero when done	

3.1.7.19 ExtinctionDetectionSetActionRamping

The *ExtinctionDetectionSetActionRamping* command sets the action when extinction is detected to ramping each mass flow controller flow setpoint up or down or not at all by the specified step within the specified duration, holding the final setpoint for the specified duration, then zeroing all setpoints.

Command ID: 0xC513
Mnemonic: ACME_EXTINCTION_DETECTION_SET_ACTION_RAMPING

Parameter	# Bytes Type	Values	Description
rampDuration	4 Float	0.0 – 99.99	Seconds
holdDuration	4 Float	0.0 – 99.99	Seconds
MFCA1RampRateSlmPerSec	4 Float		SLPM per second
MFCA1MaxSlm	4 Float		
MFCA2RampRateSlmPerSec	4 Float		SLPM per second
MFCA2MaxSlm	4 Float		
MFCA3RampRateSlmPerSec	4 Float		SLPM per second
MFCA3MaxSlm	4 Float		
zeroWhenDone	1 Quint8	0 = Don't zero when done 1 = Zero when done	

3.1.7.20 DataRecordingSetSampleRate

The *DataRecordingSetSampleRate* command sets the period/frequency of data acquisition when the DataRecordingStart command is issued. On startup the sample time is the SampleTimeMsec value in the Diagnostics group in AcmeConfig.ini. The current value is in telemetry.

Command ID: 0xC514
Mnemonic: ACME_DATA_RECORDING_SAMPLE_RATE

Parameter	# Bytes Type	Values	Description
sampleTimeMs	2 Uint16	0-65535	Duration in milliseconds between data samples.

3.1.8 ACME E-Field Group Command Details

3.1.8.1 EFieldEnable

The *EFieldEnable* command enables or disables the high voltage supply that powers the electric field mesh. This command also effectively performs a *EFieldSetVoltage* command with voltage parameter equal to zero prior to an enable and after a disable. Any waveforms are stopped by a disable.

This command controls the cube DNx-AO-333 board AO-6 (HV2_EN) signal. See **Table *18***.

The high voltage supply is an Ultravolt part number 20A12-N4-Y6-C.

Command ID: 0xC601
Mnemonic: ACME_EFIELD_ENABLE

Parameter	# Bytes Type	Values	Description
state	1 Uint8	0 = Disable 1 = Enable	Disable sets HV_EN to 0V. Enable sets HV_EN to 5V.

3.1.8.2 EFieldSetVoltage

The *EFieldSetVoltage* command controls the voltage present on the electric field mesh. The voltage parameter, which is desired voltage on the mesh, is converted to control voltage output to the cube DNx-AO-333 board AO-4 (HV_CTRL) signal. See **Table *18***. The conversion is according to the equation:

$$Control\ Voltage = \frac{-B + \sqrt{B^2 - 4C(A - voltage)}}{2C}$$

,where A, B, and C are items in AcmeConfig.ini, See **Table *32***, and depend on whether the negative or positive supply is being used. For the negative supply the names are:

EField/NegativeOutputFactorA
EField/NegativeOutputFactorB
EField/NegativeOutputFactorC

For the positive supply the names are:

EField/PositiveOutputFactorA
EField/PositiveOutputFactorB
EField/PositiveOutputFactorC

The *SystemDefineSetup* command tells the software which polarity of E-Field supply is installed.

The reverse equation is:

$$voltage = C * cv^2 + B * cv + A$$

, where cv is control voltage and A,B,andC are as above.

The EfieldEnable command must be issued first to enable the supply.

Command ID: 0x602
Mnemonic: ACME_EFIELD_SET_VOLTAGE

Parameter	# Bytes Type	Values	Description
voltage	4 Float	-10000 to 0 if negative polarity supply is installed. 0 to +10000 for positive polarity supply.	

3.1.8.3 EFieldSteppedSignal

The *EFieldSteppedSignal* command steps the electric field voltage between a specfied minimum and maximum voltage with steps of a specified size and duration and back down (or up) for a specified number of cycles. One cycle is a series of steps in one direction, therefore set cycles to a factor of 2 to perform complete up and down waveforms.

The maximum and minimum voltages are not violated even if the step voltage does not divide evenly between them.

The EfieldEnable command must be issued first to enable the supply.

Special cases:

- stepVoltage < 0 starts at maxVoltage and steps to minVoltage (and back up if cycles > 1).
- Cycles = 1 produces a single staircase.
- stepVoltage = (maxVoltage – minVoltage) produces a square waveform.

Command ID: 0xC603
Mnemonic: ACME_EFIELD_STEPPED_SIGNAL

Parameter	# Bytes Type	Values	Description
minVoltage	4 Float	-10000 to 0 if negative polarity supply is installed. 0 to +10000 for positive polarity supply.	
maxVoltage	4 Float	-10000 to 0 if negative polarity supply is installed. 0 to +10000 for positive polarity supply.	
stepVoltage	4 Float	-10000 to 10000	
stepDurationSecs	4 Float	0.005 – 999.999	
cycles	4 Uint32	1-9999	

3.1.8.4 EFieldRampedSignal

The *EFieldRampedSignal* command is essentially an *EFieldSteppedSignal* command with steps replaced by ramps of specified rate and duration.

The EfieldEnable command must be issued first to enable the supply.

Special cases:
- rampRatePerSec < 0 starts a maxVoltage and ramp/holds to minVoltage.
- Cycles = 1 produces a single ramp/hold sequence betweem min and max.
- holdDurationSec = 0 produces a triangle waveform.

Command ID: 0xC604
Mnemonic: ACME_EFIELD_RAMPED_SIGNAL

Parameter	# Bytes Type	Values	Description
minVoltage	4 Float	-10000 to 0 if negative polarity supply is installed. 0 to +10000 for positive polarity supply.	
maxVoltage	4 Float	-10000 to 0 if negative polarity supply is installed. 0 to +10000 for positive polarity supply.	
rampRatePerSec	4 Float		
rampDurationSecs	4 Float	0.005 – 999.999	
holdDurationSecs	4 Float	0.005 – 999.999	
cycles	4 Uint32	1 - 9999	

3.1.8.5 EFieldRectangularWave

The *EFieldRectangularWave* command is an *EFieldSteppedSignal* command but there is a second step with its own size and duration. The two step sizes can be the same or opposite polarity.

The EfieldEnable command must be issued first to enable the supply.

Special cases:
- If (stepAVoltage + stepBVoltage) < 0 the sequence starts at maxVoltage.
- Cycles = 1 produces a single step sequence between min and max voltage.
- stepAVoltage = -1 * stepBVoltage produces a square wave between minVoltage and minVoltage + stepAVoltage.

Command ID: 0xC605
Mnemonic: ACME_EFIELD_RECTANGULAR_WAVE

Parameter	# Bytes Type	Values	Description
minVoltage	4 Float	-10000 to 0 if negative polarity supply is installed. 0 to +10000 for positive polarity supply.	
maxVoltage	4 Float	-10000 to 0 if negative polarity supply is installed. 0 to +10000 for positive polarity supply.	
stepAVoltage	4 Float	-10000 to 10000	
stepADurationSecs	4 Float	0.005 – 999.999	
stepBVoltage	4 Float	-10000 to 10000	
stepBDurationSecs	4 Float	0.005 – 999.999	
Cycles	4 Uint32	1 - 9999	

3.1.8.6 EFieldRampedWave

The *EFieldRampedWave* command is essentially an *EFieldRectangularWave* command with ramps replacing the steps.

The EfieldEnable command must be issued first to enable the supply.

Special cases:

- rampARatePerSec OR rampBRatePerSec = 0 produces a sawtooth or reverse-sawtooth waveform.

Command ID: 0xC606
Mnemonic: ACME_EFIELD_RAMPED_WAVE

Parameter	# Bytes Type	Values	Description
minVoltage	4 Float	-10000 to 0 if negative polarity supply is installed. 0 to +10000 for positive polarity supply.	
maxVoltage	4 Float	-10000 to 0 if negative polarity supply is installed. 0 to +10000 for positive polarity supply.	
rampARatePerSec	4 Float		
rampADurationSecs	4 Float	0.0 – 999.999	
holdADurationSecs	4 Float	0.0 – 999.999	
rampBRatePerSec	4 Float		

Parameter	# Bytes Type	Values	Description
rampBDurationSecs	4 Float	0.0 – 999.999	
holdBDurationSecs	4 Float	0.0 – 999.999	
cycles	4 Uint32	1 - 9999	

3.1.9 ACME Telemetry

The IOP requests telemetry at 1 Hz from all packages in the system, including the PI Location (PIL). See Section A.2, FCF Telemetry Protocol in FCF-ICD-0076A for a description of the telemetry protocol, including all required header fields. The user data of the ACME 1 Hz telemetry message that follows has a maximum size of 1460 - 16 (FCF header size) = 1444. In the FCF header, the BCD time is filled in by the FCFCommunications applications (the base classes). In the FCF header, the state field (byte 4) is filled in by ACME according to ***Table 21***. In the tables below in the units column, the word 'raw' is used to indicate that the value is the same as the internal interface source and not converted prior to placement in the telemetry stream.

Table 3 ACME Telemetry

ACME Telemetry				
Description	Reference	Offset (bytes)	Size (bytes)	Comments
System Telemetry	*Table 4*	0	54	System-level data.
System Setup Telemetry	**Table *14***	54	18	Hardware configuration data
Igniter Telemetry	*Table 5*	72	20	Igniter data.
IPSU Telemetry	*Table **8***	92	15	Status of communication with IPSUs.
Flow Telemetry	*Table **9***	107	51	Gas flow data.
E-field Telemetry	*Table 10*	158	14	E-field high voltage data.
Diagnostic Telemetry	*Table 11*	172	139	Sensor data.
Fiber Telmetry	*Table 12*	311	13	TFP motion data
Extinction Detection Telemetry	*Table 13*	324	39	Extinction detection state and parameters
Imaging Telemetry	*Table 6*	363	22	Commanded digital camera motion data. Chamber camera data.
Imaging Motion Telemetry	*Table 7*	385	25	Digital camera motion data from GetMotionTelemetry return value command.
Log Messages		410	1034	A buffer of text for log messages.

Table 4 ACME System Telemetry

System Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Description, Units, Range	Mnemonic
Heartbeat	0	1	Uint8	0-255 then repeat	HEARTBEAT
Cube Connected	1	1	Uint8	0 = Not connected, 1 = Connected	CUBE_CONNECT
CPU Temperature	2	4	Float	°C, CPU on Lippert Cool XpressRunner-GS45, using LEMT.	CPU_TEMP
CPU Board Temperature	6	4	Float	°C, Lippert Cool XpressRunner-GS45 board, using LEMT.	CPU_BOARD_TEMP
Cube Temperature 1	10	4	Float	°C, TEMP1 on UEI Cube Power Layer	CUBE_TEMP_1
Cube Temperature 2	14	4	Float	°C, TEMP2 on UEI Cube Power Layer	CUBE_TEMP_2
Hard Drive Temperature	18	1	Uint8	°C, /dev/sda1 Airflow Temperature from smartctl output.	HARD_DRIVE_TEMP
Test Point Name	19	30	String	Current test point name	
Script State	49	1	Uint8		
Script Line Number	50	2	Uint16		
Script Duration Remaining	52	2	Uint16		

Table 5 ACME Igniter Telemetry

Igniter Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Units or Enumeration	Source Mnemonic
Igniter Arm Position	0	4	Int32	counts	IGN_MOTOR_GET_POS
Igniter Arm Commanded Position	4	4	Int32	counts	
Igniter Arm Max Position	8	4	Int32	counts	
Last Igniter Pulse Power	12	4	Float	Volts	
Last Igniter Pulse Duration	16	4	Uint32	Milliseconds	

Table 6 ACME Imaging Telemetry

Imaging Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Description, Units, Range	Source Mnemonic
End LED Bank State	0	1	Uint8	State of the end (horizontal) LED bank on the ops camera. 0 = Off. 1 = On.	
Side LED Bank State	1	1	Uint8	State of the side (vertical) LED bank on the ops camera. 0 = Off. 1 = On.	
Illumination Cycing State	2	1	Uint8	State of LED bank cycling. 0 = Not cycling. 1 = Cycling.	
Analog Camera AGC	3	1	Uint8	Automatic gain control setting of the analog camera. 0=Off. 1=6 dB, 2 = 12 dB, 3 = 21 dB, 4 = 31 dB.	
Analog Camera Shutter Speed	4	1	Uint8	Shutter speed setting of the analog camera. 0=1/60s, 1=1/100s, 2=1/250s, 3=1/500s, 4=1/1000s, 5=1/2000s, 6=1/4000s, 7=1/10000s, 8= 1/20000s, 9=1/30000s	
Commanded Zoom Position	5	4	Uint32	Last commanded position for zoom. Actual is in Imaging Motion Telemetry and in IPSU-G telemetry if ACME is not controlling the camera.	
Commanded Iris Position	9	4	Uint32	Last commanded position for iris. Actual is in Imaging Motion Telemetry and in IPSU-G telemetry if ACME is not controlling the camera.	
Commanded Focus Position	13	4	Uint32	Last commanded position for focus. Actual is in Imaging Motion Telemetry and in IPSU-G telemetry if ACME is not controlling the camera.	
Commanded Filter Position	17	4	Uint32	Last commanded position for filter. Actual is in Imaging Motion Telemetry and in IPSU-G telemetry if ACME is not controlling the camera.	
Filter Number	21	1	Uint8	0 = In between filters, 1-4	

Table 7 ACME Imaging Motion Telemetry

Imaging Motion Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Description, Units, Range	Source Mnemonic
Unused	0	1	Uint8	There are unused fields in this telemetry group because when ACME is controlling the camera motor directly it uses the same software and data structure as IPSU-G does.	

Imaging Motion Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Description, Units, Range	Source Mnemonic
Zoom Position	1	4	UInt32	Axis 1 of the PMD motion controller retrieved from the IPSU-G. Zoom lens position 0-85000 counts. 0 is zoomed out. 85000 is zoomed in.	ZOOM_ MOTOR _GET_POS
Iris Position	5	4	UInt32	Axis 2 of the PMD motion controller retrieved from the IPSU-G. Iris position 0-76000 counts. 0 is open. 76000 is closed.	IRIS_ MOTOR _GET_POS
Focus Position	9	4	UInt32	Axis 3 of the PMD motion controller retrieved from the IPSU-G. Focus position 0-101000 counts. 0 is near focus. 101000 is far focus.	FOCUS_ MOTOR _GET_POS
Filter Position	13	4	UInt32	Axis 4 of the PMD motion controller retrieved from the IPSU-G. Filter position 0-451000 counts. Filters at 0, 120000, 240000, and 360000 counts.	FILTER_ MOTOR _GET_POS
Unused	17	2	Uint16		
Unused	19	2	Uint16		
Unused	21	2	Uint16		
Unused	23	2	Uint16		

Table 8 ACME IPSU Telemetry

IPSU Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Units or Enumeration	Mnemonic
IPSU Phase	0	1	Uint8	***Table 22***	IPSU_PHASE
IPSU 1 Status	1	1	Uint8	***Table 23***	IPSU1_STATUS
IPSU 2 Status	2	1	Uint8	***Table 23***	IPSU2_STATUS
IPSU 3 Status	3	1	Uint8	***Table 23***	IPSU3_STATUS
IPSU 4 Status	4	1	Uint8	***Table 23***	IPSU4_STATUS
IPSU 5 Status	5	1	Uint8	***Table 23***	IPSU5_STATUS
IPSU 6 Status	6	1	Uint8	***Table 23***	IPSU6_STATUS

Table 9 ACME Flow Telemetry

Flow Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Units or Enumeration	Source Mnemonic
MFCA1 Static Mixer Setpoint	0	4	Float	SLPM	MFCA1_CTRL
MFCA1 Static Mixer MFC Flow	4	4	Float	SLPM	MFCA1_FLOW
MFCA1 Static Mixer Full Scale	8	4	Float	SLPM	MFCA1_FULL_SCALE
MFCA1 Static Mixer State	12	1	Uint8	***Table 24***	MFCA1_STATE
MFCA1 Static Mixer Waveform	13	1	Uint8	***Table 25***	
MFCA2 Bypass Setpoint	14	4	Float	SLPM	MFCA2_CTRL
MFCA2 Bypass Flow	18	4	Float	SLPM	MFCA2_FLOW
MFCA2 Bypass Full Scale	22	4	Float	SLPM	MFCA2_FULL_SCALE
MFCA2 Bypass State	26	1	Uint8	*Table 24*	MFCA2_STATE
MFCA2 Bypass Waveform	27	1	Uint8	***Table 25***	
MFCA3 Fuel Setpoint	28	4	Float	SLPM	MFCA3_CTRL
MFCA3 Fuel Flow	32	4	Float	SLPM	MFCA3_FLOW
MFCA3 Fuel Full Scale	36	4	Float	SLPM	MFCA3_FULL_SCALE
MFCA3 Fuel State	40	1	Uint8	*Table 24*	MFCA3_STATE
MFCA3 Fuel Waveform	41	1	Uint8	***Table 25***	
Solenoid Valve SVA1	42	1	Uint8	**Table *30***	SVA1
Solenoid Valve SVA2	43	1	Uint8	**Table *30***	SVA2
Solenoid Valve SVA3	44	1	Uint8	**Table *30***	SVA3

Flow Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Units or Enumeration	Source Mnemonic
Solenoid Valve SVA4	45	1	Uint8	**Table *30***	SVA4
Solenoid Valve SVA5	46	1	Uint8	**Table *30***	SVA5
Solenoid Valve SVA6	47	1	Uint8	**Table *30***	SVA6
Solenoid Valve SVA7	48	1	Uint8	**Table *30***	SVA7
Solenoid Valve SVA8	49	1	Uint8	**Table *30***	SVA8
Solenoid Valve SVA9	50	1	Uint8	**Table *30***	SVA9

Table 10 ACME E-Field Telemetry

E-Field Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Units or Enumeration	Source Mnemonic
High Voltage State	0	1	Uint8	***Table 26***	HV_EN
E-Field Setpoint	1	4	Float	Volts	HV_CTRL
E-Field Voltage	5	4	Float	Volts	HV_OUT
Ion Current Voltage	9	4	Float	Volts	ION_CURRENT
Waveform Type	13	1	Uint8	***Table 25***	

Table 11 ACME Diagnostic Telemetry

Diagnostic Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Units	Source Mnemonic
Data Recording Flag	0	1	Uint8	0=not recording 1=recording	
Data Recording Elapsed Time	1	4	Uint32	seconds	
Sample Time	5	2	Uint16	milliseconds	
Thermocouple A BRE Center 180[BRE]	7	4	Float	Volts	TC_A BRE_CTR_180 [BRE]
Thermocouple B BRE Offset 180 [BRE]	11	4	Float	Volts	TC_B BRE_OFF_180 [BRE]
Thermocouple C BRE Center [BRE]	15	4	Float	Volts	TC_C BRE_CTR [BRE]
Thermocouple D BRE Offset [BRE]	19	4	Float	Volts	TC_D BRE_OFF [BRE]
Thermocouple Spare	23	4	Float	Volts	TC_SPARE
Thermocouple 200 A	27	4	Float	Volts	TC_200A
Thermocouple 200 B	31	4	Float	Volts	TC_200B
Burner Thermocouple	35	4	Float	Volts	TC_BNR
Thermocouple Cold Junction	39	4	Float	Volts	TC_CJ
BRE Heat Flux Sensor - Center [BRE]	43	4	Float	Volts	BRE_FLUX_CTR [BRE]
BRE Heat Flux Sensor – Offset [BRE]	47	4	Float	Volts	BRE_FLUX_OFF [BRE]
Radiometer +60 +60 BRE Offset [BRE]	51	4	Float	Volts	RA+60 BRE_OFF+60 [BRE]
Radiometer +60 Cold Junction +60 BRE Offset Cold Junction [BRE]	55	4	Float	Volts	RA+60_CJ BRE_OFF+60CJ [BRE]
Radiometer +30 +30 BRE Offset [BRE]	59	4	Float	Volts	RA+30 BRE_OFF+30 [BRE]
Radiometer +30 Cold Junction +30 BRE Offset Cold Junction [BRE]	63	4	Float	Volts	RA+30_CJ BRE_OFF+30CJ [BRE]

Diagnostic Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Units	Source Mnemonic
Radiometer Center -10 BRE Offset [BRE]	67	4	Float	Volts	RA_CTR BRE_OFF-10 [BRE]
Radiometer Center Cold Junction -10 BRE Offset Cold Junction [BRE]	71	4	Float	Volts	RA_CTR_CJ BRE_OFF-10CJ [BRE]
Radiometer -30	75	4	Float	Volts	RA-30
Radiometer -30 Cold Junction	79	4	Float	Volts	RA-30_CJ
Radiometer -60	83	4	Float	Volts	RA-60
Radiometer -60 Cold Junction	87	4	Float	Volts	RA-60_CJ
Radiometer Offset	91	4	Float	Volts	RA_OFF
Radiometer Offset Cold Junction	95	4	Float	Volts	RA_OFF_CJ
Photomultiplier Tube 1	99	4	Float	microamps	PMT1_OUT
Photomultiplier Tube 2	103	4	Float	microamps	PMT2_OUT
Photomultiplier Tube 3	107	4	Float	microamps	PMT3_OUT
Photomultiplier Tube 1 Gain	111	4	Float	Volts	PMT2_GAIN
Photomultiplier Tube 2 Gain	115	4	Float	Volts	PMT3_GAIN
Photomultiplier Tube 3 Gain	119	4	Float	Volts	PMT4_GAIN
Photomultiplier Tube Power	123	4	Float	Volts	PMT_PWR
Pressure Transducer 1	127	4	Float	psia	PTA1_OUT
Pressure Transducer 2	131	4	Float	psia	PTA2_OUT
Pressure Transducer 3	135	4	Float	psia	PTA3_OUT

[BRE] Valid when the BRE burner is installed.

Table 12 ACME Fiber Telemetry

Fiber Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Units or Enumeration	Source Mnemonic
Fiber Position	0	4	Uint32	counts	TFP_ MOTOR _GET_POS
Fiber Commanded Position	4	4	Uint32	counts	

Fiber Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Units or Enumeration	Source Mnemonic
Fiber Max Position	8	4	Uint32	counts	
Fiber Waveform State	12	1	Uint8	***Table 27***	

Table 13 ACME Extinction Detection Telemetry

Extinction Detecton Telemetry					
Name	Offset (bytes)	Size (bytes)	Data Type	Units or Enumeration	Source Mnemonic
Extinction Detection State	0	1	Uint8	***Table 28***	
Extinction Detection Action	1	1	Uint8	***Table 29***	
Extinction Detection Averaging Duration	2	4	Float	seconds	
Extinction Detection Number of Instruments	6	1	Uint8	0-8	
Extinction Detection Plus 60 Rad Limit	7	4	Float	Volts	
Extinction Detection Plus 30 Rad Limit	11	4	Float	Volts	
Extinction Detection Ctr Rad Limit	15	4	Float	Volts	
Extinction Detection Minus 30 Rad Limit	19	4	Float	Volts	
Extinction Detection Minus 60 Rad Limit	23	4	Float	Volts	
Extinction Detection PMT1 Limit	27	4	Float	Volts	
Extinction Detection PMT2 Limit	31	4	Float	Volts	
Extinction Detection PMT3 Limit	35	4	Float	Volts	

Table 14 ACME System Setup Telemetry

System Setup Telemetry

Name	Offset (bytes)	Size (bytes)	Data Type	Units or Enumeration	Source Mnemonic
Setup Defined	0	1	Uint 8	0=Not defined, 1=Defined	ACME_ SYSTEM_DEFINE_SETUP command
Experiment Type	1	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
MFCA1 Gas Type	2	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
MFCA2 Gas Type	3	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
MFCA3 Gas Type	4	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
E-Field Config	5	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
E-Field Polarity	6	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
Igniter Arm	7	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
Burner Type	8	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
Thermocouple Config	9	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
Radiometer Config	10	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
Use IPSU 1	11	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
Use IPSU 2	12	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
Use IPSU 3	13	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
Use IPSU 4	14	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
Use IPSUG 5	15	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
Use IPSUG 6	16	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command
Fiber Config	17	1	Uint 8	**Table *34***	ACME_ SYSTEM_DEFINE_SETUP command

3.2 External Ethernet Interfaces

See also **Table *20***, Internal Ethernet Interfaces. The ‘Kernels’ string points to a hardware ethernet interface on the processor. It is used in /etc/udev/rules.d/90-net.rules to ensure that the device name is always assigned to the same connector.

Table 15 External Ethernet Interfaces

Source	Device Name	IP Address	Net Mask	Kernels	Description
Lippert SBC	eth0	172.27.253.48	255.255.0.0	0000:00:19.0	Interface to IOP

4.0 INTERNAL INTERFACES

4.1 DNx-AI-207 Board

This analog input layer in the UEI PowerDNA Cube has 16 differential input channels with an input range of +/-10 Volts and 18-bit resolution. Several gains are available but ACME is using a gain of 1.

Table 16 DNx-AI-207 Board Signals

DNx-AI-207 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AIN-0	MFCA1_FLOW	MFCA1 Flow Rate	0 to 5 V	Static mixer/oxygen mass flow controller flow rate. 0 to MFCA1_FULL_SCALE in SLPM. Flow rate (SLPM) = Voltage / 5.0 * MFCA1_FULL_SCALE. Device is Teledyne Hastings HFC-D-302. Full scale depends on installed device.
AIN-1	MFCA2_FLOW	MFCA2 Flow Rate	0 to 5 V	Nitrogen bypass/inert mass flow controller flow rate. 0 to MFCA2_FULL_SCALE in SLPM. Flow rate (SLPM) = Voltage / 5.0 * MFCA2_FULL_SCALE. Device is Teledyne Hastings HFC-D-302. Full scale depends on installed device.
AIN-2	MFCA3_FLOW	MFCA3 Flow Rate	0 to 5 V	Fuel mass flow controller flow rate. 0 to MFCA2_FULL_SCALE in SLPM. Flow rate (SLPM) = Voltage / 5.0 * MFCA3_FULL_SCALE. Device is Teledyne Hastings HFC-D-302. Full scale depends on installed device.
AIN-3	ION_CURRENT	Ion Current	-1.5 to 1.5V	Measurement of ion current between E-Field grid and burner. Ion Current (A) = 1000 * Voltage/162, where R_{sense} is resistance of shunt resistor and R_{DA} is the data acquisition resistance. -25 to 25 µA..
AIN-4	PTA1_OUT	PTA1 Pressure	0.5 to 4.5 V	Outer Coflow pressure. Converted in flight software to 0 to 145 psia using Pressure (psia) = (Voltage – 0.5) * 14.5038 * 10 / 4. Device is Measurement Specialties EPRB-2-VR33-10B-/CM1/ATR.
AIN-5	PTA2_OUT	PTA2 Pressure	0.5 to 4.5 V	Inner Coflow/Burner pressure. Converted in flight software to to 0 to 145 psia using Pressure (psia) = (Voltage – 0.5) * 14.5038 * 10 / 4. Device is Measurement Specialties EPRB-2-VR33-10B-/CM1/ATR.
AIN-6	PTA3_OUT	PTA3 Pressure	0.5 to 4.5 V	Chamber pressure. Converted in flight software to to 0 to 145 psia using Pressure (psia) = (Voltage – 0.5) * 14.5038 * 10 / 4. Device is Measurement Specialties EPRB-2-VR33-10B-/CM1/ATR.

DNx-AI-207 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AIN-7	HV_OUT	E-Field Voltage	-4 to 4 V	Indirect measurement of E-Field high voltage supply. -10000 to +10000 V. E-Field (V) = Voltage * 2486.1795 + 141.9733. (Emperical equation).
AIN-8	PMT1_OUT	Photomultiplier Tube 1 Current	0 to 4 V	0 to 4 µA. Current (µA) = Voltage * 1. Depends on PMT1_GAIN, Table 18. Device is Hamamatsu H10722-10-MOD2. Looks through a 310 nm bandpass interference filter
AIN-9	PMT2_OUT	Photomultiplier Tube 2 Current	0 to 4 V	0 to 4 µA. Current (µA) = Voltage * 1. Depends on PMT2_GAIN, Table 18. Device is Hamamatsu H10722-10-MOD2. Looks through a broadband filter with a fused silica glass window and a neutral density coating.
AIN-10	PMT3_OUT	Photomultiplier Tube 3 Current	0 to 4 V	0 to 4 µA. Current (µA) = Voltage * 1. Depends on PMT3_GAIN, Table 18. Device is Hamamatsu H10722-10-MOD2. Looks through a 430 nm bandpass interference filter.

4.2 DNx-AI-225 Board

This analog input layer in the UEI PowerDNA Cube has 25 differential input channels with an input range of +/-1.25 Volts and 24-bit resolution. °

Table 17 DNx-AI-225 Board Signals

DNx-AI-225 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
	TC_A	Rake Thermocouple A	-6.45 to 55 mV	Near or far array Rake A Type K thermocouple, -200 to 1250 C.
AIN-0	BRE_CTR_180 BRE	BRE Center 180	-6.45 to 55 mV	BRE Center 180 (South) Type K thermocouple, 0 to 500 C.
	TC_B	Rake Thermocouple B	0 to 14 mV	Near or far array Rake B Type B thermocouple, 0 to 1700 C.
AIN-1	BRE_OFF_180 BRE	BRE Offset 180	-6.45 to 55 mV	BRE Offset 180 (Far South) Type K thermocouple, 0 to 500 C.
AIN-2	TC_C	Rake Thermocouple C	0 to 14 mV	Near or far array Rake C Type B thermocouple, 0 to 1700 C.

DNx-AI-225 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
	BRE_CTR [BRE]	BRE Center	-6.45 to 55 mV	BRE Center Type K thermocouple on Medtherm 8-2.688-10SB-0-0-36-21427KSA.
	TC_D	Rake Thermocouple D	0 to 14 mV	Near or far array Rake D Type K thermocouple, -200 to 1250 C.
AIN-3	BRE_OFF [BRE]	BRE Offset	-6.45 to 55 mV	BRE Offset Type K thermocouple on Medtherm 8-2.688-10SB-0-0-36-21427KSA.
AIN-4	TC_200A	200mm Thermocouple A	-6.45 to 55 mV	Igniter side Type K thermocouple, 200 mm from axis. -200 to 1250 C.
AIN-5	TC_200B	200mm Thermocouple B	-6.45 to 55 mV	Relay side Type K thermocouple, 200 mm from axis. -200 to 1250 C.
AIN-6	TC_BNR	Burner Thermocouple	-6.45 to 55 mV	Type K thermocouple on burner. -200 to 1250 C. Not valid when BRE burner is installed.
	RA+60	Radiometer +60	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 1500. Device is Dexter Research Center (DRC) XP-0055.
AIN-7	BRE_OFF+60 [BRE]	+60 BRE Offset	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 1000. Device is DRC XP-0249.
	RA+60_CJ	Radiometer +60 Cold Junction		
AIN-8	BRE_OFF+60CJ [BRE]	+60 BRE Offset Cold Junction	0 to 2.5 V	150 to -75 Celcius. 1.57 V at 25C. Device is TI LM20.
	RA+30	Radiometer +30	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 1500. Device is DRC XP-0055.
AIN-9	BRE_OFF+30 [BRE]	+30 BRE Offset	0 to 2.5 V	Thermopile output (V) = (Voltage-1.25) / 1000. Device is DRC XP-0249.
	RA+30_CJ	Radiometer +30 Cold Junction		
AIN-10	BRE_OFF+30CJ [BRE]	+30 BRE Offset Cold Junction	0 to 2.5 V	150 to -75 Celcius. 1.57 V at 25C. Device is TI LM20.
	RA_CTR	Radiometer Center	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 500. Device is DRC XP-0156.
AIN-11	BRE_OFF-10 [BRE]	-10 BRE Offset	0 to 2.5 V	Thermopile output (V) = (Voltage-1.25) / 1000. Device is DRC XP-0249.

DNx-AI-225 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AIN-12	RA_CTR_CJ BRE_OFF-10CJ [BRE]	Radiometer Center Cold Junction -10 BRE Offset Cold Junction	0 to 2.5 V	150 to -75 Celcius. 1.57 V at 25C. Device is TI LM20.
AIN-13	RA-30	Radiometer -30	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 1500. Device is DRC XP-0055.
AIN-14	RA-30_CJ	Radiometer -30 Cold Junction	0 to 2.5 V	150 to -75 Celcius. 1.57 V at 25C. Device is TI LM20.
AIN-15	RA-60	Radiometer -60	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 1500. Device is DRC XP-0055.
AIN-16	RA-60_CJ	Radiometer -60 Cold Junction	0 to 2.5 V	150 to -75 Celcius. 1.57 V at 25C. Device is TI LM20.
AIN-17	RA_OFF	Radiometer Offset	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 1000. Device is DRC XP-0869. This is located on the downstream relay board.
AIN-18	RA_OFF_CJ	Radiometer Offset Cold Junction	0 to 2.5 V	150 to -75 Celcius. 1.57 V at 25C. Device is Texas Instruments LM20.
AIN-19	TC_SPARE	Spare Thermocouple	N/A	To use, an additional thermocouple would need to be hard wired to the Cold Junction Compensation Assembly. This is located on the downstream relay board.
AIN-20	BRE_FLUX_CTR [BRE]	BRE Heat Flux Sensor - Center	0 to 1.25 V	Heat flux sensor output on Medtherm 8-2.688-10SB-0-0-36-21427KSA. Valid only with BRE burner installed. 1mV = 1 Btu/ft^2
AIN-21	BRE_FLUX_OFF [BRE]	BRE Heat Flux Sensor - Offset	0 to 1.25 V	Heat flux sensor output on Medtherm 8-2.688-10SB-0-0-36-21427KSA. Valid only with BRE burner installed. 1mV = 1 Btu/ft^2
AIN-22	N/A	Not Used	N/A	N/A
AIN-23	N/A	Not Used	N/A	N/A
AIN-24	TC_CJ	Thermocouple Cold Junction	0.803 to 1.10 V	Temperature on the Cold Junction Compensation Assembly. 0 to 100 C. Device is Analog Devices AD590. Temperature (Celcius) = (Voltage/0.00295) – 273.15.

[BRE] Valid when the BRE burner is installed.

4.3 DNx-AO-333 Board

This analog output layer in the UEI PowerDNA Cube has 32 channels with an output range of +/-10 Volts and 16-bit resolution.

Table 18 DNx-AO-333 Board

DNx-AO-333 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AO-0	MFCA1_CTRL	MFCA1 Control	0 to 5 V	Static mixer/oxygen mass flow controller setpoint. 0 to MFCA1_FULL_SCALE in SLPM. Full scale depends on which Teledyne Hastings HFC-D-302 is installed. Output voltage = Setpoint(SLPM) * 5 / MFCA1_FULL_SCALE. Or Setpoint (SLPM) = Output voltage * MFCA1_FULL_SCALE / 5. See Table 19.
AO-1	MFCA2_CTRL	MFCA2 Control	0 to 5 V	Nitrogen bypass/inert mass flow controller setpoint. 0 to MFCA2_FULL_SCALE in SLPM. Full scale depends on which Teledyne Hastings HFC-D-302 is installed. Output voltage = Setpoint(SLPM) * 5 / MFCA2_FULL_SCALE. Or Setpoint (SLPM) = Output voltage * MFCA2_FULL_SCALE / 5. See Table 19.
AO-2	MFCA3_CTRL	MFCA3 Control	0 to 5 V	Fuel mass flow controller setpoint. 0 to MFCA3_FULL_SCALE in SLPM. Full scale depends on which Teledyne Hastings HFC-D-302 is installed. Output voltage = Setpoint(SLPM) * 5 / MFCA3_FULL_SCALE. Or Setpoint (SLPM) = Output voltage * MFCA3_FULL_SCALE / 5. See Table 19.
AO-3	PMT_PWR	PMT Power	0 or 5 V	Power to three photomultipler tubes. 0 V = off. 5 V = on.
AO-4	HV_CTRL	E-Field HV Control	0 to 10 V	Sets E-field HV supply output between about -10000 and 10000 V. EField See EFieldSetVoltage command, section 3.1.8.2, for equation..
AO-5	HV_EN	E-Field HV Enable	0 or 5 V	E-Field high voltage supply enable. 0 V = disabled, 5 V = enabled
AO-6	N/A	Not Used	0 or 5 V	N/A
AO-7	IGN_EN	Igniter Enable	0 or 5 V	0 V = disabled. 5 V enabled. Software sets IGN_CTRL then pulses IGN_EN.
AO-8	IGN_CTRL	Igniter Control	-0.6 to +0.8V	Sets voltage at the igniter between 0.46 to 12 V.
AO-9	MFCA1_PWR	MFCA1 Power	0 or 5 V	Power to static mixer/oxygen MFC. 0 V = unpowered. 5 V = powered.
AO-10	MFCA2_PWR	MFCA2 Power	0 or 5 V	Power to nitrogen bypass/inert MFC. 0 V = unpowered. 5 V = powered.

DNx-AO-333 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AO-11	MFCA3_PWR	MFCA3 Power	0 or 5 V	Power to fuel MFC. 0 V = unpowered. 5 V = powered.
AO-12	RB_UP_EN	Relay Board Upstream Enable	0 or 5 V	0 V = disables signals. 5 V = state of signal output depends on signal value. Applicable signals: SV1, SV2, SV3, SV4, SV5, SV6, SV7, SV8, SV9, PMT_PWR, MFCA1_PWR, MFCA2_PWR, MFCA3_PWR
AO-13	TFP_MOTOR_RST	TFP Motor Reset	0 or 5 V	0 V to rest TFP stepper motor. 5 V to enable motor.
AO-14	IGN_MOTOR_RST	Igniter Motor Reset	0 or 5 V	0 V to reset igniter stepper motor. 5 V to enable motor.
AO-15	SVA4	Solenoid Valve A4	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-16	SVA5	Solenoid Valve A5	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-17	SVA6	Solenoid Valve A6	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-18	SVA7	Solenoid Valve A7	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-19	SVA8	Solenoid Valve A8	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-20	SVA9	Solenoid Valve A9	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-21	SVA1	Solenoid Valve A1	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-22	SVA2	Solenoid Valve A2	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-23	SVA3	Solenoid Valve A3	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-24	N/A	Not Used	0 or 5 V	N/A
AO-25	PMT1_GAIN	PMT1 Gain	4.63 to 10 V	For recommended device control voltage of 0.5 to 1.1 V. Gain range is $5x10^3$ to $3.5x10^6$. Gain=$2x10^6$ * (Output voltage/9.2645)^8.643856. Or Output voltage = 10^(0.115689 * log(Gain) + 0.23786213). Device is Hamamatsu H10722-10-MOD2.

DNx-AO-333 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AO-26	PMT2_GAIN	PMT2 Gain	4.63 to 10 V	For recommended device control voltage of 0.5 to 1.1 V. Gain range is $5x10^3$ to $3.5x10^6$. Gain=$2x10^6$ * (Output voltage/9.2645)^8.643856. Or Output voltage = 10^(0.115689 * log(Gain) + 0.23786213). Device is Hamamatsu H10722-10-MOD2.
AO-27	PMT3_GAIN	PMT3 Gain	4.63 to 10	For recommended device control voltage of 0.5 to 1.1 V. Gain range is $5x10^3$ to $3.5x10^6$. Gain=$2x10^6$ * (Output voltage/9.2645)^8.643856. Or Output voltage = 10^(0.115689 * log(Gain) + 0.23786213). Device is Hamamatsu H10722-10-MOD2.
AO-28	LED_SIDE	LED Side Bank	0 or 5 V	Bank of 3 white LEDs to side of analog camera turning mirror. 0 V = off. 5 V = on.
AO-29	LED_END	LED End Bank	0 or 5 V	Bank of 3 white LEDs past the end of the analog camera turning mirror. 0 V = off. 5 V = on.
AO-30	RB_DOWN_EN	Relay Board Downstream Enable	0 or 5 V	0 V = disables signals. 5 V = state of signal output depends on signal value. Applicable signals: HV1, HV2, LED1, LED2, TFP_RST, IGN_RST, IGN_EN
AO-31	N/A	Not Used	0 or 5 V	N/A

4.4 Serial Communication

Table 19 Serial Communication

Serial Communication				
Source	Device Name	Mnemonic	Signal Range	Description
Lippert SBC	/dev/ttyS0	MFCA1_FULL_SCALE	Depends on what MFC is installed. 0.5, 20 SLPM.	Full scale flow of static mixer/oxygen mass flow controller in SLPM. Pulled from gas list item 18 in the result of the SFL (System Full List) command to device.
Lippert SBC RXD1/TXD1	/dev/ttyS0	MFCA2_FULL_SCALE	Depends on what MFC is installed. 0.01, 0.02, 0.05, 0.1, 0.5, 2, 5 SLPM.	Full scale flow of nitrogen bypass/Inert mass flow controller in SLPM. Pulled from gas list item 18 in the result of the SFL (System Full List) command to device.
Lippert SBC RXD1/TXD1	/dev/ttyS0	MFCA3_FULL_SCALE	Depends on what MFC is installed. 0.05, 0.1, 0.2, 0.5, 1 SLPM.	Full scale flow of fuel mass flow controller in SLPM. Pulled from gas list item 18 in the result of the SFL (System Full List) command to device.
Lippert SBC RXD1/TXD1	/dev/ttyS0	MFCA1_STATE	0-10. See *Table 24.*	State of the oxygen/static mixer mass flow controller. Result of a MS (get state) command.
Lippert SBC RXD1/TXD1	/dev/ttyS0	MFCA2_STATE	0-10. See *Table 24.*	State of the nitrogen bypass mass flow controller. Result of a MS (get state) command.
Lippert SBC RXD1/TXD1	/dev/ttyS0	MFCA3_STATE	0-10. See *Table 24.*	State of the fuel mass flow controller. Result of an MS (get state) command.
ConnectTech serial card P5	/dev/CTI4	CUBE_SERIAL	See DNA-PPCx-1G PowerDNA Gigabit Ethernet I/O Cube User Manual.	UEI cube serial connection. Can be used to retrieve version numbers, set IP addresses, and update firmware.
ConnectTech serial card P6	/dev/CTI5	ANALOG_CAM_AGC_MODE	On or Off.	Turn automatic gain control on or off
ConnectTech serial card P6	/dev/CTI5	ANALOG_CAM_GAIN	6, 12, 21, and 31 dB.	Set the automatic gain control gain.

Serial Communication				
Source	Device Name	Mnemonic	Signal Range	Description
ConnectTech serial card P6	/dev/CTI5	ANALOG_CAM_SHUTTER	1/60, 1/100, 1/250, 1/500, 1/1000, 1/2000, 1/4000, 1/10000, 1/20000, 1/30000 seconds.	Set the shutter speed.
ConnectTech serial card P7	/dev/CTI6 and /dev/ttyS2	IGN_MOTOR_SET_POS	In counts. 0 is the retracted position. Maximum extended position depends on registry values Igniter/IgniterLimits and Igniter/IgniterMaxCount and the burner type last specified by a System Define Setup command. Default maximum is 2500 counts.	Axis 1 control of motion control board in AVP controlling igniter motor in CIA.
ConnectTech serial card P7	/dev/CTI6 and /dev/ttyS2	IGN_MOTOR_GET_POS	See Signal Range for IGN_MOTION_SET_POS	Axis 1 position reporting of motion control board in AVP controlling igniter motor in CIA.
ConnectTech serial card P7	/dev/CTI6 and /dev/ttyS2	TFP_ MOTOR _SET_POS	In counts. 0=retracted. Maximum=closer to burner. Maximum depends on burner type and registry values TfpMaxCount and TfpToBurnerOffsets. See Set TFP Position command for calculation of maximum.	Axis 2 control of motion control board in AVP controlling TFP (thin filament pyrometry array) motor in CIA.
ConnectTech serial card P7	/dev/CTI6 and /dev/ttyS2	TFP_ MOTOR _GET_POS	See Signal Range for TFP_MOTION_SET_POS	Axis 2 position reporting of motion control board in AVP controlling TFP motor in CIA.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	ZOOM_ MOTOR _SET_POS	0-85000 counts. 0 is zoomed out. 85000 is zoomed in.	Not valid when digital camera is connected to the IPSU-G. Axis 1 control of motion control board in digital camera package.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	ZOOM_ MOTOR _GET_POS	0-85000 counts. 0 is zoomed out. 85000 is zoomed in.	Not valid when digital camera is connected to the IPSU-G. Axis 1 position reporting of motion control board in digital camera package.

Serial Communication				
Source	Device Name	Mnemonic	Signal Range	Description
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	IRIS_ MOTOR _SET_POS	0-76000 counts. 0 is open. 76000 is closed.	Not valid when digital camera is connected to the IPSU-G. Axis 2 control of motion control board in digital camera package.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	IRIS_ MOTOR _GET_POS	0-76000 counts. 0 is open. 76000 is closed.	Not valid when digital camera is connected to the IPSU-G. Axis 2 position reporting of motion control board in digital camera package.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	FOCUS_ MOTOR _SET_POS	0-101000 counts. 0 is near focus. 101000 is far focus.	Not valid when digital camera is connected to the IPSU-G. Axis 3 control of motion control board in digital camera package.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	FOCUS_ MOTOR _GET_POS	0-101000 counts. 0 is near focus. 101000 is far focus.	Not valid when digital camera is connected to the IPSU-G. Axis 3 position reporting of motion control board in digital camera package.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	FILTER_ MOTOR _SET_POS	0-451000 counts. Filters at 0, 120000, 240000, and 360000 counts.	Not valid when digital camera is connected to the IPSU-G. Axis 4 control of motion control board in digital camera package.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	FILTER_ MOTOR _GET_POS	0-451000 counts. Filters at 0, 120000, 240000, and 360000 counts.	Not valid when digital camera is connected to the IPSU-G. Axis 4 position reporting of motion control board in digital camera package.

4.5 Internal Ethernet Interfaces

See also **Table *15***, External Ethernet Interfaces where the interface to the IOP is listed. The 'Kernels' string points to a hardware ethernet interface on the processor. It is used in /etc/udev/rules.d/90-net.rules to ensure that the device name is always assigned to the same connector.

Table 20 Internal Ethernet Interfaces

Source	Device Name	IP Address	Net Mask	Kernels	Description
Lippert SBC	eth1	192.168.10.1	255.255.0.0	0000:06:00.0	Interface to data camera when connected directly to ACME and not IPSU-G.
Lippert SBC	eth2	192.168.100.1	255.255.255.0	0000:08:00.0	Interface to UEI Cube.
ACME Camera Package	DeviceIPAddress	192.168.10.100	255.255.0.0		Camera IP address. Changeable using the CLIpConfig command line utility included in ACME software install.
UEI Cube	ip2	192.168.100.102	255.255.255.0		Cube main ethernet port. Labeled NIC1. (yes, for ip2)
UEI Cube	ip	192.168.100.2	255.255.255.0		Cube secondary/diagnostic ethernet port. Labeled NIC2. NOT CONNECTED.

6.0 ENUMERATIONS

Table 21 ACME System States

ACME System States		
Value	State	Description
0	NULL	State at the start of software execution.
1	INITIALIZE	The software is initializing all hardware and sub-systems.
2	OP_IDLE	The software is ready to accept commands.
3	OP_EXPERIMENT	Automated experiments can be run in this state.
4	OP_MAINTENANCE	Maintenance activities can be run in this state.
5	OP_UPLINK_DOWNLINK	Currently unused.
6	OFF_NOMINAL	An error has occurred and sub-systems have been safed.
7	SAFED	Going to this state readies the system for a shutdown or reboot command.

Table 22 IPSU Phases

IPSU Phases		
Value	State	Description
0	None	No test point-related commands are pending to IPSUs or IPSU-Gs.
1	Opening Test Point	An open test point command has been sent to all selected IPSUs and IPSU-Gs.
2	Enabling Archiving	An enable archiving command has been sent to all selected IPSUs.
3	Disabling Video	A disable Video command has been sent to all selected IPSUs.
4	Starting Test Point	A start test point command has been sent to all selected IPSUs and IPSU-Gs.
5	Stopping Test Point	A stop test point command has been sent to all selected IPSUs and IPSU-Gs.
6	Closing Test POint	A close test point command has been sent to all selected IPSUs and IPSU-Gs.

Table 23 IPSU Status

IPSU Status		
Value	State	Description
0	None	No test point-related commands have been sent to the IPSU or IPSU-G yet.
1	Command Sent	Command has been sent to IPSU or IPSU-G.
2	Command Failed	Last command to IPSU or IPSU-G failed. For an IPSU this means a -1 in the subCommand field of the returned command. For an IPSU-G this means that the fcfSource in the returned command was not CMD_complete = 0x81.
3	Command Successful	Last command to IPSU or IPSU-G was successful. For an IPSU this means not a -1 in the subCommand field of the returned command. For an IPSU-G this means that the fcfSource in the returned command was CMD_complete = 0x81.
4	Command Timed Out	The IPSU or IPSU-G did not respond to the last command sent before the timeout period was up. Timeout period is registry value Ipsu/IpsuPhaseTimerPeriodMsec. See *Table 32.*
5	Inactive	This is an inactive IPSU as determined by the SystemDefineSetup command.

.

Table 24 Mass Flow Controller State

Mass Flow Controller States Source: Teledyne Hastings Digital 300 Software Guide Rev D. Document # 167-082010		
Value	State	Description
0	OFF	The MFC is off. An ACME state, not retrieved from MFC.
1	INIT	Initialization in progress, disables auto flow control, disables MFM alarms. This clears all alarms when beginning INIT state.
2	IDLE	MFM alarms are cleared and disabled (certain alarms such as instrument failures are not cleared and remain enabled). Indicated flow values are available. Flow totalizing is disabled. MFC flow control stopped, valve goes to user-defined default position. Many configuration commands are only allowed in the IDLE state
3	RESERVED	
4	OPERATE	Executing; MFM and MFC alarms and flow totalizing are enabled. Automatic flow control enabled if otherwise permitted.
5	ABORT	Network command sets ABORT state to permit diagnosis and recovery commands to be entered if/as necessary before restoring instrument to IDLE, OPERATE or INIT state. MFC flow control stopped, valve set to user-defined default position.

Mass Flow Controller States Source: Teledyne Hastings Digital 300 Software Guide Rev D. Document # 167-082010		
Value	State	Description
6	SFAIL	Recoverable error detected such as critical data not set, missing, or out of range. Reset using power down, the reset button, or setting state to INIT (=1) are the only escapes.
7	CAL	Calibration state disables auto flow control,clears and disables MFM and MFC alarms. Network commands to perform flow calibration are permitted.
8	TEST	Network commands to perform certain tests are permitted. May disable alarms, may disable auto flow control, depending upon specific test invoked. Tests in TEST state are generally factory-only.
9	RECOVER	Setting RECOVER state while in ABORT state initiates return to IDLE state.
10	TUNE	Network command to perform certain flow and flow control tests are permitted. May disable alarms, may disable auto flow control, depending upon specific test invoked.

Table 25 Waveform Enumeration

Waveform Enumeration	
Value	Waveform
0	None
1	Stepped
2	Ramped
3	Rectangular Wave
4	Ramped Wave

Table 26 Enabled Disabled Enumeration

Enabled Disabled Enumeration	
Value	State
0	Disabled
1	Enabled

Table 27 TFP Waveform State Enumeration

TFP Waveform Enumeration	
Value	Waveform
0	None
1	Ramp
2	RampedWave

Table 28 Extinction Detection State Enumeration

Extinction Detection State Enumeration	
Value	State
0	Disabled
1	Enabled
2	Detected

Table 29 Extinction Detection Action Enumeration

Extinction Detection Action Enumeration	
Value	State
0	Zero
1	Step
2	Ramp
3	Step Hold
4	Ramp Hold

Table 30 Solenoid Valve State Enumeration

Solenoid Valve State	
Value	State
0	Disabled (Stop flow)
1	Enabled (Allow flow)

Table 31 Script State Enumeration

Script State Enumeration	
Value	State
0	None
1	Loaded
2	Running
3	Paused
4	Aborted
5	Complete

7.0 CONFIGURATION FILES

7.1 AcmeConfig.ini

This is also known as the registry file. It is a text file with the format shown below. In this document and as an argument to the SetRegistryItem command the keys are referenced as Group/Key. For example, to change Value2 in the example below issue the command SetRegistryItem("GroupA/Key2") followed by the command SetRegistryValue("NewValue"). The table below shows the valid keys and values, including the value in the released version of AcmeConfig.ini.

```
 [GroupA]
Key1=Value1
Key2=Value2
…
[GroupB]
Key3=Value3
…
; Any comments are preceded with a semi-colon
```

Table 32 AcmeConfig.ini Contents

AcmeConfig.ini Contents				
Group	Key	Valid Values	Value	Description
AvtCamera	FileFormatTiff	true, false	false	Save images in TIFF format or raw. True = TIFF. False = raw. Only used when ACME is connected directly to the digital camera.
AvtCamera	SerialNumber		02-2143A-17086	Serial number of the digital camera. Needed for the software to connect to the camera. Only used when ACME is connected directly to the camera.
AvtCameraMotion	Filter1Position	Uint32	10000	Position where filter 1 is centered in the view.
AvtCameraMotion	Filter2Position	Uint32	120000	Position where filter 2 is centered in the view.
AvtCameraMotion	Filter3Position	Uint32	235000	Position where filter 3 is centered in the view.
AvtCameraMotion	Filter4Position	Uint32	350000	Position where filter 4 is centered in the view.

AcmeConfig.ini Contents				
Group	Key	Valid Values	Value	Description
System	UseAnalogCamera	true,false	true	Enable serial control of the analog camera in the CIA.
System	UseCiaMotionBoard	true,false	true	Connect to and use the motion controller in the CIA controlling the igniter arm and the TFP positioning.
System	UseCube	true,false	true	Use the analog input and output layers on the UEI Cube.
System	UseDataCamera	true,false	false	Enable control of the digital data camera directly from ACME.
System	UseFcfComms	true,false	true	Automatically start FcfCommunications (Base Classes) application during initialization.
System	UseLippert	true,false	true	Enable retrieval of CPU and CPU Board temperatures from CPU using LEMT.
System	UseSmartmon	true,false	true	Enable retrieval of hard drive temperatures using smartctl.

AcmeConfig.ini Contents				
Group	Key	Valid Values	Value	Description
Diagnostics	BreHeader	Comma-separated words, no spaces	"Time, MFCA1StaticMixerFlow, MFCA2BypassFlow, MFCA3FuelFlow, IonCurrent, PTA1, PTA2, PTA3, EfieldVoltage, PMT1, PMT2, PMT3, BRECenter180, BREOffset180, BRECenter, BREOffset, TC200A, TC200B, TCBurner, RadiometerBREOffset+60, RadiometerBREOffset+60CJ, RadiometerBREOffset+30, RadiometerBREOffset+30CJ, RadiometerBREOffset-10, RadiometerBREOffset-10CJ, Ch13, Ch14, Ch15, Ch16, RadiometerOffset, RadiometerOffsetCJ, TCE, BREHeatFluxCenter, BREHeatFluxOffset, TCCJ, EfieldSetpointOut"	Specifies the header string for the diagnostic data file when the experiement type communicated by the *SystemDefineSetup* command is BRE. Note: Value in previous column has spaces for this document. Actual value does not have spaces.

AcmeConfig.ini Contents				
Group	Key	Valid Values	Value	Description
Diagnostics	Header	Comma-separated words, no spaces	"Time, MFCA1StaticMixerFlow ,MFCA2BypassFlow, MFCA3FuelFlow, IonCurrent ,PTA1, PTA2, PTA3, EfieldVoltage ,PMT1 ,PMT2, PMT3, TCA, TCB ,TCC, TCD, TC200A, TC200B,TCBurner, Radiometer+60, Radiometer+60CJ, Radiometer+30, Radiometer+30CJ, RadiometerCenter ,RadiometerCenterCJ, Radiometer-30, Radiometer-30CJ,Radiometer-60, Radiometer-60CJ, RadiometerOffset, RadiometerOffsetCJ, TCE, Ch20, Ch21, TCCJ, EfieldSetpointOut"	Specifies the header string for the diagnostic data file when the experiement type communicated by the *SystemDefineSetup* command is not BRE. Note: Value in previous column has spaces for this document. Actual value does not have spaces.
Diagnostics	SampleTimeMsec	Uint32	8	Duration in msec between diagnostic data samplings for the diagnostic data file. 8 msec is equivalent to 125 Hz.
Diagnostics	ExtinctionDetection SampleTimeMsec	Uint32	100	Time in milliseconds between data samples for averaging to detect extinction.
Diagnostics	ExtinctionDetection UseSimDataFile	Bool	false	When enabled takes data from sim data file instead of analog input boards.
Diagnostics	ExtinctionDetection SimDataFileName	String	/payload/app/config/ extinctionDetectionSimData.csv	Sim extinction data. Format is csv, 8 columnds of data in order of configure command plus a comment column. First line is a header.

AcmeConfig.ini Contents				
Group	Key	Valid Values	Value	Description
Diagnostics	ExtinctionDetection ReverseLimitCheck	Bool	false	When false, extinction is detected when values go below limits. When true, detection is when values go above limits.
EField	NegativeOutputFactorA	Double	-10886.5	Used to convert between desired E-Field voltage and control voltage when negative supply is used. See EFieldSetVoltage command, section 3.1.8.2, for equation.
EField	NegativeOutputFactorB	Double	1048.203	See NegativeOutputFactorA description.
EField	NegativeOutputFactorC	Double	2.7365	See NegativeOutputFactorA description.
EField	NegativeMinVoltage	Float	0	This is enforced so that the output to the cube does not exceed 9.99V.
EField	NegativeMaxVoltage	Float	-10000	
EField	PositiveOutputFactorA	Double	-8.79	Used to convert between desired E-Field voltage and control voltage when positive supply is isntalled. See EFieldSetVoltage command, section 3.1.8.2, for equation.
EField	PositiveOutputFactorB	Double	1074.01	See PositiveOutputFactorA description.
EField	PositiveOutputFactorC	Double	-1.594	See PositiveOutputFactorA description.
EField	PositiveMaxVoltage	Float	0	
EField	PositiveMinVoltage	Float	10000	This is enforced so that the output to the cube does not exceed 9.99V.
EField	RampResolutionMsec	Uint32	1	This is the resolution of the signal generator.
FCF	RvcTimeoutMsec	Uint32	10000	Maximum time in msec that the FcfCommunications (ie. Base Classes) application waits for returns from RCV commands. This is argument 4 passed to FcfCommunications.
Flow	MFCA1StaticMixer FullScaleSlm	Float	20	This is the full scale flow of the MFC. Only used if the full scale flow was not obtained from the configuration information request (MFC ‘slm’ command) after power-on.

AcmeConfig.ini Contents				
Group	Key	Valid Values	Value	Description
Flow	MFCA2Bypass FullScaleSlm	Float	0.1	This is the full scale flow of the MFC. Only used if the full scale flow was not obtained from the configuration information request (MFC ‘slm’ command) after power-on.
Flow	MFCA3Fuel FullScaleSlm	Float	0.1	This is the full scale flow of the MFC. Only used if the full scale flow was not obtained from the configuration information request (MFC ‘slm’ command) after power-on.
Flow	MfcBootTimeMsec	Uint32	10000	Time to wait after powering on mass flow controllers before opening serial port to the MFCs and requesting configuration information (MFC ‘slm’ command).
Flow	MfcRampResolutionMsec	Uint32	100	This is the resolution of the signal generator.
Flow	SimulateMfcs	Bool	false	Simulates the mass flow controllers if true.
Flow	WriteMfcSetpointsToFile	Bool	true	If a test point is open, whenever any MFC changes setpoint a timestamped row of setpoints is written to the file, testPointName_mfcSetpoitns.csv.
Igniter	BurnerLimits	Comma-separated Uint32s	"2500, 2410, 2386, 2362, 2510, 2492, 2486, 2420, 2314"	List of maximum igniter positions per burner. First value corresponds to burner type 0, second to burner type 1, etc. See burnerType parameter in SystemDefineSetup command for list of burner types.
Igniter	AccelerationCountsPer SecSquared	Uint32	5000	This is the acceleration used to move the igniter arm when velocity is not specified in the command.
Igniter	CheckMotionPeriodMsec	Uint32	100	While in motion how often to report position in verbose logging.
Igniter	ConversionFactorA	Float	-0.6558	Used to convert desired voltage at igniter to control voltage. See IgniterPulse command in section 3.1.5.2.
Igniter	ConversionFactorB	Float	0.1213	See description for ConversionFactorA.

AcmeConfig.ini Contents				
Group	Key	Valid Values	Value	Description
Igniter	DecelerationCountsPer SecSquared	Uint32	5000	This is the deceleration used to move the igniter arm when velocity is not specified in the command.
Igniter	HomeAccelerationCounts PerSecSquared	Uint32	5000	This is the acceleration used to move the igniter arm to home position when it is not specified in the command.
Igniter	HomeDecelerationCounts PerSecSquared	Uint32	5000	This is the deceleration used to move the igniter arm to home position when it is not specified in the command.
Igniter	HomeVelocityCountsPerSec	Uint32	2000	This is the velocity used to move the igniter arm to home position when it is not specified in the command.
Igniter	JerkCountsPerSecCubed	Uint32	0	The motion jerk parameter for all movements.
Igniter	MaxPosition	Uint32	2500	If IgniterLimits is not defined for the current burner then this value is used as the maximum igniter position.
Igniter	MaxPowerVolts	Float	-0.11	The maximum control voltage setpoint allowed to the igniter. -0.11 corresponds to 4.5 V.
Igniter	MinPowerVolts	Float	-0.6	The minimum control voltage setpoint allowed to the igniter.
Igniter	PositionError	Uint32	0	The position error. See PMD documentation.
Igniter	Profile	Uint16	0	0=Trapezoidal, 1=VelocityContouring, 2=Curve, 3=Electronic Gear
Igniter	RetryOnError	Bool	False	If there is a motion error (i.e. slippage) then if this is true the motor will go slow until it recovers.
Igniter	VelocityCountsPerSec	Uint32	2000	This is the velocity used to move the igniter arm when velocity is not specified in the command.
Ipsu	HandleIpsuRvcs	Bool	true	Can turn off handling of command responses and use timeout instead by making this false.

AcmeConfig.ini Contents				
Group	Key	Valid Values	Value	Description
Ipsu	IpsuCommandResponse TimoutMsec	Uint32	30000	Duration in msec to wait for an IPSU or IPSU-G command response.
Ipsu	IpsuForDataCamera	5-6	5	This is the number of the IPSU-G that is controlling the ACME digital data camera.
Ipsu	IpsuForDataCameraCamera Number	Uint32	3	For the IPSU-G specified by IpsuForDataCamera, this is the camera number assigned to the ACME digital data camera.
Ipsu	IpsuForDataCameraMotion	5-6	5	This is the number of the IPSU-G that is controlling the ACME data camera motors: zoom, iris, focus, and filter.
Ipsu	IpsuForDataCameraMotion CommandGroup	Uint32	91 (x5B)	Command group assigned to ACME motion commands on the IPSU-G specified by IpsuForDataCameraMotion. Used to populate move camera axis and get position commands.
Ipsu	IpsuForDataCameraMotion EnablePositionRequests	bool	true	Whether or not to request data camera motion position data from the IPSU-G. Positions are only requested between motion start and detection of motion stopped when positions no longer change.
Ipsu	IpsuForDataCameraMotion ModuleIdentifier	Uint32	3	For the IPSU-G specified by IpsuForDataCameraMotion, this is the external module number assigned to the ACME motion controller.
Ipsu	IpsuForDataCameraMotion ModuleTelemetryIdentifier	Uint32	2	For GetMotionTelemetry commands sent to the IPSU-G for filter and lens positions this is the camera number.
Ipsu	IpsuForDataCameraMotion PositionRequestPeriodMsec	Uint32	1500	How often to request positions from the IPSU-G when motion is in progress. Stops when values do not change.

AcmeConfig.ini Contents				
Group	Key	Valid Values	Value	Description
Ipsu	IpsugAutoStartStopArchive	Bool	true	Can turn off start and stop of archive on IPSU-Gs during StartIpsus and StopIpsus command by setting this to false.
Ipsu	IpsugReopenExistingTest Point	Bool	false	This is used as the reopen parameter in the TestPoint Open command issued to IPSU-G.
Ipsu	IpsuWaitForResponse	Bool	1=true	Can turn off waiting for all IPSUs to respond for a phase by setting to 0 or false.
Intensifier	CommandDelayMsec	Uint16	200	Delay between intensifier commands to the IPSU-G for IntensifierCameraGate and IntensifiedCameraSafe commands.
Intensifier	CommandGroup	Uint16	89 (0x59)	Command group for intensifier commands to the IPSU-G.
Intensifier	FilterAxis	Uint16	4	Intensifier motion axis.
Intensifier	ModuleIdentifier	Uint16	1	External module id for intensifier on IPSU-G.
Intensifier	MotionCommandGroup	Uint16	93 (0x5D)	Command group for intensifier motion commands to the IPSU-G.
Intensifier	MotionDelaySecs	Uint16	10	Time to wait between motion commands (ie. Home, wait, go to position)
Intensifier	MotionModuleIdentifier	Uint16	2	External module id for intensifier motion on IPSU-G.
Intensifier	SafeFilterPosition	Uint32	1960	Used during IntensifiedCameraSafe command.
Intensifier	SafeGainCounts	Uint32	2048	Used during IntensifiedCameraSafe command.
Intensifier	SafeGateWidthCounts	Uint16	0	Used during IntensifiedCameraSafe command.
OpsCamera	InitalGain	0-4	3=21 dB	On software startup the analo ops camera is set to use this gain. The meaning of the values is shown in section 3.1.4.27.
OpsCamera	InitialShutter	0-9	2=1/250	On software startup the analo ops camera is set to use this shutter speed. The meaning of the values is shown in section3.1.4.28.

AcmeConfig.ini Contents				
Group	Key	Valid Values	Value	Description
Fiber	AccelerationCounts PerSecSquared	Uint32	1000	This is the acceleration used to move the TFP fiber array when velocity is not specified in the command.
Fiber	CheckMotionPeriodMsec	Uint32	100	While in motion how often to report position in verbose logging.
Fiber	DecelerationCounts PerSecSquared	Uint32	0	This is the deceleration used to move the TFP fiber array when velocity is not specified in the command.
Fiber	FiberToBurnerOffsets	Comma-separated Uint32s	"0, 6996, 7245, 7496, 4606, 4921, 4921, 4921, 4921"	List of TFP offsets from TfpMaxCount for each burner. First value corresponds to burner type 0, second to burner type 1, etc. See burnerType parameter in *SystemDefineSetup* command for list of burner types.
Fiber	HomeAccelerationCounts PerSecSquared	Uint32	1000	This is the acceleration used to move the TFP fiber array to home position when it is not specified in the command.
Fiber	HomeDecelerationCounts PerSecSquared	Uint32	0	This is the deceleration used to move the TFP fiber array to home position when it is not specified in the command.
Fiber	HomeVelocityCountsPerSec	Uint32	1200	This is the velocity used to move the TFP fiber array to home position when it is not specified in the command and when moving apart from a command, such as after startup and some state changes.
Fiber	JerkCountsPerSecCubed	Uint32	0	The motion jerk parameter for all movements.
Fiber	MaxPosition	Uint32	23450	Maximum position for the TFP (thin filament pyrometry array) for any burner.
Fiber	PositionError	Uint32	0	The position error. See PMD documentation.
Fiber	Profile	Uint16	0	0=Trapezoidal, 1=VelocityContouring, 2=Curve, 3=Electronic Gear

AcmeConfig.ini Contents				
Group	Key	Valid Values	Value	Description
Fiber	RetryOnError	Bool	false	If there is a motion error (i.e. slippage) then if this is true the motor will go slow until it recovers.
UeiCube	DataAcqIpAddress		192.168.100.102	Cube IP address.

Note 1: For ACME to control Camera Package change NO_EXT_MOTION_ACME to EXT_MOTION_ACME.

7.2 AcmeLogging.ini

This file located in /payload/app/config allows filter levels to be configured so that more or less information shows up in the telemetry and file message logs. In the message log each message contains a logging category, for example "acme.diag" or "zt.fcf". To turn off a category make a rule in AcmeLogging.ini. For example, "acme.diag=false" turns off any message with the acme.diag category. Comments in the file indicate available categories.

7.3 Motion_ACME.ini

This file located in /payload/app/config contains configuration for motion control of the camera package when ACME is controlling the camera motion instead of IPSU-G. It specifies the camera serial port (serialPortNumber=3 means use /dev/ttyS3) and parameters for each axis. While no longer used it may still be included to show what IPSU-G should have for this file.

7.4 Script Files

Script files are used to conduct automated operations using the commands: *SetScriptDirectory*, *LoadScript*, *RunScript*, *AbortScript*, *PauseScript*, *ResumeScript*, and *ChangeScriptParameter*. They are named with a .js extension (for java script) and are typically stored in the /payload/app/scripts/ directory or any sub-directory of /payload/app/scripts, but can be anywhere as long as the *SetScriptDirectory* and/or *LoadScript* commands specify the location.

The language used to write script files is technically ECMAScript-262 (http://www.ecma-international.org/publications/files/ECMA-ST/Ecma-262.pdf). A brief summary can be currently found at https://wiki.qt.io/JavaScript.

Many ACME functions and variables have been made visible to the scripts. All commands and their parameters can be used. All current telemetry is available. Public methods in controllers derived from XSystemController can be called. Any other objects passed to the QT macro Q_DECLARE_METATYPE(object) in the ACME source code can be used.

Several helper functions are preloaded to the script engine and can also be used. print(message) logs a message. sendCommand(command) sends the specified command to the command handler. uSleep(time) pauses script execution for the specified number of milliseconds.

More helper functions are loaded from the script file "/payload/app/scripts/helpers.js" if it exists.

Here is an example of a basic ACME script:

```
// BasicScript.js

// When RunScript command is received this says to execute the run function below
AcmeAvionicsController.runScript.connect(run);

//This variable must be named parameters so the ground and flight software can manipulate the values.
var parameters = { parameter1:1.0, parameter2:4, parameter3:true };

// from helpers.js.  This tells the ground software the parameters and values so they can be
// displayed and modified using the ChangeScriptParameter command
printParams();
```

```
function run()
{
  //The keyword ‘running’ is important for the ground software.
  print("BasicScript running");

  if ( parameters.parameter1 != 5 )
  {
        // do something
  }

  // examples of filling in command parameters and sending to main program
  FlowSetSolenoidValveState.solendoidValve = 3;
  FlowSetSolenoidValveState.state = 1;
  sendCommand(FlowSetSolenoidValveState);

  // example of looking at current telemetry
  If ( FlowTelemetry.MFCA3FuelFlowSlpm > parameters.parameter3)
  {
        // do something
  }

  // script can pause itself. No parameters.
 sendCommand(PauseScript);
 print(“Script is paused.  Press resume when ready.”);

  // now operator must issue ResumeScript command from ground

  // call a function in this file
 doSomethingElse();

  // the keyword ‘completed’ is important for the ground software.
  Print(“BasicScript completed”);
}

Function doSomethingElse()
{
        // more code
}
```

8.0 OUTPUT FILES

8.1 Data Output File

Data output files are created using the *SystemOpenTestPoint* command which creates a file named and stored at /payload/sci/testpointName_x/testpointName_x_data.csv, where testpointName is the parameter received in the SystemOpenTestPoint command and _x is optional to avoid duplicate files, where x = 1,2,3, etc.

The file contains a two-line header. The first line has the date and time, documentation of the setup received in a *SystemDefineSetup* command, and the PMT gains. The second line has the column headings for the subsequent rows of data. The text of the column headings is defined by the Diagnostics/Header registry entry or the Diagnostics/BreHeader entry depending on whether the *SystemDefineSetup* command experimentType parameter was BRE or not. The column headings should match the data which is the following:

- A timestamp in the format HH:MM:SS:mmm, where mmm is milliseconds.
- 11 channels of analog input from the cube DNx-AI-207 board in the order shown in ***Table 16***.
- 25 channels of analog input from the cube DNx-AI-225 board in the order shown in ***Table 17***. (The names of many of these values change depending on whether it is a BRE test or not, hence the two options for header.)
- Channel AO-4 of the DNx-AO-333 analog output board. This is the E-Field control voltage.

Rows of comma-separated data is added to the file between *DataRecordingStart* and *DataRecordingStop* commands at a rate defined by the Diagnostics/SampleTimeMsec registry entry, see ***Table 32***, defaulting to 8 msec for 125 Hz sampling.

8.2 Mass Flow Controller Setpoint File

If the registry value Flow/WriteMfcSetpointsToFile is true when the *SystemOpenTestPoint* command is executed then a file of MFC setpoints is created alongside the data output file. The name of the file will be /payload/sci/testpointName_x/testpointName_x_mfcSetpoints.csv. The first row of text contains the same header as the data output file. The second row contains the headers, “Time,MFCA1Setpoint(V),MFCA2Setpoint(V),MFCA3Setpoint(V)”. Subsequent rows are written whenever one of the three MFCs change setpoints and includes the changed setpoint and the current setpoints of the other two MFCs, and a timestamp with millisecond resolution. The file is close when the *SystemCloseTestPoint* command is executed.

8.3 Message Log File

Software messages are logged to files named and stored at /payload/logs/YYYY-MM-DD_hh-mmACME.log, where YYYY is year, MM is month, DD is day of month, hh is hour, and mm is minutes. The date in the name is when the log file is created, soon after software startup.

Each message line in the file begins with a timestamp and a logging category. The timestamp is of the format YYYY-MM-DD DDD hh:mm:ss:mmm, where DDD is day of the year, mmm is milliseconds, and the rest are self-explanatory. See section 1.1 for information about the logging category and how to filter them out if desired.

8.4 Command Log File

The message log file, section 8.3, includes messages showing commands received and their parameters. Command log files contain only the details of each command received. The files are named and stored at /payload/logs/YYYY-MM-DD_hh-mmACMECommands.log. The format is for each command received a timestamped line of text with the command name followed by timestamped lines of text showing each parameter name and parameter value, including header information.

9.0 DIRECTORY STRUCTURE

The files described in the previous two sections are stored on the solid state drive in the following locations.

Table 33 Directory Structure

Directory Strucure	
Path	Contents
/payload	Root directory for ACME files. The script, startup.sh, looks here during each software startup to see if there is new software in a tar file to install.
/payload/app	Root directory for ACME application files.
/payload/app/bin	AcmeMp and FcfCommunications executables are located here.
/payload/app/config	Configurations files: AcmeConfig.ini AcmeLogging.ini
/payload/app/scripts	Science Qt script files including the automatically loaded helpers.js script.
/payload/app/scripts/shell	System bash scripts for startup, software updates, and directory listings.
/payload/logs	Message log files and command log files.
/payload/sci	Root directory for ACME science files.
/payload/sci/YYDDDtt	Data output files and Mass Flow Controller Setpoint files. The names of the directories are what is specified in the TestPointOpen commands, tentatively to follow the convention of YY=year (ie. 17 for 2017), DDD=GMT day (001-365), and tt=test points of the day (ie. A1, A2, B1, …).

10.0 DATA FILE CONVERSION

This section describes how to convert the headers and sensor readings in the data output file to engineering values. The raw data is in units of volts read directly from the UEI data acquisition system. Converions of the data depend on whether the experiment type is BRE or not. See sub-section 10.2 for non-BRE or sub-section 10.3 for BRE.

10.1 Metadata Header

The first row of the data output file is a comma-separate list of metadata documenting the system configuration when the data was collected.

Table 34 Data File Metadata Header

Data File Metadata Header	
Name	Possible Values
Timestamp	Formatted as year-month-day GMTday hr:min:sec:millisecond
Experiment	0 Flame Design 1 Coflow 2 S-Flame 3 E-Field 4 BRE
Test Point	Name of test point
MFCA1 Gas	0 Oxygen/Nitrogen (21/79) 1 Oxygen/Nitrogen (30/70) 2 Oxygen/Nitrogen (40/60) 3 Oxygen/Nitrogen (85/15) 4 Oxygen/Helium (30/70)
MFCA1 Full Scale	0 to 20 SLPM
MFCA2 Gas	0 Nitrogen 1 Helium
MFCA2 Full Scale	0 to 20 SLPM
MFCA3 Gas	0 Methane 1 Methane/Nitrogen (30/70) 2 Methane/Nitrogen (40/60) 3 Methane/Nitrogen (45/55) 4 Ethylene 5 Ethylene/Nitrogen (30/70) 6 Ethylene/Nitrogen (50/50) 7 Hydrogen 8 Hydrogen/Methane (44.4/55.6) 9 Hydrogen/Methane (55.6/44.4) 10 Hydrogen/Methane (66.7/33.3) 11 Hydrogen/Methane/Helium (20/25/55) 12 Hydrogen/Methane/Helium (25/20/55) 13 Hydrogen/Methane/Helium (30/15/55) 14 Hydrogen/Helium (45/55) 15 Methane/Helium (45/55) 16 Ethylene/Helium (20/80) 17 Ethylene/Helium (25/75)

	18 Ethylene/Helium (30/70)
MFCA3 Full Scale	0 to 20 SLPM
EField	0 Mesh Removed 1 Mesh in 35 mm position 2 Mesh in 40 mm position
EFieldPol	0 Not installed 1 Negative 2 Positive
Igniter	0 0 mm 1 10 mm 2 30 mm
Burner	0 Spherical (1/4") 1 Spherical (3/8") 2 Spherical (1/2") 3 Coflow 4 Gas Jet(1.3mm) 5 Gas Jet(2.1mm) 6 BRE (25mm) 7 BRE (50mm)
Thermocouple	0 Near 1 Far 2 Removed
Radiometer	0 Standard 1 BRE
IPSU 1	0 Do not archive images 1 Archive images during test point (HiBMs)
IPSU 2	0 Do not archive images 1 Archive images during test point
IPSU 3	0 Do not archive images 1 Archive images during test point
IPSU 4	0 Do not archive images 1 Archive images during test point
IPSU 5	0 Do not archive images 1 Archive images during test point (ACME camera and/or Intensified Camera)
IPSU 6	0 Do not archive images 1 Archive images during test point
Fiber	0 Not installed 1 Installed
PMT 1 Gain	0 to 10 Volts
PMT 2 Gain	0 to 10 Volts
PMT 3 Gain	0 to 10 Volts

10.2 Data Conversions (non-BRE)

Each paragraph name in this section refers to a column in the data output file. The Voltage variable in each equation is the data value.

10.2.1 Time

Description: Timestamp in the format hour:min:sec:msec. The date can be found the the first row header.

10.2.2 MFCA1StaticMixerFlow

Description: Measured flow from MFCA1 sourced from the CIR static mixer manifold. This is the oxidizer.

Device: Teledyne Hastings HFC-D-302

Conversion: Flow rate (SLPM) = Voltage / 5.0 * MFCA1 Full Scale

,where MFCA1 Full Scale is indicated in the meta data header.

10.2.3 MFCA2BypassFlow

Description: Measured flow from MFCA2 sourced from the CIR nitrogen or nitrogen bypass manifold. This is the inert.

Device: Teledyne Hastings HFC-D-302

Conversion: Flow rate (SLPM) = Voltage / 5.0 * MFCA2 Full Scale

,where MFCA2 Full Scale is indicated in the meta data header.

10.2.4 MFCA3FuelFlow

Description: Measured flow from MFCA3 sourced from the CIR fuel manifold.

Device: Teledyne Hastings HFC-D-302

Conversion: Flow rate (SLPM) = Voltage / 5.0 * MFCA3 Full Scale

,where MFCA3 Full Scale is indicated in the meta data header.

10.2.5 IonCurrent

Description: Measurement of ion current between E-Field grid and burner

Conversion: Ion Current (microAmps) = 1000 * Voltage / 162

10.2.6 PTA1

Description: Inner coflow/burner pressure.

Device: Measurement Specialties EPRB-2-VR33-10B-/CM1/ATR

Conversion: Pressure (psia) = (Voltage – 0.5) * 14.5038 * 10 / 4

Pressure (atm) = (Voltage – 0.5) * 14.5038 * 10 / 4 / 14.69595

10.2.7 PTA2

Description: Outer coflow burner pressure.

Device: Measurement Specialties EPRB-2-VR33-10B-/CM1/ATR

Conversion: Pressure (psia) = (Voltage – 0.5) * 14.5038 * 10 / 4

Pressure (atm) = (Voltage – 0.5) * 14.5038 * 10 / 4 / 14.69595

10.2.8 PTA3

Description: Chamber pressure

Device: Measurement Specialties EPRB-2-VR33-10B-/CM1/ATR

Conversion: Pressure (psia) = (Voltage – 0.5) * 14.5038 * 10 / 4

Pressure (atm) = (Voltage – 0.5) * 14.5038 * 10 / 4 / 14.69595

10.2.9 EfieldVoltage

Description: Indirect measurement of the E-Field high voltage supply.

Conversion source: Determined empirically.

Conversion: E-Field (V) = Voltage * 2486.1795 + 141.9733

10.2.10 PMT1

Description: Photomultiplier tube with a 310 nm bandpass interference filter

Device: Hamamatsu H10722-10-MOD2

Conversion source: ACME-RPT-0013, PMT Characterization Report, Rev -

Conversion: Depends on PMT gain.

Gain = 3V Irradiance (W/cm^2) = $1.991881x^2 - 0.005211x + 0.000200$

Gain = 4V Irradiance (W/cm^2) = $196.835980x^3 + 0.345545x^2 + 0.519914x + 0.000150$

Gain = 5V Irradiance (W/cm^2) = $1{,}104.055224x^3 + 34.407322x^2 + 2.770232x + 0.000277$

Gain = 5.5V Irradiance (W/cm^2) = $-58{,}849.143398x^4 + 11{,}868.553546x^3 - 224.738423x^2 + 10.187086x + 0.000282$

Gain = 6V Irradiance (W/cm^2) = $-55{,}617.012418x^4 + 22{,}186.536267x^3 - 645.571330x^2 + 22.895739x + 0.000400$

Gain = 7V Irradiance (W/cm^2) = $3{,}308{,}609.058167x^4 - 166{,}821.231989x^3 + 3{,}599.519526x^2 + 39.920446x + 0.003750$

Gain = 8V Irradiance (W/cm^2) = $395{,}393.992726x^3 - 7{,}363.493027x^2 + 253.097826x + 0.002500$

10.2.11 PMT2

Description: Photomultiplier tube with a broadband filter with a fused silica glass window and a neutral density coating.

Device: Hamamatsu H10722-10-MOD2

Conversion source: ACME-RPT-0013, PMT Characterization Report, Rev -

Conversion: depends on PMT gain.

Gain = 2V: Irradiance (W/cm^2) = $8.585779x^2 + 0.156690x + 0.000105$

Gain = 2.5V: Irradiance (W/cm^2) = $-43.824964x^3 + 169.992980x^2 + 1.665894x - 0.000060$

Gain = 3V: Irradiance (W/cm^2) = $291{,}761.896381x^5 - 88{,}314.764348x^4 + 8{,}801.002262x^3 + 65.786376x^2 + 3.172557x + 0.000010$

Gain = 3.5V: Irradiance (W/cm^2) = $459{,}377.096347x^4 - 39{,}905.504054x^3 + 1{,}814.542253x^2 + 15.361730x + 0.000260$

Gain = 4V; Irradiance (W/cm^2) = $2{,}092.641446x^2 + 40.832842x - 0.000930$

Gain = 5V; Irradiance (W/cm^2) = $728{,}809{,}327.944092x^4 - 11{,}745{,}484.619257x^3 + 60{,}902.178128x^2 + 271.865031x - 0.007000$

Gain = 6V; Irradiance (W/cm^2) = $71{,}216.045377x^2 + 1{,}600.544829x - 0.043000$

10.2.12 PMT3

Description: Photomultiplier tube with a 430 nm bandpass interference filter.

Device: Hamamatsu H10722-10-MOD2

Conversion source: ACME-RPT-0013, PMT Characterization Report, Rev -

Conversion: depends on PMT gain.

Gain = 3V: Irradiance (W/cm^2) = $-1{,}764.688010x^4 + 508.863505x^3 - 15.508597x^2 + 0.393871x + 0.000070$

Gain = 4V: Irradiance (W/cm^2) = $-14{,}425.292616x^4 + 4{,}236.185250x^3 - 46.099632x^2 + 2.012753x + 0.000600$

Gain = 4.5V: Irradiance (W/cm^2) = $-2{,}955{,}659.559357x^5 + 543{,}397.470716x^4 - 19{,}411.363291x^3 + 505.687643x^2 + 5.655574x + 0.002100$

Gain = 5V: Irradiance (W/cm^2) = $911{,}016.817407x^4 - 44{,}193.276271x^3 + 1{,}087.860546x^2 + 8.544647x + 0.003600$

Gain = 6V: Irradiance (W/cm^2) = $7{,}785.072945x^3 + 2{,}717.730343x^2 + 50.108622x + 0.019400$

Gain = 7V: Irradiance (W/cm^2) = $703{,}610.180561x^3 - 11{,}656.854323x^2 + 337.786210x + 0.074000$

Gain = 8V: Irradiance (W/cm^2) = $165{,}632.254213x^2 + 451.923332x + 0.285000$

10.2.13 TCA

Description: Thermocouple at position A on the near or far thermcouple rake.

Device: Marlin 92420280-107 for near assembly, Marlin 92420280-087 for far assembly

Conversion: Use the Type K thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.2.14 TCB

Description: Thermocouple at position B on the near or far thermcouple rake.

Device: Marlin 92420281-053 for near assembly, Marlin 92420281-042 for far assembly

Conversion: Use the Type B thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.2.15 TCC

Description: Thermocouple at position C on the near or far thermcouple rake.

Device: Marlin 92420281-048 for near assembly, Marlin 92420281-034 for far assembly

Conversion: Use the Type B thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.2.16 TCD

Description: Thermocouple at position D on the near or far thermcouple rake.

Device: Marlin 92420280-087 for near assembly, Marlin 92420280-087 for far assembly

Conversion: Use the Type K thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.2.17 TC200A

Description: Igniter side Type K thermocouple, 200 mm from axis.

Device: Marlin 9242083-052

Conversion: Use the Type K thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.2.18 TC200B

Description: Relay side Type K thermocouple, 200 mm from axis.

Device: Marlin 92420283-038

Conversion: Use the Type K thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.2.19 TCBurner

Description: Thermocouple on the burner

Device: Depends on burner:

1.3 mm gas jet burner	Marlin 92420308-667
2.1 mm gas jet burner	Marlin 92420308-667
1/4" spherical burner	Marlin 92420308-705
3/8" spherical burner	Marlin 92420308-705
1/2" spherical burner	Marlin 92420308-705
Coflow burner	No thermocouple

Conversion: Use the Type K thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.2.20 Radiometer+60

Description: Radiometer at the 60mm position on the standard radiometer assembly.

Device: Dexter Research Center XP-0055 Thermopile Detector

Conversion Source: ACME-RPT-011, Radiometer Characterization Report, Rev -.

Conversion: Irradiance (W/cm^2) = 0.002622 * Voltage - 0.003370

10.2.21 Radiometer+60CJ

Description: Cold junction temperature sensor associated with the 60mm radiometer.

Device: TI LM20

Conversion: $Temperature\ (C) = -1481.96 + \sqrt{2196200 + 257700(1.8639 - Voltage)}$

10.2.22 Radiometer+30

Description: Radiometer at the 30mm position on the standard radiometer assembly.

Device: Dexter Research Center XP-0055 Thermopile Detector

Conversion Source: ACME-RPT-011, Radiometer Characterization Report, Rev -.

Conversion: Irradiance (W/cm^2) = 0.001794 * Voltage - 0.002309

10.2.23 Radiometer+30CJ

Description: Cold junction temperature sensor associated with the 30mm radiometer.

Device: TI LM20

Conversion: $Temperature\ (C) = -1481.96 + \sqrt{2196200 + 257700(1.8639 - Voltage)}$

10.2.24 RadiometerCenter

Description: Radiometer at the center position on the standard radiometer assembly.

Device: Dexter Research Center XP-0156 Thermopile Detector

Conversion Source: ACME-RPT-011, Radiometer Characterization Report, Rev -.

Conversion: Irradiance (W/cm^2) = 0.002894 * Voltage - 0.003932

10.2.25 RadiometerCenterCJ

Description: Cold junction temperature sensor associated with the center radiometer.

Device: TI LM20

Conversion: $Temperature\ (C) = -1481.96 + \sqrt{2196200 + 257700(1.8639 - Voltage)}$

10.2.26 Radiometer-30

Description: Radiometer at the -30mm position on the standard radiometer assembly.

Device: Dexter Research Center XP-0055 Thermopile Detector

Conversion Source: ACME-RPT-011, Radiometer Characterization Report, Rev -.

Conversion: Irradiance (W/cm^2) = 0.002397 * Voltage - 0.003077

10.2.27 Radiometer-30CJ

Description: Cold junction temperature sensor associated with the -30mm radiometer.

Device: TI LM20

Conversion: $Temperature\ (C) = -1481.96 + \sqrt{2196200 + 257700(1.8639 - Voltage)}$

10.2.28 Radiometer-60

Description: Radiometer at the -60mm position on the standard radiometer assembly.

Device: Dexter Research Center XP-0055 Thermopile Detector

Conversion Source: ACME-RPT-011, Radiometer Characterization Report, Rev -.

Conversion: Irradiance (W/cm^2) = 0.002591 * Voltage - 0.003300

10.2.29 Radiometer-60CJ

Description: Cold junction temperature sensor associated with the -60mm radiometer.

Device: TI LM20

Conversion: $Temperature\ (C) = -1481.96 + \sqrt{2196200 + 257700(1.8639 - Voltage)}$

10.2.30 RadiometerOffset

Description: Radiometer located on the downstream relay board.

Device: Dexter Research Center DX-0869 Thermopile Detector

Conversion Source: ACME-RPT-011, Radiometer Characterization Report, Rev -.

Conversion: Irradiance (W/cm^2) = 0.001798 * Voltage - 0.002307

10.2.31 RadiometerOffsetCJ

Description: Cold junction temperature sensor associated with the offset radiometer.

Device: TI LM20

Conversion: $Temperature\ (C) = -1481.96 + \sqrt{2196200 + 257700(1.8639 - Voltage)}$

10.2.32 TCE

Description: Not used – spare

10.2.33 Ch20

Description: Not used during non-BRE experiments

10.2.34 Ch21

Description: Not used during non-BRE experiments

10.2.35 TCCJ

Description: Temperature sensor located on the cold junction compensation assembly used for cold junction compensation of thermocouples.

Device: Analog Devices AD590

Conversion: Temperature (C) = 338.98305 * Voltage - 273.15

10.2.36 EfieldSetpointOut

Description: Control voltage output to the E-Field supply to achieve E-Field voltage from 0 to +/- 10kV.

Conversion: E-Field Voltage (V) = C * EfieldSetpointOut^2 + B * EfieldSetpointOut + A,

where A, B,and C depend on supply polarity:

Negative polarity	Positive polarity
A = -10886.5	A = -8.79
B = 1048.203	B = 1074.01
C = 2.7365	C = -1.594

The avionics software uses the inverse of this equation to determine the control voltage for the commanded E-Field voltage:

$$EfieldSetpointOut = \frac{-\mathrm{B} + \sqrt{B^2 - 4\mathrm{C}(\mathrm{A} - \mathrm{EFieldVoltage})}}{2C}$$

10.3 Data Conversions (BRE)

When the experiment type is BRE as set by the SystemDefineSetup command some data columns change name and conversion to reflect the connection of the BRE burner and the BRE radiometer array and the removal of the thermcouple rake.

10.3.1 Time

Description: Timestamp in the format hour:min:sec:msec. The date can be found the the first row header.

10.3.2 MFCA1StaticMixerFlow

Description: Measured flow from MFCA1 sourced from the CIR static mixer manifold. This is the oxidizer.

Device: Teledyne Hastings HFC-D-302

Conversion: Flow rate (SLPM) = Voltage / 5.0 * MFCA1 Full Scale

,where MFCA1 Full Scale is indicated in the meta data header.

10.3.3 MFCA2BypassFlow

Description: Measured flow from MFCA2 sourced from the CIR nitrogen or nitrogen bypass manifold. This is the inert.

Device: Teledyne Hastings HFC-D-302

Conversion: Flow rate (SLPM) = Voltage / 5.0 * MFCA2 Full Scale

,where MFCA2 Full Scale is indicated in the meta data header.

10.3.4 MFCA3FuelFlow

Description: Measured flow from MFCA3 sourced from the CIR fuel manifold.

Device: Teledyne Hastings HFC-D-302

Conversion: Flow rate (SLPM) = Voltage / 5.0 * MFCA3 Full Scale

,where MFCA3 Full Scale is indicated in the meta data header.

10.3.5 IonCurrent

Description: Measurement of ion current between E-Field grid and burner

Conversion: Ion Current (microAmps) = 1000 * Voltage / 162

10.3.6 PTA1

Description: Inner coflow/burner pressure.

Device: Measurement Specialties EPRB-2-VR33-10B-/CM1/ATR

Conversion: Pressure (psia) = (Voltage – 0.5) * 14.5038 * 10 / 4

Pressure (atm) = (Voltage – 0.5) * 14.5038 * 10 / 4 / 14.69595

10.3.7 PTA2

Description: Outer coflow burner pressure.

Device: Measurement Specialties EPRB-2-VR33-10B-/CM1/ATR

Conversion: Pressure (psia) = (Voltage – 0.5) * 14.5038 * 10 / 4

Pressure (atm) = (Voltage – 0.5) * 14.5038 * 10 / 4 / 14.69595

10.3.8 PTA3

Description: Chamber pressure

Device: Measurement Specialties EPRB-2-VR33-10B-/CM1/ATR

Conversion: Pressure (psia) = (Voltage – 0.5) * 14.5038 * 10 / 4

Pressure (atm) = (Voltage – 0.5) * 14.5038 * 10 / 4 / 14.69595

10.3.9 EfieldVoltage

Description: Indirect measurement of the E-Field high voltage supply.

Conversion source: Determined empirically.

Conversion: E-Field (V) = Voltage * 2486.1795 + 141.9733

10.3.10 PMT1

Description: Photomultiplier tube with a 310 nm bandpass interference filter

Device: Hamamatsu H10722-10-MOD2

Conversion source: ACME-RPT-0013, PMT Characterization Report, Rev -

Conversion: Depends on PMT gain.

Gain = 3V Irradiance (W/cm^2) = $1.991881x^2 - 0.005211x + 0.000200$

Gain = 4V Irradiance (W/cm^2) = $196.835980x^3 + 0.345545x^2 + 0.519914x + 0.000150$

Gain = 5V Irradiance (W/cm^2) = $1{,}104.055224x^3 + 34.407322x^2 + 2.770232x + 0.000277$

Gain = 5.5V Irradiance (W/cm^2) = $-58{,}849.143398x^4 + 11{,}868.553546x^3 - 224.738423x^2 + 10.187086x + 0.000282$

Gain = 6V Irradiance (W/cm^2) = -55,617.012418x^4 + 22,186.536267x^3 - 645.571330x^2 + 22.895739x + 0.000400

Gain = 7V Irradiance (W/cm^2) = 3,308,609.058167x^4 - 166,821.231989x^3 + 3,599.519526x^2 + 39.920446x + 0.003750

Gain = 8V Irradiance (W/cm^2) = 395,393.992726x^3 - 7,363.493027x^2 + 253.097826x + 0.002500

10.3.11 PMT2

Description: Photomultiplier tube with a broadband filter with a fused silica glass window and a neutral density coating.

Device: Hamamatsu H10722-10-MOD2

Conversion source: ACME-RPT-0013, PMT Characterization Report, Rev -

Conversion: depends on PMT gain.

Gain = 2V: Irradiance (W/cm^2) = 8.585779x^2 + 0.156690x + 0.000105

Gain = 2.5V: Irradiance (W/cm^2) = -43.824964x^3 + 169.992980x^2 + 1.665894x - 0.000060

Gain = 3V: Irradiance (W/cm^2) = 291,761.896381x^5 - 88,314.764348x^4 + 8,801.002262x^3 + 65.786376x^2 + 3.172557x + 0.000010

Gain = 3.5V: Irradiance (W/cm^2) = 459,377.096347x^4 - 39,905.504054x^3 + 1,814.542253x^2 + 15.361730x + 0.000260

Gain = 4V; Irradiance (W/cm^2) = 2,092.641446x^2 + 40.832842x - 0.000930

Gain = 5V; Irradiance (W/cm^2) = 728,809,327.944092x^4 - 11,745,484.619257x^3 + 60,902.178128x^2 + 271.865031x - 0.007000

Gain = 6V; Irradiance (W/cm^2) = 71,216.045377x^2 + 1,600.544829x - 0.043000

10.3.12 PMT3

Description: Photomultiplier tube with a 430 nm bandpass interference filter.

Device: Hamamatsu H10722-10-MOD2

Conversion source: ACME-RPT-0013, PMT Characterization Report, Rev -

Conversion: depends on PMT gain.

Gain = 3V: Irradiance (W/cm^2) = -1,764.688010x^4 + 508.863505x^3 - 15.508597x^2 + 0.393871x + 0.000070

Gain = 4V: Irradiance (W/cm^2) = -14,425.292616x^4 + 4,236.185250x^3 - 46.099632x^2 + 2.012753x + 0.000600

Gain = 4.5V: Irradiance (W/cm^2) = -2,955,659.559357x^5 + 543,397.470716x^4 - 19,411.363291x^3 + 505.687643x^2 + 5.655574x + 0.002100

Gain = 5V: Irradiance (W/cm^2) = $911{,}016.817407x^4 - 44{,}193.276271x^3 + 1{,}087.860546x^2 + 8.544647x + 0.003600$

Gain = 6V: Irradiance (W/cm^2) = $7{,}785.072945x^3 + 2{,}717.730343x^2 + 50.108622x + 0.019400$

Gain = 7V: Irradiance (W/cm^2) = $703{,}610.180561x^3 - 11{,}656.854323x^2 + 337.786210x + 0.074000$

Gain = 8V: Irradiance (W/cm^2) = $165{,}632.254213x^2 + 451.923332x + 0.285000$

10.3.13 BRECenter180

Description: Thermocouple at the south position on the BRE burner.

Device: Marlin 92420321

Conversion: Use the Type K thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.3.14 BREOffset180

Description: Thermocouple at the far south position on the BRE burner.

Device: Marlin 92420321

Conversion: Use the Type K thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.3.15 BRECenter

Description: Thermocouple on the BRE burner center heat flux transducer.

Device: Medtherm 8-2.688-10SB-0-0-36-21427KSA.

Conversion: Use the Type K thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.3.16 BREOffset

Description: Thermocouple on the BRE burner offset heat flux transducer.

Device: Medtherm 8-2.688-10SB-0-0-36-21427KSA.

Conversion: Use the Type K thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.3.17 TC200A

Description: Igniter side Type K thermocouple, 200 mm from axis.

Device: Marlin 9242083-052

Conversion: Use the Type K thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.3.18 TC200B

Description: Relay side Type K thermocouple, 200 mm from axis.

Device: Marlin 92420283-038

Conversion: Use the Type K thermocouple conversion in Section 0 using the converted TCCJ column, Section 10.2.35, for cold junction temperature (C).

10.3.19 TCBurner

Description: Thermocouple on the burner. No connection when the BRE burner is installed.

10.3.20 RadiometerBREOffset+60

Description: Radiometer at the 60mm position on the BRE radiometer assembly.

Device: Dexter Research Center XP-0249 Thermopile Detector

Conversion Source: ACME-RPT-012, BRE Radiometer Characterization Report, Rev -.

Conversion: Irradiance (W/cm^2) = 0.001140 * Voltage - 0.001481

10.3.21 RadiometerBREOffset+60CJ

Description: Cold junction temperature sensor associated with the 60mm BRE radiometer.

Device: TI LM20

Conversion: $Temperature\ (C) = -1481.96 + \sqrt{2196200 + 257700(1.8639 - Voltage)}$

10.3.22 RadiometerBREOffset+30

Description: Radiometer at the 30mm position on the BRE radiometer assembly.

Device: Dexter Research Center XP-0249 Thermopile Detector

Conversion Source: ACME-RPT-012, BRE Radiometer Characterization Report, Rev -.

Conversion: Irradiance (W/cm^2) = 0.001127 * Voltage - 0.001579

10.3.23 RadiometerBREOffset+30CJ

Description: Cold junction temperature sensor associated with the 30mm BRE radiometer.

Device: TI LM20

Conversion: $Temperature\ (C) = -1481.96 + \sqrt{2196200 + 257700(1.8639 - Voltage)}$

10.3.24 RadiometerBREOffset-10

Description: Radiometer at the -10mm position on the BRE radiometer assembly.

Device: Dexter Research Center XP-0249 Thermopile Detector

Conversion Source: ACME-RPT-012, BRE Radiometer Characterization Report, Rev -.

Conversion: Irradiance (W/cm^2) = 0.001188 * Voltage - 0.001583

10.3.25 RadiometerBREOffset-10CJ

Description: Cold junction temperature sensor associated with the -10mm BRE radiometer.

Device: TI LM20

Conversion: $Temperature\ (C) = -1481.96 + \sqrt{2196200 + 257700(1.8639 - Voltage)}$

10.3.26 Ch13

Description: Not used during BRE experiments

10.3.27 Ch14

Description: Not used during BRE experiments

10.3.28 Ch15

Description: Not used during BRE experiments

10.3.29 Ch16

Description: Not used during BRE experiments

10.3.30 RadiometerOffset

Description: Radiometer located on the downstream relay board.

Device: Dexter Research Center DX-0869 Thermopile Detector

Conversion Source: ACME-RPT-011, Radiometer Characterization Report, Rev -.

Conversion: Irradiance (W/cm^2) = 0.001798 * Voltage - 0.002307

10.3.31 RadiometerOffsetCJ

Description: Cold junction temperature sensor associated with the offset radiometer.

Device: TI LM20

Conversion: $Temperature\ (C) = -1481.96 + \sqrt{2196200 + 257700(1.8639 - Voltage)}$

10.3.32 TCE

Description: Not used – spare

10.3.33 BREHeatFluxCenter

Description: BRE burner center heat flux sensor.

Device: Medtherm 8-2.688-10SB-0-0-36-21427KSA.

Conversion Source: Device quote indicates that output is 10 BTU/ft^2 at 10 mV.

Conversion: Irradiance (W/cm^2) = 1135.6538527 * Voltage

10.3.34 BREHeatFluxOffset

Description: BRE burner offset heat flux sensor.

Device: Medtherm 8-2.688-10SB-0-0-36-21427KSA.

Conversion Source: Device quote indicates that output is 10 BTU/ft^2 at 10 mV.

Conversion: Irradiance (W/cm^2) = 1135.6538527 * Voltage

10.3.35 TCCJ

Description: Temperature sensor located on the cold junction compensation assembly used for cold junction compensation of thermocouples.

Device: Analog Devices AD590

Conversion: Temperature (C) = 338.98305 * Voltage - 273.15

10.3.36 EfieldSetpointOut

Description: Control voltage output to the E-Field supply to achieve E-Field voltage from 0 to +/- 10kV.

Conversion: E-Field Voltage (V) = C * EfieldSetpointOut^2 + B * EfieldSetpointOut + A,

where A, B,and C depend on supply polarity:

Negative polarity	Positive polarity
A = -10886.5	A = -8.79
B = 1048.203	B = 1074.01
C = 2.7365	C = -1.594

The avionics software uses the inverse of this equation to determine the control voltage for the commanded E-Field voltage:

$$EfieldSetpointOut = \frac{-\mathrm{B} + \sqrt{B^2 - 4\mathrm{C}(\mathrm{A} - \mathrm{EFieldVoltage})}}{2C}$$

10.4 Type K Thermocouple Conversion

Input to this conversion is thermocouple output in millivolts and cold junction temperature in Celcius. First convert cold junction temperature (C), cjTemp, to millivolts in regard to a type K thermocouple:

$$cjMv = -0.176004136860\text{E}-01 + 0.389212049750\text{E}-01 * \text{cjTemp} + 0.185587700320\text{E}-04 * cjTemp^2 - 0.994575928740\text{E}-07 * cjTemp^3 + 0.318409457190\text{E}-09 * cjTemp^4 - 0.560728448890\text{E}-12 * cjTemp^5 + 0.560750590590\text{E}-15 * cjTemp^6 - 0.320207200030\text{E}-18 * cjTemp^7 + 0.971511471520\text{E}-22 * cjTemp^8 - 0.121047212750\text{E}-25 * cjTemp^9 + a0 * e^{a1(cjTemp-a2)^2}$$

,where

a0 = 0.118597600000E+00
a1 = -0.118343200000E-03
a2 = 0.126968600000E+03

Next add cjMv to thermocouple output: totalMv = cjMv + tcMv

If totalMv is between 0 and 20.644 mV

$$Temperature\ (C) = 25.08355 * \text{totalMv} + 0.07860106 * totalMv^2 - 0.2503131 * totalMv^3 + 0.08315270 * totalMv^4 - 0.01228034 * totalMv^5 + 0.0009804036 * totalMv^6 - 0.00004413030 * totalMv^7 + 0.000001057734 * totalMv^8 - 0.00000001052755 * totalMv^9$$

If totalMv is between 20.644 and 54.886 mV

$$Temperature\ (C) = -131.8058 + 48.30222 * \text{totalMv} - 1.646031 * totalMv^2 + 0.05464731 * totalMv^3 - 0.0009650715 * totalMv^4 + 0.000008802193 * totalMv^5 - 0.00000003110810 * totalMv^6$$

If totalMv is less than 0 then the temperature is between -200 and 0 C and is likely invalid for this application. If totalMv is greater than 54.886 then the response is out of range of the thermocouple.

10.5 Type B Thermocouple Conversion

Input to this conversion is thermocouple output in millivolts and cold junction temperature in Celcius. First convert cold junction temperature (C), cjTemp, to millivolts in regard to a type B thermocouple:

If cjTemp < 630 C use:

$$cjMv = -0.246508183460\text{E}-03 * \text{cjTemp} + 0.590404211710\text{E}-05 * cjTemp^2 - 0.132579316360\text{E}-08 * cjTemp^3 + 0.156682919010\text{E}-11 * cjTemp^4 - 0.169445292400\text{E}-14 * cjTemp^5 + 0.629903470940\text{E}-18 * cjTemp^6$$

If cjTemp >= 630 C use:

$$cjMv = -0.389381686210\text{E}+01 + 0.285717474700\text{E}-01 * \text{cjTemp} - 0.848851047850\text{E}-04 * cjTemp^2 + 0.157852801640\text{E}-06 * cjTemp^3 - 0.168353448640\text{E}-09 * cjTemp^4 + 0.111097940130\text{E}-12 * cjTemp^5 - 0.445154310330\text{E}-16 * cjTemp^6 + 0.989756408210\text{E}-20 * cjTemp^7 - 0.937913302890\text{E}-24 * cjTemp^8$$

Next add cjMv to thermocouple output. totalMv = cjMv + tcMv

If totalMv is between 0.291 and 2.431 mV use:

$$Temperature\ (C) = 98.423321 + 699.71500 * \text{totalMv} - 847.65304 * totalMv^2 + 1005.2644 * totalMv^3 - 833.45952 * totalMv^4 + 455.08542 * totalMv^5 - 155.23037 * totalMv^6 + 29.886750 * totalMv^7 - 2.4742860 * totalMv^8$$

If totalMv is between 2.431 and 13.820 mV

$$Temperature\ (C) = 213.15071 + 285.10504 * \text{totalMv} - 52.742887 * totalMv^2 + 9.9160804 * totalMv^3 - 1.2965303 * totalMv^4 + 0.11195870 * totalMv^5 - 0.0060625199 * totalMv^6 + 0.00018661696 * totalMv^7 - 0.0000024878585 * totalMv^8$$

If totalMv is less than 0.291 mV or greater than 13.820 mV then the response is out of range of the thermcouple.

11.0 ACRONYMS

ACME	Advanced Combustion via Microgravity Experiments
AGC	Automatic Gain Control
AGS	ACME Ground Software
BASH	Bourne Again SHell
BRE	Burning Rate Emulator
CIR	Combustion Integrated Rack
CLD	Coflow Laminar Diffusion
CSCI	Computer Software Configuration Item
ECMA	European Computer Manufacturers Association
E-Field	Electric Field
FCF	Fluids and Combustion Facility
FTP	File Transfer Protocol
IDD	Interface Design Description
IOP	Input Output Processor
IPSU	Image Processing and Storage Unit
IPSU-G	Image Processing and Storage Unit with GigE Interfaces
LEMT	Lippert Enhanced Management Technology
MFC	Mass Flow Controller
MP	Main Processor
PI	Prinicpal Investigator
PMD	Performance Motion Devices
PMT	Photomultiplier
ROI	Region Of Interest
SBC	Single Board Computer
SLPM	Standard Liters Per Minut
SV	Solenoid Valve
TFP	Thin Filament Pyrometry
TSC	Telescience Center
UEI	United Electronic Industries